

Some Conventions and Notations

- If not specified otherwise, A will always denote a commutative ring with unity.
- We assume that every ring homomorphism maps the unity to the unity.
- We denote the nilradical as $\mathfrak{N}$, the Jacobson radical as $\mathfrak{J}$, provided that the ambient ring is clear.
- Sometimes, we use $[n]$ to denote the set $\{1, \dots, n\}$ and use $[[n]]$ to denote $\{0, \dots, n\}$.
- The term *either A or B ...* is **only** used to describe ***exactly one of A and B ...***. The term *A or B ...* means ***at least one of A and B ...***.
- Sometimes we use $\diamond$ to denote ring actions.

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1 Chapter 1 Rings and Ideals

1.1 Exercises

1.1.1 Basics

1.1. Let x be a nilpotent element of a ring A . Show that $1 + x$ is a unit of A . And deduce that the sum of a nilpotent element and a unit is a unit.

Proof. x is nilpotent $\Rightarrow x \in \mathfrak{R} \Rightarrow x \in \mathfrak{J}$. Suppose that $1 + x$ is a non-unit, then $\exists$ a maximal ideal $\mathfrak{m} \triangleleft A$ s.t. $1 + x \in \mathfrak{m}$. And since $x \in \mathfrak{m}$, we must have $1 \in \mathfrak{m}$, a contradiction. Let u be an arbitrary unit of A . Suppose $u + x$ is a non-unit, then by the same argument, $\exists$ a maximal ideal $\mathfrak{m}$ s.t. $u \in \mathfrak{m}$, a contradiction.

1.2. Let $f = a_0 + a_1x + \dots + a_nx^n \in A[x]$. Prove that:

(a). f is a unit in $A[x]$ if and only if a_0 is a unit in A and $a_1, \dots, a_n$ are nilpotent.

Proof. Suppose $g = b_0 + b_1x + \dots + b_mx^m$ is the inverse of f . I claim that $a_n^{r+1}b_{m-r} = 0$. Induction hypothesis: Assume $a_n^{k+1}b_{m-k} = 0$ for any $k \in [[r-1]]$. Base case: $a_nb_m = 0$. The coefficient of the degree $m+n-r$ term of $f \cdot g$ is:

$$a_nb_{m-r} + \sum_{k=0}^{r-1} a_{n-(r-k)}b_{m-k} = 0$$

Multiplying by a_n^r on both sides of the above equation kills every term except $a_n^{r+1}b_{m-r}$. Hence, by taking $r = m$ we have $a_n^{m+1}b_0 = 0$, which implies a_n is nilpotent since b_0 is clearly a unit. Thus proves $a_1, \dots, a_n$ are all nilpotent by induction. To show the converse direction, note that $a_1, \dots, a_n$ are nilpotent, $\Rightarrow a_1x, \dots, a_nx^n$ are nilpotent, which implies the sum $a_1x + \dots + a_nx^n$ is nilpotent; then, applying 1.1. shows that f is a unit.

(b). f is nilpotent $\Leftrightarrow a_0, a_1, \dots, a_n$ are nilpotent.

Proof. The direct proof for both sides are straightforward. We try an alternative way: Consider the canonical projection $\phi : A \rightarrow A/\mathfrak{R}$. Note that $(A/\mathfrak{R})[x]$ is nilpotent-free except 0. This is because if $h \neq 0 \in (A/\mathfrak{R})[x]$ is nilpotent, then the leading term of h is nilpotent in $A/\mathfrak{R}$, a contradiction. Therefore, f is nilpotent $\Rightarrow \bar{f} := \phi(f) = 0$, which forces every term of $\bar{f}$ to be zero since $(A/\mathfrak{R})[x]$ is a free $A/\mathfrak{R}$ -module. Hence $a_0, \dots, a_n$ are all in $\mathfrak{R}$.

(c). f is a zero-divisor $\Leftrightarrow \exists a \neq 0 \in A$ s.t. $af = 0$.

Proof. Take non-zero $g = b_0 + \dots + b_mx^m \in A[x]$ s.t. $f \cdot g = 0$ with least degree. Then $a_n \cdot g \cdot f = 0$. Note that a_ng is a polynomial of degree less than g since $a_nb_m = 0$, thus implying $a_ng = 0$. Now suppose $a_{n-k}g = 0, \forall k = 0, \dots, r-1$, then $b_ia_{n-k} = 0, \forall i \in [[n]]$. Note that the coefficient of the degree $m+n-r$ term of $f \cdot g$ is:

$$b_ma_{n-r} + \sum_{k=0}^{r-1} b_{m-(r-k)}a_{n-k} = 0$$

Therefore, $b_m a_{n-r} = 0$. Hence implies $a_{n-r} g = 0$ by the same reasoning above. Therefore, each coefficient b_j annihilates every coefficient a_i of f . If $\deg g > 0$, deleting the leading term of g would still annihilate f , contradicting the minimality of $\deg g$. This implies $g = b_0$. The converse direction is just the definition of a zero-divisor.

Remark 1. This tells us if $f \in A[x]$ is a zero-divisor, let g be the polynomial as defined above. Then g must be a constant, i.e., an element in A .

(d). f is said to be **primitive** if $\langle a_0, a_1, \dots, a_n \rangle = \langle 1 \rangle$. Prove that if $f, g \in A[x]$, $f \cdot g$ is primitive $\Leftrightarrow f$ and g are primitive.

Proof. $\Rightarrow$: Let $g = b_0 + \dots + b_m x^m$. Let $\mathfrak{a} := \langle a_0, \dots, a_n \rangle$, $\mathfrak{b} := \langle b_0, \dots, b_m \rangle$. Let $\mathfrak{c}$ denote the ideal generated by the coefficients of $f \cdot g$. Since fg is primitive, we have $\mathfrak{c} = \langle 1 \rangle$. However, clearly $\mathfrak{c} \subseteq \mathfrak{a}\mathfrak{b} \subseteq \mathfrak{a} \cap \mathfrak{b}$. Thus, it follows $\mathfrak{a} = \mathfrak{b} = \langle 1 \rangle$. $\Leftarrow$: Suppose $\mathfrak{c} \neq \langle 1 \rangle$. Then there exists a maximal ideal $\mathfrak{m} \triangleleft A$ s.t. $\mathfrak{c} \subseteq \mathfrak{m}$. Consider the quotient $A/\mathfrak{m}$, we have $0 = \overline{fg} = \overline{f} \cdot \overline{g}$. But since $A/\mathfrak{m}$ is a field, we have no zero-divisors in $(A/\mathfrak{m})[x]$. Therefore, at least one of $\overline{g}$ and $\overline{f}$ is 0, implying $\mathfrak{b} \subseteq \mathfrak{m} \neq \langle 1 \rangle$, or $\mathfrak{a} \subseteq \mathfrak{m} \neq \langle 1 \rangle$. Therefore, f and g cannot be both primitive.

1.3. Generalize the results of 1.2. to a polynomial ring $A[x_1, \dots, x_r]$ in several indeterminates.

Let $f \in A[x_1, \dots, x_r]$, the coefficients of f in A are denoted as $a_0, \dots, a_N$ where a_0 is the constant term. Let's denote the ring $A[x_1, \dots, x_{r-1}]$ as R , then we may consider $A[x_1, \dots, x_r]$ as $R[x_r]$ and write $f = r_0 + r_1 x_r + \dots + r_n x_r^n$ where $r_0, \dots, r_n \in R$. We first generalize the result (b) in 1.2. I claim:

(b'). f is nilpotent $\Leftrightarrow$ all the coefficients of f in A are nilpotent.

Proof. From 1.2.(b) we know that $r_0, \dots, r_n$ are all nilpotent in R . But each $r_i, i \in [[n]]$ is in $A[x_1, \dots, x_{r-2}][x_{r-1}]$. Therefore, by inductively using 1.2.(b) we eventually see that every coefficient of f is nilpotent in A . The converse direction is immediate.

(a'). f is a unit in $A[x_1, \dots, x_r] \Leftrightarrow a_0$ is a unit and $a_1, \dots, a_n$ are nilpotent.

Proof. If f is a unit, then by 1.2.(a), we know (i) r_0 is a unit in R , and hence by inductively applying 1.2.(a), we have a_0 is a unit in A . (ii) $r_1, \dots, r_n$ are nilpotent in R , and hence by (b'), all the coefficients of them are nilpotent in A , which follows that $a_1, \dots, a_N$ are nilpotent. The converse direction follows from the same reasoning in 1.1(a).

(c'). f is a zero-divisor $\Leftrightarrow \exists a \neq 0 \in A$ s.t. $af = 0$.

Proof. $\Rightarrow$: By 1.2.(c) we know that $\exists r \neq 0 \in R$ s.t. $rf = 0$. Then consider the ring homomorphism $\psi : R[x_r] \rightarrow R$ by sending $x_r \mapsto 1_A$. Then ψ preserves zero-divisors since it's injective. Consider $f' := \psi(f) \in R$, then by 1.2.(c) again, we have a non-zero r' in the coefficient ring of R s.t. $r'f' = 0$, follows that $r'f = 0$. Therefore, by applying 1.2.(c) inductively, we eventually have some non-zero $a \in A$ s.t. $af = 0$.

(d'). $f, g \in A[x_1, \dots, x_r]$, then fg is primitive $\Leftrightarrow f$ and g are primitive.

Proof. This is a direct generalization of 1.2.(d), since $A[x_1, \dots, x_r]$ is a free A -module with a basis of all monomials $\{x_1^{\alpha_1} \cdots x_r^{\alpha_n}\}$ where $\alpha_i \in \mathbb{N}$, and primitivity only concerns the ideal generated by the coefficients.

1.4. In the ring $A[x]$, $\mathfrak{J} = \mathfrak{R}$.

Proof. We know $\mathfrak{R} \subseteq \mathfrak{J}$. Hence, suppose $f \in \mathfrak{J}$, it suffices to show that f is nilpotent. By proposition 1.9. we know $1 + xf$ is a unit in $A[x]$. But this implies, by 1.2.(a), all the coefficients of f are nilpotent in A , which further implies, by 1.2.(b), f is nilpotent.

1.5. Let's consider the ring of formal power series $A[[x]]$. Let $f = \sum_{i=0}^{\infty} a_i x^i$, Show that:

(a). f is a unit in $A[[x]] \Leftrightarrow a_0$ is a unit in A .

Proof. Suppose a_0 is a unit. It suffices to construct a $g = b_0 + b_1 x + \dots \in A[[x]]$ s.t. $fg = 1$. We construct the $\{b_j\}_{j=0}^{\infty}$ by the following:

$$b_0 = a_0^{-1} \quad b_1 = -a_1 b_0 \cdot a_0^{-1} \quad b_2 = -(a_2 b_0 + a_1 b_1) \cdot a_0^{-1} \quad \dots \quad b_j = -\sum_{i=0}^{j-1} a_{j-i} b_i \cdot a_0^{-1} \quad \dots$$

The converse direction is immediate.

(b). f is nilpotent $\Rightarrow a_i$ is nilpotent for all $i \geq 0$.

Proof. Apply the exact same reasoning in 1.2.(b).

(c). $f \in \mathfrak{J}(A[[x]]) \Leftrightarrow a_0 \in \mathfrak{J}(A)$.

Proof. $f \in \mathfrak{J}(A[[x]]) \xLeftrightarrow{\text{Prop 1.9.}} 1 + fg = 1 + a_0 b_0 + (a_0 b_1 + a_1 b_0)x + \dots$ is a unit in $A[[x]]$, $\forall g = b_0 + b_1 x + \dots \xLeftrightarrow{1.5.(a)} 1 + a_0 b_0$ is a unit in $A \xLeftrightarrow{\text{Prop 1.9.}} a_0 \in \mathfrak{J}(A)$.

(d). The contraction of a maximal ideal $\mathfrak{m} \triangleleft A[[x]]$, denoted as $\mathfrak{m}^c$, is a maximal ideal of A , and $\mathfrak{m}$ is generated by $\mathfrak{m}^c$ and x .

Proof. Let $\psi : A \hookrightarrow A[[x]]$ be the canonical embedding. Note that since $A \cong A[[x]]/\langle x \rangle$, we have the canonical projection $\phi : A[[x]] \twoheadrightarrow A$. Therefore, $\phi \circ \psi = \text{id}_A$.

$$\text{id} : A \xhookrightarrow{\psi} A[[x]] \twoheadrightarrow_{\phi} A$$

Now I claim that $\mathfrak{m}^c = \phi(\mathfrak{m})$. $\subseteq$: $\forall a \in \mathfrak{m}^c \Rightarrow \psi(a) \in \mathfrak{m} \Rightarrow a = \phi(\psi(a)) \in \phi(\mathfrak{m})$. $\supseteq$: $\forall a \in \phi(\mathfrak{m})$, pick any $f \in \mathfrak{m}$ with $\phi(f) = a$. Write $f = a + xh$ where $h \in A[[x]]$. Since $a = f - xh \in \mathfrak{m}$, we have $a = \phi^{-1}(a) \in \mathfrak{m}^c$. Note that if $\mathfrak{m} \triangleleft A[[x]]$ is maximal, then $\phi(\mathfrak{m})$ is a maximal ideal in A by the correspondence theorem. So is $\mathfrak{m}^c$. To show $\mathfrak{m} = \langle \mathfrak{m}^c, x \rangle$, note that $\supseteq$ is clear since $\mathfrak{m}$ is maximal. $\subseteq$: Take any $f = a_0 + xg \in \mathfrak{m}$. Note that $a_0 \in \mathfrak{m}^c$, hence $f \in \langle \mathfrak{m}^c, x \rangle$.

(e). Every prime ideal of A is the contraction of a prime ideal of $A[[x]]$.

Proof. Consider the composition $\phi \circ \psi = \text{id}_A$ defined in (d) again. Take any prime ideal $\mathfrak{p} \triangleleft A$. It suffices to show that, $\phi^{-1}(\mathfrak{p}) \triangleleft A[[x]]$ is prime. Take arbitrary $f = a_0 + a_1x + \dots$, $g = b_0 + b_1x + \dots$ s.t. $fg = a_0b_0 + xh \in \phi(\mathfrak{p})$ for some $h \in A[[x]]$. Note that $\phi(f)\phi(g) \in \mathfrak{p}$ implies $a_0 \in \mathfrak{p}$ or $b_0 \in \mathfrak{p}$, which implies $f \in \phi^{-1}(\mathfrak{p})$ or $g \in \phi^{-1}(\mathfrak{p})$. Thus completes the proof.

1.6. A ring s.t. every ideal not contained in $\mathfrak{R}$ contains a non-zero idempotent (that is, an element e s.t. $e^2 = e \neq 0$). Prove that $\mathfrak{R} = \mathfrak{J}$.

Proof. It suffices to show $\mathfrak{R} \subseteq \mathfrak{J}$. First, I claim that for any integer $n \geq 2$, $e^n = e$. Base case: $e^2 = e$. For $n > 2$, $e^n = e^{n-1}e = e^2 = e$ by induction. Therefore, $\mathfrak{R}$ can not contain any non-zero idempotent. Now, suppose the contrary that $\mathfrak{J} \subsetneq \mathfrak{R}$. Then $\exists e \neq 0 \in \mathfrak{J}$ s.t. $e^2 = e$. But this implies $e \in \mathfrak{R}$, a contradiction.

1.7. Let A be a ring in which every element x satisfies $x^n = x$ for some $n > 1$ (depending on x). Show that every prime ideal in A is maximal.

Proof. Take any prime ideal $\mathfrak{p} \triangleleft A$. Pick any $\bar{x} \neq \bar{0} \in A/\mathfrak{p}$, we know that $\exists n > 1$ s.t. $\bar{x}^n - \bar{x} = \bar{x}(\bar{x}^{n-1} - 1) = 0$. Note that $A/\mathfrak{p}$ is an integral domain, hence $\bar{x}^{n-1} - 1 = 0$, implying that $\bar{x}^{-1} = \bar{x}^{n-2}$. Since $\bar{x}$ is arbitrarily chosen, $A/\mathfrak{p}$ is a field, follows that $\mathfrak{p}$ is maximal in A .

1.8. Let A be a ring $\neq 0$. Show that the set of prime ideals of A has minimal elements with respect to inclusion.

Proof. Let Σ be the set of all prime ideals of A . Clearly Σ is not empty since every ring has a maximal ideal which is prime. Define a partial ordering $\preceq$ on Σ as: $\mathfrak{p} \preceq \mathfrak{q} \Leftrightarrow \mathfrak{q} \subseteq \mathfrak{p}$. Then by Zorn's lemma, it suffices to show that every ascending chain in Σ has an upper bound. Take any chain $\mathfrak{p}_1 \preceq \mathfrak{p}_2 \preceq \dots$, denoted as $\{\mathfrak{p}_i\}_{i \in I}$ where I is the index set. I claim that $\bigcap_{i \in I} \mathfrak{p}_i$, denoted as $\mathfrak{a}$, is a desired one. it's clear that $\mathfrak{a} \subseteq \mathfrak{p}_i \forall i \in I$, so it remains to show that $\mathfrak{a} \in \Sigma$. Take any $x, y \in A$ but $\notin \mathfrak{a}$, then there exists a $\mathfrak{p}_k$ where $k \in I$ s.t. $x, y \notin \mathfrak{p}_k$. Hence $xy \notin \mathfrak{p}_k$ since $\mathfrak{p}_k$ is prime. This implies $xy \notin \mathfrak{a}$ since $\mathfrak{a} \subseteq \mathfrak{p}_k$. Therefore, $\mathfrak{a}$ is prime and hence in Σ , as desired.

Remark 2. The minimal prime ideal (defined as in 1.8.) is obviously not necessarily unique. And when they are not unique, the $\mathfrak{R}$ can not be prime. In fact, the intersection of distinct minimal prime ideals can not be prime. Suppose $\mathfrak{p}_1, \dots, \mathfrak{p}_n$ are distinct minimal primes. Let $\mathfrak{b} = \bigcap_{i=1}^n \mathfrak{p}_i$, then $\mathfrak{p}_1 \cdots \mathfrak{p}_n \subseteq \mathfrak{b}$. $\mathfrak{b}$ prime would imply one of those $\mathfrak{p}_i = \mathfrak{b}$, contradicts the minimality.

1.9. Let $\mathfrak{a} \neq \langle 1 \rangle \triangleleft A$. Show that $\mathfrak{a} = \text{rad}(\mathfrak{a}) \Leftrightarrow \mathfrak{a}$ is an intersection of prime ideals.

Proof. This is an immediate corollary following from the fact (Prop 1.14.): $\text{rad}(\mathfrak{a}) =$ the intersection of all prime ideals that contain $\mathfrak{a}$.

1.10. Let A be a ring, show that the following are equivalent:

- (i) . A has exactly one prime ideal.
- (ii). Every element of A is either a unit or a nilpotent.
- (iii). $A/\mathfrak{R}$ is a field.

Proof. (i) $\Rightarrow$ (ii): Since every ring has at least one maximal ideal, which is prime, we know that the unique prime ideal in A must be maximal, which is also the nilradical of A , denoted as $\mathfrak{N}$. If $e \in \mathfrak{N}$, then it is nilpotent; if not, then by Corollary 1.5. we know it is a unit. (ii) $\Rightarrow$ (iii): Every non-zero element in $A/\mathfrak{N}$ is a unit, hence $A/\mathfrak{N}$ is a field. (iii) $\Rightarrow$ (i): $\mathfrak{N}$ is maximal $\Rightarrow \forall$ prime ideal $\mathfrak{p}$ of A equals $\mathfrak{N}$, done.

1.11. A ring A is **Boolean** if $x^2 = x, \forall x \in A$. In a Boolean ring A , show that:

- (i). $2x = 0, \forall x \in A$.
- (ii). Every prime ideal $\mathfrak{p}$ is maximal, and $A/\mathfrak{p}$ is a field with 2 elements.
- (iii). Every finitely generated ideal in A is principal.

Proof. (i): Note that $x = x^2 = (-x)^2 = -x^2 = -x$. Hence $2x = x + x = 0$. (ii): $\mathfrak{p}$ is maximal follows directly from 1.7. Take $\forall x \notin \mathfrak{p}$, consider $\bar{x} \in A/\mathfrak{p}$, we have $\bar{0} = \bar{x}^2 - \bar{x} = \bar{x}(\bar{x} - \bar{1})$. Hence $\bar{x} = \bar{1}$, meaning that $A/\mathfrak{p}$ only has 2 elements: $\bar{0}, \bar{1}$. (iii): It suffices to show that every ideal generated by 2 elements can be generated by a single element. Let $\mathfrak{a} = \langle x, y \rangle$. I claim that $\mathfrak{a} = \langle x + y + xy \rangle$. $\supseteq$ is trivial. To show $\subseteq$, it suffices to show that $x + y + xy$ generates x . Note that $x(x + y + xy) = x^2 + 2xy = x$, so we are done.

1.12. A local ring contains no idempotent $\neq 0, 1$.

Proof. Let e be an idempotent $\neq 1, 0$. If e is a unit, then $e = e^{-1}e^2 = e^{-1}e = 1$, hence e is a non-unit. Then by Corollary 1.5., $e \in \mathfrak{m}$ for some maximal ideal $\mathfrak{m}$. But since A is a local ring, the only maximal ideal is its Jacobson radical $\mathfrak{J}$, so $e \in \mathfrak{J}$. This implies, by Prop 1.9., $1 + e$ is a unit. However, note that $1 + e$ is also idempotent since $(1 + e)^2 = 1 + 2e + e^2 = 1 + e$. This means $1 + e = 1$, so $e = 0$, a contradiction. Alternatively, note that $1 - e$ is also an idempotent $\neq 1, 0$, hence a non-unit and $\in \mathfrak{J}$. But $e + (1 - e) = 1$ implies $\mathfrak{J}$ is the entire ring, contradiction.

Construction of an algebraic closure of a field (E. Artin).

1.13. Let K be a field and let Σ be the set of all irreducible monic polynomials f in one indeterminate with coefficients in K . Let A be the polynomial ring over K generated by indeterminates x_f , one for each $f \in \Sigma$. Let $\mathfrak{a}$ be the ideal of A generated by the polynomials $f(x_f)$ for all $f \in \Sigma$. Show that $\mathfrak{a} \neq \langle 1 \rangle$.

Proof. Suppose the contrary that $\mathfrak{a} = \langle 1 \rangle$. Then there exist finitely many irreducible monic polynomials $f_1, \dots, f_n \in \Sigma$ and polynomials $g_1, \dots, g_n \in A$ s.t.

$$g_1(x_{f_1}, \dots, x_{f_n})f_1(x_1) + \dots + g_n(x_{f_1}, \dots, x_{f_n})f_n(x_n) = 1$$

Since this equality is polynomial identity, it must hold true under any evaluation of variables in any extension field of K . For each f_i , let L_i be its splitting field, in which f_i has at least one root, say α_i . Let L be the compositum of all L_i and consider the evaluation homomorphism: $A \rightarrow L$ by sending $x_{f_i} \mapsto \alpha_i$. Then we have $0 = 1$ in L , a contradiction.

Remark 3. Note that for any field K , we always have:

$$f_1, \dots, f_n \text{ have common root(s)} \implies \langle f_1, \dots, f_n \rangle \neq \langle 1 \rangle$$

This is because of the same reasoning: if $1 = g_1 f_1 + \dots + g_n f_n$ for some $g_1, \dots, g_n$ in the polynomial ring. Then $f_1(\alpha) = \dots = f_n(\alpha) = 0$ for some $\alpha \in K^n$ would yield a contradiction $0 = 1$. Note that the converse direction is not true in general. E.x., take $f = x^2 - 2 \in \mathbb{Q}[x]$. But if K is algebraically closed, it is indeed an 'if and only if', due to the Weak Nullstellensatz.

Let $\mathfrak{m}$ be a maximal ideal of A containing $\mathfrak{a}$, and let $K_1 = A/\mathfrak{m}$. Then K_1 is an extension field of K in which each $f \in \Sigma$ has a root. Repeat the construction with K_1 in place of K , obtaining K_2 , and so on. Let $L = \cup_{i=1}^{\infty} K_i$. Then L is a field in which each $f \in \Sigma$ splits completely into linear factors. Let $\overline{K}$ be the set of all elements of L which are algebraic over K . Then $\overline{K}$ is an algebraic closure of K .

Remark 4. Before proving the above statement, let's first make some comments:

► Why isn't K_1 the algebraic closure of K already? Because as we are introducing new algebraic elements over K , we have to ensure that polynomials over the newly constructed field also have roots (in order to form an algebraic closure), but we didn't care about this in this step. In other words, $\overline{K_1} \neq K_1$.

► How can L contain elements not algebraic over K ? Note that we are indeed doing something like constructing:

$$K \subset K(\alpha_1, \alpha_2, \dots) =: K_1 \subset K_1(\beta_1, \beta_2, \dots) =: K_2 \subset \dots$$

where each extension is clearly algebraic. However, since we are doing it infinitely many times, and infinite extension may be transcendental, L may (but not necessarily) contain transcendental elements.

Proof. It suffices to show that each polynomial in $\overline{K}[x]$ has a root in $\overline{K}$. Let $K_0 := K$, by the above construction, we know that any polynomial $f \in \cup_{j=0}^i K_i[x]$ has a root in K_{i+1} , $\forall i = 0, 1, 2, \dots$. Therefore, $\forall f \in \overline{K}[x], \exists i \in \mathbb{N}$ s.t. $f \in \cup_{j=0}^i K_i[x]$, implying f has a root in K_{i+1} , but $K_{i+1} \subset \overline{K}$ follows that f has a root in $\overline{K}$, as desired. Or simply, f has a root in L , and $L \subseteq \overline{K}$, so f has a root in $\overline{K}$.

1.14. In a ring A , let Σ be the set of all ideals in which every element is a zero-divisor. Show that Σ has maximal elements and that every maximal element of Σ is a prime ideal. Hence the set of zero-divisors in A is a union of prime ideals.

Proof. To show that Σ has maximal elements, we use Zorn's lemma. First note that Σ is not empty since $\langle 0 \rangle \in \Sigma$. Then, take any chain, ordered by inclusion, say $\{\mathfrak{a}_j\}_\alpha$ in Σ . Note that $\mathfrak{b} := \cup_\alpha \mathfrak{a}_j$ is an ideal and clearly an upper bound, so it remains to show that every element in $\mathfrak{b}$ is a zero-divisor. Take $\forall x \in \mathfrak{b}$, we know $x \in \mathfrak{a}_i$ for some i indexed by α , hence x is a zero-divisor. Let $\mathfrak{b}$ be a maximal element in Σ . (A special case: If $\mathfrak{b} = \langle 0 \rangle$, then the only zero-divisor in A is 0, i.e., $A = A/\langle 0 \rangle$ is an integral domain, hence $\mathfrak{b} = \langle 0 \rangle$ is a prime ideal). For the general case, we mimic the proof of Prop.1.8.: Take arbitrary $x, y \notin \mathfrak{b}$, then $\mathfrak{b} + \langle x \rangle, \mathfrak{b} + \langle y \rangle \notin \Sigma$. Now I claim that if $p, q \neq 0 \in A$ are not zero-divisors, then pq cannot be a zero-divisor. Because, suppose the contrary, then $\exists r \neq 0 \in A$ s.t. $pqr = 0$. q is not a zero-divisor $\Rightarrow qr \neq 0 \Rightarrow p$ is a zero-divisor, contradiction. Therefore, $\exists p, q \in \mathfrak{b} + \langle x \rangle, \mathfrak{b} + \langle y \rangle$ respectively s.t. $pq \in (\mathfrak{b} + \langle x \rangle)(\mathfrak{b} + \langle y \rangle) \subseteq \mathfrak{b} + \langle xy \rangle$ but $pq \notin \mathfrak{b}$. This implies $xy \notin \mathfrak{b}$, follows that $\mathfrak{b}$ is prime, as desired.

1.1.2 The prime spectrum of a ring

1.15. Let X be the set of all prime ideals of the ring A . For each subset $E \subseteq A$, let $V(E)$ denote the set of all prime ideals of A which contain E . Prove that:

(a). Let $\mathfrak{a}$ be the ideal generated by E , then $V(E) = V(\mathfrak{a}) = V(\text{rad}(\mathfrak{a}))$.

Proof. To show $V(E) = V(\mathfrak{a})$, Note that $\supseteq$ is trivial. $\subseteq$: For any $\mathfrak{p} \in V(E) \Rightarrow \mathfrak{p} \supseteq E \Rightarrow \mathfrak{p} \supseteq \mathfrak{a} \Rightarrow \mathfrak{p} \in V(\mathfrak{a})$. $V(\mathfrak{a}) = V(\text{rad}(\mathfrak{a}))$ is immediate by Prop 1.14.: $\text{rad}(\mathfrak{a}) = \bigcap_{\mathfrak{p} \in V(\mathfrak{a})} \mathfrak{p}$.

(b). $V(0) = X$, $V(A) = \emptyset$.

Proof. Any prime ideal contains 0, and prime ideals cannot be the entire ring, provided that A is non-trivial. In fact, we have: $V(E) = \emptyset \Leftrightarrow E$ generates 1_A , since Corollary 1.4.

(c). If $\{E_i\}_{i \in I}$ is any family of subsets of A , then:

$$V\left(\bigcup_{i \in I} E_i\right) = \bigcap_{i \in I} V(E_i)$$

Proof. $\mathfrak{p} \in \text{LHS} \Leftrightarrow \mathfrak{p}$ contains every $E_{i \in I} \Leftrightarrow \mathfrak{p} \in \text{RHS}$.

(d). $V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}\mathfrak{b}) = V(\mathfrak{a}) \cup V(\mathfrak{b})$. $\forall \mathfrak{a}, \mathfrak{b} \triangleleft A$.

Proof. To show the first equation, note that by Exer 1.13.(iii) and (a), we have: $V(\mathfrak{a} \cap \mathfrak{b}) = V(\text{rad}(\mathfrak{a} \cap \mathfrak{b})) = V(\text{rad}(\mathfrak{a}\mathfrak{b})) = V(\mathfrak{a}\mathfrak{b})$. $V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}) \cup V(\mathfrak{b})$ follows immediately from Prop 1.11.(ii).

These results show that the sets $V(E)$ satisfy the axioms for closed sets in a topological space. The resulting topology is called the **Zariski topology**. The topological space X is called the **prime spectrum** of A , denoted as $\text{Spec}(A)$.

1.17. For each $f \in A$, let X_f denote the complement of $V(f)$ in $X = \text{Spec}(A)$. Show that the sets X_f form a basis of open sets for Zariski topology and:

Proof. To show that $\{X_f\}$ forms a basis that **generates** the Zariski topology, we need to show that for any $x \in U$ where U is a non-empty open set, there exists a X_f s.t. $x \in X_f \subseteq U$. Let $U = X \setminus V(E)$, and $\mathfrak{p} \in U$. Then taking $f \in E \setminus V(E)$ satisfies $\mathfrak{p} \in X_f \subseteq X \setminus V(E)$.

(a). $X_f \cap X_g = X_{fg}$.

Proof. $\mathfrak{p} \in X_f \cap X_g \Leftrightarrow \mathfrak{p} \notin V(f)$ and $\mathfrak{p} \notin V(g) \Leftrightarrow f \notin \mathfrak{p}$ and $g \notin \mathfrak{p} \Leftrightarrow fg \notin \mathfrak{p} \Leftrightarrow \mathfrak{p} \notin V(fg) \Leftrightarrow \mathfrak{p} \in X_{fg}$.

(b). $X_f = \emptyset \Leftrightarrow f$ is nilpotent.

Proof. f is nilpotent $\Leftrightarrow f \in \mathfrak{N} \Leftrightarrow f \in \mathfrak{p}, \forall \mathfrak{p}$ prime $\Leftrightarrow \mathfrak{p} \in V(f), \forall \mathfrak{p} \Leftrightarrow V(f) = X \Leftrightarrow X_f = \emptyset$.

(c). $X_f = X \Leftrightarrow f$ is a unit.

Proof. f is a unit $\Leftrightarrow f \notin \mathfrak{p}, \forall \mathfrak{p}$ prime $\Leftrightarrow \mathfrak{p} \notin V(f), \forall \mathfrak{p} \Leftrightarrow V(f) = \emptyset \Leftrightarrow X_f = X$.

Remark 5. Be careful! Note that (a) and (c) together only imply that $\{X_f\}$ forms a basis (for some topology on X), but it does not directly imply that this topology is indeed the Zariski topology.

(d). $X_f = X_g \Leftrightarrow \text{rad}(\langle f \rangle) = \text{rad}(\langle g \rangle)$.

Proof. $X_f = X_g \Leftrightarrow V(f) = V(g) \Leftrightarrow V(\text{rad}(\langle f \rangle)) = V(\text{rad}(\langle g \rangle)) \Leftrightarrow \text{rad}(\langle f \rangle) = \text{rad}(\langle g \rangle)$.

(e). X is quasi-compact (that is, every open covering of X has a finite sub-covering).

Proof. It is enough to consider a covering of X by basic open sets X_{f_i} ($i \in I$). Note that:

$$\mathfrak{p} \in \bigcup_{i \in I} X_{f_i} \Leftrightarrow \exists i \in I \text{ s.t. } \mathfrak{p} \notin V(f_i) \Leftrightarrow \mathfrak{p} \not\supseteq \langle \{f_i\}_{i \in I} \rangle \Leftrightarrow \mathfrak{p} \in X \setminus V(\langle \{f_i\}_{i \in I} \rangle)$$

Therefore, if $\{X_{f_i}\}_{i \in I}$ covers X , then $V(\langle \{f_i\}_{i \in I} \rangle) = \emptyset$. By 1.15.(b) we have $\{f_i\}_{i \in I}$ generates 1_A . This implies, $\exists$ a finite subset of I , say indexed by $[n]$, s.t. $1_A = \sum_{i=1}^n g_i f_i$ where $g_i \in A$. Then $\{X_{f_i}\}_{i \in [n]}$ covers X , as desired.

(f). More generally, each X_f is quasi-compact.

Proof. Let $\mathfrak{a}$ be the ideal generated by $\{f_i\}_{i \in I}$. Then I claim:

$$X_f \subseteq \bigcup_{i \in I} X_{f_i} \Leftrightarrow f \in \text{rad}(\mathfrak{a})$$

Proof:

$$X_f \subseteq \bigcup_{i \in I} X_{f_i} \Leftrightarrow V(f) \supseteq \bigcap_{i \in I} V(f_i) \stackrel{1.15.(c)}{=} V\left(\bigcup_{i \in I} f_i\right) \stackrel{1.15.(a)}{=} V(\text{rad}(\mathfrak{a})) \Leftrightarrow f \in \text{rad}(\mathfrak{a})$$

Therefore, suppose $\{X_{f_i}\}_{i \in I}$ covers X_f , then $\exists m \in \mathbb{N}$ s.t. $f^m \in \mathfrak{a} \Rightarrow \exists [n] \subseteq I$ s.t. $f^m = \sum_{i=1}^n g_i f_i$ where $g_i \in A \Rightarrow f \in \text{rad}(\mathfrak{b})$ where $\mathfrak{b}$ is generated by $\{f_i\}_{i \in [n]}$. This implies $X_f \subseteq \bigcup_{i \in [n]} X_{f_i}$, as we desired. Note that by taking $f = 1_A$, we simply prove (e).

(g) An open subset of X is quasi-compact if and only if it is a finite union of sets X_f . The sets X_f are called **basic open sets** of $X = \text{Spec}(A)$.

Proof. $\Leftarrow$ is immediate since each basic open set is quasi-compact. For $\Rightarrow$, we can write $Y \subseteq X$ as an arbitrary union of basic open sets, and then take the finite subcover since Y is quasi-compact.

1.18. For psychological reason it is sometimes convenient to denote a prime ideal of A by a letter such as x or y when thinking of it as a point of $X = \text{Spec}(A)$. When thinking of x as a prime ideal of A , we denote it by $\mathfrak{p}_x$. Show that:

(b). $\overline{\{x\}} = V(\mathfrak{p}_x)$

Proof. Note that $V(E) \supseteq \{x\} \Leftrightarrow \mathfrak{p}_x \in V(E) \Leftrightarrow \mathfrak{p}_x \supseteq E$. Therefore,

$$y \in \overline{\{x\}} \Leftrightarrow \mathfrak{p}_y \in V(E), \forall E \subseteq \mathfrak{p}_x \Leftrightarrow \mathfrak{p}_y \in \bigcap_{E \subseteq \mathfrak{p}_x} V(E) \Leftrightarrow \mathfrak{p}_y \in V\left(\bigcup_{E \subseteq \mathfrak{p}_x} E\right) = V(\mathfrak{p}_x)$$

(a). *the set $\{x\}$ is closed in $\text{Spec}(A) \Leftrightarrow \mathfrak{p}_x$ is maximal.*

Proof. $\overline{\{x\}} = \{x\} \stackrel{(b)}{\Leftrightarrow} \{x\} = V(\mathfrak{p}_x) \Leftrightarrow \mathfrak{p}_x$ is maximal.

Remark 6. Be careful! This does not imply that "If $\mathfrak{p}_x$ is not maximal, then $\{x\}$ is open". In fact, it is very common that a set in a given topology is neither open nor closed. In Zariski topology, singletons can very rarely be open.

(c). $y \in \overline{\{x\}} \Leftrightarrow \mathfrak{p}_x \subseteq \mathfrak{p}_y$

Proof. $y \in \overline{\{x\}} \Leftrightarrow \mathfrak{p}_y \in V(\mathfrak{p}_x) \Leftrightarrow \mathfrak{p}_x \subseteq \mathfrak{p}_y$.

(d). *X is a T_0 -space.*

Proof. Say x, y are distinct points in X . This means: $\mathfrak{p}_x \not\subseteq \mathfrak{p}_y$, or $\mathfrak{p}_y \not\subseteq \mathfrak{p}_x$. Suppose the first case, by (c) we have: $y \notin \overline{\{x\}}$, follows that y is in the open set $X \setminus \overline{\{x\}}$, which does not contain x . Together with the other case, by symmetry, the proof is complete.

1.19. *Show that $\text{Spec}(A)$ is irreducible if and only if the nilradical $\mathfrak{R}$ is prime.*

Proof. $\Leftarrow$: X is not empty since $\mathfrak{R} \in \text{Spec}(A)$. Take any non-empty open basis element X_f, X_g , it suffices to show that $X_f \cup X_g \neq \emptyset$ since every non-zero open set contains a non-zero open basis element as a subset. Note that by 1.17.(b), f, g are not nilpotent; hence $f, g \notin \mathfrak{R}$, it follows that fg is not nilpotent since $\mathfrak{R}$ is prime. Therefore, by 1.17.(a), $X_f \cup X_g = X_{fg} \neq \emptyset$. $\Rightarrow$: $\text{Spec}(A)$ is irreducible $\Rightarrow X$ is not empty and every pair of non-empty open basis elements intersect. That is, take any $f, g \notin \mathfrak{R}$, $X_{fg} = X_f \cup X_g \neq \emptyset \Rightarrow fg \notin \mathfrak{R} \Rightarrow \mathfrak{R}$ is prime, as desired.

1.20. *Let X be a topological space.*

(a). *If Y is a subspace of X , then the closure $\overline{Y}$ in X is irreducible.*

Proof. Note that every non-empty open set U in $\overline{Y}$ must touch Y , since otherwise, $\overline{Y} \setminus U = \overline{Y} \cap (X \setminus U)$ would be a closed set in X containing Y but strictly contained in $\overline{Y}$. Now suppose $\overline{Y}$ is reducible, then $\exists$ non-empty open sets $U, V \subseteq \overline{Y}$ s.t. $U \cap V = \emptyset$. But this implies $U \cap Y$ and $V \cap Y$, 2 open sets in Y , intersect. A contradiction.

(b). *Every irreducible subspace of X is contained in a maximal irreducible subspace.*

Proof. Let Y be an arbitrary irreducible subspace of X . Let Σ be the collection of all irreducible subspaces of X containing Y . First, Σ is non-empty since $Y \in \Sigma$. Then, I claim that, for

any ascending chain $Y = Y_0 \subseteq Y_1 \subseteq Y_2 \subseteq \dots$ indexed by I , the set $\overline{\cup_{i \in I} Y_i}$, denoted as Z , is an upper bound that lives in Σ . Then by Zorn's lemma, we are done. By the same reasoning in (a), any non-empty open set in Z must touch $\cup_{i \in I} Y_i$, and suppose U, V are such open sets that do not intersect, then since they both touch $\cup_{i \in I} Y_i$, $\exists Y_k$ where $k \in I$ s.t. $U \cap Y_k$ and $V \cap Y_k$ are open sets in Y_k but do not intersect. This contradicts the fact that Y_k is irreducible. Therefore, $U \cap V \neq \emptyset$, as we desired.

(c). *The maximal irreducible subspaces are closed and cover X . They are called **irreducible components** of X . What are the irreducible components of a Hausdorff space?*

Proof. Let $Y \subseteq X$ be an irreducible subspace. If $Y = X$, then Y is closed. If $Y \subsetneq X$, suppose Y is open, then we have $Y \subsetneq \overline{Y} \subseteq X$. This contradicts the maximality of Y since $\overline{Y}$ is always irreducible by (a). To show maximal irreducible subspaces cover X , take any $x \in X$, $\exists$ an open neighborhood $U \ni x$. We know $\overline{U}$ is irreducible and hence contained in a maximal irreducible subspace by (b). Therefore, x is contained in a maximal irreducible subspace. Since x is chosen arbitrarily, we are done. Let X be a Hausdorff space; it is straightforward to see that any subspace $Y \subseteq X$ (inherits the subspace topology of course) is still Hausdorff. Therefore, suppose $a, b \in Y$ are distinct points, then $\exists$ open neighborhoods $U \ni a, V \ni b$ s.t. $U \cap V = \emptyset$. So any irreducible subspace cannot have distinct elements. And singletons are clearly irreducible, hence are also maximal.

Remark 7. As a side product, we proved that in a Hausdorff space, the singletons are closed. A direct way to prove this: Let X be a Hausdorff topological space. Fix an arbitrary point $x \in X$. Since X is Hausdorff, for any other point $a \in X$, there exists an open neighborhood U_a that does not touch x . Therefore, $\bigcup_{a \in X \setminus \{x\}} U_a = X \setminus \{x\}$, which is an open set. It follows that $\{x\}$ is closed.

(d). *The irreducible components of $X = \text{Spec}(A)$ are the closed sets $V(\mathfrak{p})$ where $\mathfrak{p}$ is a minimal prime ideal.*

Proof. First, I claim that for any prime ideal $\mathfrak{p}$, $V(\mathfrak{p})$ is irreducible as a subspace of X . To show this, it suffices to show that for any open basis element X_f and X_g that touch $V(\mathfrak{p})$, we always have: $X_f \cap X_g = X_{fg}$ also touches $V(\mathfrak{p})$. Since $\mathfrak{p}$ prime, we have:

$$X_f \cap V(\mathfrak{p}) \neq \emptyset \wedge X_g \cap V(\mathfrak{p}) \neq \emptyset \iff f \notin \mathfrak{p} \wedge g \notin \mathfrak{p} \iff fg \notin \mathfrak{p} \iff X_{fg} \cap V(\mathfrak{p}) \neq \emptyset$$

Second, having the above conclusion, we know that if $\mathfrak{p}$ is a minimal prime, then $V(\mathfrak{p})$ is a maximal irreducible subspace due to the reverse inclusion. To show the converse, let $Y \subseteq X$ be an irreducible subspace. By (c) we know that Y has to be closed, hence of the form $V(\mathfrak{a})$. Suppose $\mathfrak{a}$ is not prime, then $\exists f, g \in A$ s.t. $f \notin \mathfrak{a} \wedge g \notin \mathfrak{a}$. But $fg \in \mathfrak{a}$. This implies, by our reasoning above, X_f and X_g touch $V(\mathfrak{a})$, but $X_f \cap X_g$ does not intersect in $V(\mathfrak{a})$, contradicting our assumption that $V(\mathfrak{a})$ is irreducible. Therefore, every irreducible component of X has to be of the form $V(\mathfrak{p})$ where $\mathfrak{p}$ is prime, as desired.

Remark 8. This result is consistent with the result in 1.19., i.e., we have:

$$\text{Spec}(A) = V(\mathfrak{R}) \text{ is irreducible} \iff \mathfrak{R} \text{ is prime} \iff A \text{ has a unique minimal prime ideal}$$

1.21. *Let $\phi : A \rightarrow B$ be a ring homomorphism. Let $X = \text{Spec}(A), Y = \text{Spec}(B)$. If $\mathfrak{q} \in Y$, then*

$\phi^{-1}(\mathfrak{q}) \in X$. Hence ϕ induces a mapping $\phi^* : Y \rightarrow X$:

$$\begin{aligned}\phi^* : \text{Spec}(B) &\longrightarrow \text{Spec}(A) \\ \mathfrak{q} &\longmapsto \phi^{-1}(\mathfrak{q}) = \mathfrak{q}^c\end{aligned}$$

(a). If $f \in A$ then $\phi^{*-1}(X_f) = Y_{\phi(f)}$, and hence that ϕ^* is continuous.

Proof:

$$\mathfrak{q} \notin \phi^{*-1}(X_f) \Leftrightarrow \phi^*(\mathfrak{q}) = \mathfrak{q}^c \in V(f) \Leftrightarrow f \in \mathfrak{q}^c \Leftrightarrow \phi(f) \in \mathfrak{q} \Leftrightarrow \mathfrak{q} \in V(\phi(f)) \Leftrightarrow \mathfrak{q} \notin Y_{\phi(f)}$$

(b). If $\mathfrak{a}$ is an ideal of A , then $\phi^{*-1}(V(\mathfrak{a})) = V(\mathfrak{a}^e)$

Proof:

$$\mathfrak{q} \in \phi^{*-1}(V(\mathfrak{a})) \Leftrightarrow \phi^*(\mathfrak{q}) = \mathfrak{q}^c \in V(\mathfrak{a}) \Leftrightarrow \mathfrak{q}^c \supseteq \mathfrak{a} \Leftrightarrow \mathfrak{q} \supseteq \phi(\mathfrak{a}) \Leftrightarrow \mathfrak{q} \in V(\langle \phi(\mathfrak{a}) \rangle) = V(\mathfrak{a}^e)$$

(c). If $\mathfrak{b}$ is an ideal of B , then $\overline{\phi^*(V(\mathfrak{b}))} = V(\mathfrak{b}^c)$

Proof:

$$\overline{\phi^*(V(\mathfrak{b}))} = \bigcap_{\mathfrak{p} \in \phi^*(V(\mathfrak{b}))} V(\mathfrak{p}) = V\left(\bigcup_{\mathfrak{p} \in \phi^*(V(\mathfrak{b}))} \mathfrak{p}\right) = V\left(\bigcup_{\mathfrak{q} \in V(\mathfrak{b})} \phi^{-1}(\mathfrak{q})\right) = V(\text{rad}(\phi^{-1}(\mathfrak{b}))) = V(\mathfrak{b}^c)$$

(d). If ϕ is surjective, then ϕ^* is a homeomorphism of Y onto the closed subset $V(\ker(\phi))$ of X . In particular, $\text{Spec}(A)$ and $\text{Spec}(A/\mathfrak{R})$ are naturally homeomorphic.

Proof. Since ϕ is surjective, we know that there is a one-to-one correspondence between prime ideals in B and prime ideals in A that contain $\ker(\phi)$. Hence, there is a bijection between $\text{Spec}(B)$ and $V(\ker(\phi))$ as follows:

$$\begin{aligned}\text{Spec}(B) &\longleftrightarrow V(\ker(\phi)) \\ \phi^* : \mathfrak{q} &\longmapsto \mathfrak{q}^c = \phi^{-1}(\mathfrak{q}) \\ \phi(\mathfrak{p}) = \mathfrak{p}^e &\longleftarrow \mathfrak{p} : \phi^{*-1}\end{aligned}$$

In (a) we have shown that ϕ^* is continuous; it remains to show that ϕ^{*-1} is also continuous:

$$\mathfrak{p} \in \phi^*(Y_f) \Leftrightarrow \mathfrak{p} \in \phi^{-1}(Y_f) \Leftrightarrow \mathfrak{p}^e \subseteq Y_f \Leftrightarrow f \notin \mathfrak{p}^e \Leftrightarrow \phi^{-1}(f) \notin \mathfrak{p} \Leftrightarrow \mathfrak{p} \in X_{\phi^{-1}(f)}$$

This holds for arbitrary prime $\mathfrak{p}$, hence $\phi^*(Y_f) = X_{\phi^{-1}(f)}$, for any open basis element $Y_f \subseteq \text{Spec}(B)$, implying that ϕ^{*-1} is continuous. Note that $\phi : A \rightarrow A/\mathfrak{R}$ is a surjective homomorphism with $\ker(\phi) = \mathfrak{R}$, therefore, $\phi^* : \text{Spec}(A/\mathfrak{R}) \rightarrow V(\ker(\phi)) = \text{Spec}(A)$ is naturally a homeomorphism.

(e). If ϕ is injective, then $\phi^*(Y)$ is dense in X . Actually, $\phi^*(Y)$ is dense in $X \Leftrightarrow \ker(\phi) \subseteq \mathfrak{R}$.

Proof. By (c) we have:

$$\phi^*(Y) \text{ is dense in } X \Leftrightarrow \overline{\phi^*(Y)} = \overline{\phi^*(\langle 0 \rangle)} = V(\langle 0 \rangle^c) = V(\ker(\phi)) = X \Leftrightarrow \ker(\phi) \subseteq \mathfrak{R}$$

Alternatively, more essentially, $\phi^*(Y)$ is dense in $X \Leftrightarrow \forall V(\mathfrak{a}) \supseteq \phi^*(Y), V(\mathfrak{a}) = \text{Spec}(A) \Leftrightarrow \forall \mathfrak{a} \subseteq \ker(\phi), \mathfrak{a} \subseteq \mathfrak{R} \Leftrightarrow \ker(\phi) \subseteq \mathfrak{R}$.

(f). Let $\psi : B \rightarrow C$ be another ring homomorphism. Then $(\psi \circ \phi)^* = \phi^* \circ \psi^*$.

Proof. For any $\mathfrak{r} \in \text{Spec}(C)$ we have:

$$(\psi \circ \phi)^*(\mathfrak{r}) = (\psi \circ \phi)^{-1}(\mathfrak{r}) = \phi^{-1}(\psi^{-1}(\mathfrak{r})) = \phi^*(\psi^*(\mathfrak{r})) = \phi^* \circ \psi^*(\mathfrak{r})$$

Remark 9. From the above discussions, we can see that we naturally have a (contravariant) functor (provided that ϕ^* is a homeomorphism) :

$$F : \mathbf{ComRing} \longrightarrow \mathbf{Top}$$

where given any object A in $\mathbf{ComRing}$, $F(A) = \text{Spec}(A)$ in $\mathbf{Top}$; given any morphism ϕ in $\mathbf{ComRing}$, $F(\phi) = \phi^*$ in $\mathbf{Top}$; Satisfying (i). $F(\text{id}_A) = \text{id}_{F(A)}$; (ii) $F(\psi \circ \phi) = F(\phi) \circ F(\psi)$.

(g). Let A be an integral domain with just one non-zero prime ideal $\mathfrak{p}$, and let K be a field of fractions of A . Let $B = A/\mathfrak{p} \times K$. Define $\phi : A \rightarrow B$ by $x \mapsto (\bar{x}, x)$. Show that ϕ^* is bijective but not a homeomorphism.

Proof. Note that both $A/\mathfrak{p}$ and K are fields, and hence B has 2 prime ideals, namely $\mathfrak{q}_1, \mathfrak{q}_2$ where $\mathfrak{q}_1 = 0_{A/\mathfrak{p}} \times K$ and $\mathfrak{q}_2 = A/\mathfrak{p} \times 0_K$. Verify that $\phi^{-1}(\mathfrak{q}_1) = \mathfrak{p}$ and $\phi^{-1}(\mathfrak{q}_2) = \langle 0 \rangle$. Therefore, ϕ^* is a bijection since $\mathfrak{p}$ and $\langle 0 \rangle$ are the only 2 prime ideals of A . ϕ^* is clearly not a homeomorphism since $\{\mathfrak{q}_1\} = V(\mathfrak{q}_1)$ is open in $\text{Spec}(B)$ but $\phi^*(\{\mathfrak{q}_1\}) = \{\mathfrak{p}\}$ is not open in $\text{Spec}(A)$.

1.22. Let $A = \prod_{i=1}^n A_i$ be the direct product of ring A_i . Show that $\text{Spec}(A)$ is the disjoint union of open (and closed) subspaces X_i , where X_i is canonically homeomorphic with $\text{Spec}(A_i)$.

Proof. Note that every prime ideal of A is of the form $\prod_{i=1}^n \mathfrak{a}_i$ where

$$\mathfrak{a}_i = \begin{cases} A_j, & \text{if } i \neq j \\ \mathfrak{p}, \text{ where } \mathfrak{p} \text{ is prime in } A_j & \text{if } i = j \end{cases}$$

Therefore, let X_i be the collection of primes in A of the form $\prod_{j \neq i} A_j \times \mathfrak{p}$ where $\mathfrak{p}$ is a prime in A_i . Clearly $\sqcup_{i=1}^n X_i = \text{Spec}(A)$. Let $\phi_i : A \rightarrow A_i$ be the i -th projection, $i \in [n]$. Then, by 1.21.(d), ϕ_i^* is a homeomorphism: $\text{Spec}(A_i) \rightarrow V(\ker(\phi_i)) = V(\prod_{j \neq i} A_j \times \langle 0_{A_i} \rangle) = X_i$, as desired.

Conversely, let A be any ring. Show that the followings are equivalent:

- (i). $X = \text{Spec}(A)$ is disconnected.
- (ii). $A \cong A_1 \times A_2$ where neither of A_1, A_2 is the zero ring.
- (iii). A contains an idempotent $\neq 0, 1$.

Proof. (i) $\Rightarrow$ (iii): X is disconnected $\Rightarrow \exists$ proper ideals $\mathfrak{a}, \mathfrak{b} \triangleleft A$ s.t. $\text{Spec}(A) = V(\mathfrak{a}) \sqcup V(\mathfrak{b})$ and $\mathfrak{a} + \mathfrak{b} = \langle 1 \rangle$. Note that $\mathfrak{a} + \mathfrak{b} = \langle 1 \rangle$ implies $\mathfrak{a}\mathfrak{b} = \mathfrak{a} \cap \mathfrak{b}$. Therefore, $V(\mathfrak{a}) \sqcup V(\mathfrak{b}) = V(\mathfrak{a}\mathfrak{b}) = V(\mathfrak{a} \cap \mathfrak{b}) = \text{Spec}(A)$, it follows that $\mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{R}$. Therefore, $\exists$ non-zero element $g \in \mathfrak{a}, h \in \mathfrak{b}$ s.t. $g + h = 1$ and $\exists m \in \mathbb{N}$ s.t. $(gh)^m = 0$. (I was stuck at here for a very long time until I came up with the

following trick:) Note that we may assume that $\mathfrak{a}, \mathfrak{b}$ are radical ideals since $V(\mathfrak{a}) = V(\text{rad}(\mathfrak{a}))$. Then, by Prop 1.16., $\mathfrak{a} + \mathfrak{b} = \langle 1 \rangle$ implies $\langle g^m \rangle + \langle h^m \rangle = \langle 1 \rangle$ since $\text{rad}(\langle g^m \rangle) = \mathfrak{a}, \text{rad}(\langle h^m \rangle) = \mathfrak{b}$. Therefore, $\exists a, b \in A$ s.t. $ag^m + bh^m = 1$, which means $(ag^m)^2 = ag^m \cdot (1 - bh^m) = ag^m$, as we desired.

(iii) $\Rightarrow$ (ii): Suppose $e \in A$ is an idempotent element $\neq 0, 1$. By 1.12. we know that so is $1 - e$. Let $\mathfrak{a} := \langle e \rangle$ and $\mathfrak{b} := \langle 1 - e \rangle$. Note that $\mathfrak{a}$ and $\mathfrak{b}$ are co-prime and hence $\mathfrak{a} \cap \mathfrak{b} = \mathfrak{a}\mathfrak{b} = \langle e(1 - e) \rangle = \langle 0 \rangle$. Now consider the homomorphism defined in Prop 1.10.:

$$\phi : A \rightarrow A/\mathfrak{a} \times A/\mathfrak{b}$$

Note that by Prop 1.10. we know $\mathfrak{a} + \mathfrak{b} = \langle 1 \rangle \Leftrightarrow \phi$ is surjective; $\mathfrak{a} \cap \mathfrak{b} = \langle 0 \rangle \Leftrightarrow \phi$ is injective. Therefore, ϕ is an isomorphism, as we desire.

(ii) $\Rightarrow$ (i) has been shown in the first part of this question.

Remark 10. Now we can summarize and draw the structure of the ring A and $\text{Spec}(A)$ that we discussed above. Once A has an idempotent element $e \neq 1, 0$, it has a dual element of $e : 1 - e$, which is also idempotent and $\neq 1, 0$. Let $\mathfrak{a} := \langle e \rangle$ and $\mathfrak{b} := \langle 1 - e \rangle$. They satisfy:

$$\mathfrak{a} + \mathfrak{b} = \langle 1 \rangle \quad \text{and} \quad \mathfrak{a} \cap \mathfrak{b} = \langle 0 \rangle$$

This, by Prop 1.10., induces a ring isomorphism ϕ defined as above. Therefore, $\phi^* : \text{Spec}(A/\mathfrak{a} \times A/\mathfrak{b}) \rightarrow \text{Spec}(A)$ is a homeomorphism and:

$$\text{Spec}(A/\mathfrak{a} \times A/\mathfrak{b}) \cong \text{Spec}(A) \cong V(\mathfrak{a}) \sqcup V(\mathfrak{b})$$

Note that therefore, by 1.12., if A is a local ring, then $\text{Spec}(A)$ is connected and A cannot be properly written as $A_1 \times A_2$ where neither of A_1, A_2 is the zero ring.

1.23. Let A be a Boolean ring, and let $X = \text{Spec}(A)$.

(a). For each $f \in A$, X_f is both open and closed.

Proof. By 1.22. we know that for any $f \in A$, since $f^2 = f$, we have $\text{Spec}(A) = V(f) \sqcup V(1 - f)$. Therefore, $\text{Spec}(A) = X_{1-f} \sqcup X_f$, and hence X_f is both open and closed. (In particular, if $f = 1$ or 0 , then $X_f = X$ or $\emptyset$, resp., as we have seen before.)

(b). Let $f_1, \dots, f_n \in A$. Show that $X_{f_1} \cup \dots \cup X_{f_n} = X_f$ for some $f \in A$.

Proof. We have: $\exists f \in A$ s.t.

$$\bigcup_{i=1}^n X_{f_i} = \bigcup_{i=1}^n (X \setminus V(f_i)) = X \setminus \bigcap_{i=1}^n V(f_i) = X \setminus V(\langle f_1 \rangle \cup \dots \cup \langle f_n \rangle) \stackrel{1.11.(c)}{=} X \setminus V(f) = X_f$$

(c). The sets X_f are the only subsets of X which are both open and closed.

Proof. Follow the hint: Suppose $Y \subseteq X$ is both open and closed. Since Y is open, it is a union of basic open sets X_f . Since Y is closed, it is quasi-compact (Thm 26.2. in [Mun00]), and

hence a finite union of X_f . Then apply (b).

(d). X is a compact Hausdorff space.

Proof. By (a), for arbitrary distinct points $x, y \in X$, it suffices to show that $\exists f \in A$ s.t. X_f separates x and y . Suppose the contrary, wlog., say $\forall X_f \ni x \Rightarrow X_f \ni y$. This implies $\forall f \not\in \mathfrak{p}_x \Rightarrow f \not\in \mathfrak{p}_y$, i.e., $\mathfrak{p}_y \subsetneq \mathfrak{p}_x$. However, by 1.11.(b), every prime ideal in A is maximal. So we get a contradiction, as desired.

1.24. Let L be a lattice, in which the sup and inf of 2 elements a, b are denoted by $a \vee b$ and $a \wedge b$ resp. L is a **Boolean lattice (or Boolean algebra)** if

- (i). L has a least element and a greatest element (denoted by $0, 1$ resp).
- (ii). Each of $\vee, \wedge$ is distributive over the other.
- (iii). Each $a \in L$ has a unique "complement" $a' \in L$ s.t. $a \vee a' = 1$ and $a \wedge a' = 0$.

Let L be a Boolean lattice. Define addition and multiplication in L by the rules:

$$a + b = (a \wedge b') \vee (a' \wedge b) \quad \text{and} \quad ab = a \wedge b$$

Verify that L becomes a Boolean ring, say $A(L)$.

Proof. It is straightforward to verify that $A(L)$ is a ring, where 0 and 1 are the identity and unity respectively, satisfying $0' = 1$ and $1' = 0$. And $-a = a'$, for any $a \in A(L)$. To show that $A(L)$ is boolean, note that $a^2 = a \wedge a = a, \forall a \in A(L)$, so we are done.

Conversely, starting from a Boolean ring A , define an ordering on A as follows:

$$a \preceq b \text{ if and only if } a = ab$$

Show that A is a boolean lattice w.r.t. this ordering.

Proof. Follow the hint. Define $a \vee b := a + b + ab$ and $a \wedge b := ab$. Clearly $\vee$ and $\wedge$ satisfy the associativity and commutativity. Verify the absorption law:

$$\begin{aligned} a \vee (a \wedge b) &= a \vee (ab) = a + ab + a^2b = a + 2ab = a \\ a \wedge (a \vee b) &= a \wedge (a + b + ab) = a^2 + ab + a^2b = a^2 + 2ab = a \end{aligned}$$

Hence, A is a lattice. It remains to show that A is Boolean. Verify:

- (i). $a = a \cdot 1 \Rightarrow a \preceq 1, \forall a \in A$, so 1 is the greatest element. $0 = a \cdot 0 \Rightarrow 0 \preceq a, \forall a \in A$, so 0 is the least element.
- (ii). It is straightforward to verify the distributive laws for $\vee$ and $\wedge$.
- (iii). Let $a' := b$. Then $a \wedge b = ab = 0$. $a \vee b = a + b + ab = a + b = 1$, hence $b = 1 - a$.

1.25. From the last 2 exercises deduce the Stone's theorem, that every Boolean lattice is isomorphic to the lattice of open-and -closed subsets of some compact Hausdorff topological space.

Proof. Given a Boolean lattice $(L(\wedge, \vee), \preceq)$, by 1.24., there is a corresponding Boolean ring, say $A(L)$. By 1.23., we know that $\text{Spec}(A(L))$ is a compact Hausdorff space. Let B denote the

collection of all open basis elements X_f of $\text{Spec}(A(L))$. Verify that $(B(\cap, \cup), \subseteq)$ is a Boolean lattice, and there is a bijection between $\phi : L \rightarrow B$ given by $f \mapsto X_f$. So we have the following diagram:

$$\begin{array}{ccc} L(\wedge, \vee) & \xrightarrow{\phi : f \mapsto X_f} & B(\cap, \cup) \\ & \searrow \quad \nearrow & \\ & \text{Spec}(A(L)) & \end{array}$$

It remains to show that ϕ is a lattice homomorphism, i.e., ϕ is order-preserving. Note that:

$$f \preceq g \Leftrightarrow f = fg \Leftrightarrow X_f = X_{fg} \Leftrightarrow X_f = X_f \cap X_g \Leftrightarrow X_f \subseteq X_g$$

as we desired.

1.26. Let A be a ring. The subspace of $\text{Spec}(A)$ consisting of the maximal ideals of A , with the induced topology, is called the **maximal spectrum** of A and is denoted by $\text{Max}(A)$. For arbitrary comm. rings it does not have the nice functorial properties of $\text{Spec}(A)$, because the inverse image of a maximal ideal under a ring homomorphism need not to be maximal.

Let X be a compact Hausdorff space and let $C(X)$ denote the ring of all continuous functions $\{f : X \rightarrow \mathbb{R}\}$. For each $x \in X$, let $\mathfrak{m}_x$ be the set of all $f \in C(X)$ s.t. $f(x) = 0$. Note that $\mathfrak{m}_x$ is maximal. Let $\tilde{X}$ denote $\text{Max}(C(X))$, we have therefore defined a mapping: $\mu : X \rightarrow \tilde{X}$, namely $x \mapsto \mathfrak{m}_x$. Show that μ is a homeomorphism of X onto $\tilde{X}$.

Proof. The hint almost gives the entire proof. I will restate it and complete it:

(i). Let $\mathfrak{m}$ be any maximal ideal of $C(X)$. Let:

$$V_{\mathfrak{m}} := \{x \in X : f(x) = 0, \forall f \in \mathfrak{m}\}$$

Suppose $V_{\mathfrak{m}} = \emptyset$. Then for any $x \in X$, $\exists f_x \in \mathfrak{m}$ s.t. $f_x(x) \neq 0$. This implies, by continuity of f_x , $\exists$ an open neighborhood of x , say U_x , s.t. $\forall x \in U_x, f_x(x) \neq 0$. Since X is compact, X can be covered by finitely many of those open sets, say $U_{x_1}, \dots, U_{x_n}$. Let

$$f = f_{x_1}^2 + \dots + f_{x_n}^2 \in \mathfrak{m}$$

We therefore have, $\forall x \in X, f(x) \neq 0$. This implies $f \in \mathfrak{m}$ is a unit in $C(X)$, contradicting the maximality of $\mathfrak{m}$. Thus, $V_{\mathfrak{m}} \neq \emptyset$, meaning that $\exists x \in X$ s.t. for all $f \in \mathfrak{m}$, $f(x) = 0$. This implies $\mathfrak{m} \subseteq \mathfrak{m}_x$, and hence $\mathfrak{m} = \mathfrak{m}_x$ since the maximality, thus proving the surjectivity of μ .

(ii). Since X is compact Hausdorff, it is normal (Thm 32.3. in [Mun00]). Hence, the Urysohn's lemma implies that for any distinct points $x, y \in X$, there exists a continuous function $f \in C(X)$ that separates x and y on the real line. (Here we implicitly use the fact the singletons are closed in a Hausdorff space, which we have shown in 1.20.) Therefore, $\mathfrak{m}_x \neq \mathfrak{m}_y$, thus proving the injectivity of μ .

(iii). It remains to show that μ is continuous. For a given $f \in C(X)$, let

$$U_f := \{x \in X : f(x) \neq 0\} \quad \text{and} \quad \tilde{U}_f := \{\mathfrak{m} \in \tilde{X} : f \notin \mathfrak{m}\}$$

I claim that the U_f (resp. $\tilde{U}_f$) form a basis of the topology of X (resp. $\tilde{X}$). Consider U_f first: Clearly the U_f are open since f is continuous and $\mathbb{R} \setminus \{0\}$ is open. Take $\forall a \in X$ and any neighborhood $V \ni a$, since X is Hausdorff, $\{a\}$ and $Y := X \setminus V$ are disjoint closed sets. Therefore, by Urysohn's lemma again, $\exists f \in C(X)$ s.t. $f(Y) = 0$ and $f(a) = 1$. Note that since $f(a) \neq 0$, $a \in U_f$; and $U_f \cap Y = \emptyset$. So we have: $x \in U_f \subseteq V$, thus proving that the U_f form an open basis of X . Then consider $\tilde{U}_f$: For any $f \in C(X)$, note that $\tilde{U}_f = X_f \cap \tilde{X}$ where X_f is a basic open set of $\text{Spec}(C(X))$. Thus, the $\tilde{U}_f$ form a basis of $\tilde{X}$. It remains to show that $\mu(U_f) = \tilde{U}_f$: For any $\mathfrak{m}_x \in \tilde{X}$:

$$\mathfrak{m}_x \in \tilde{U}_f \Leftrightarrow f \notin \mathfrak{m}_x \Leftrightarrow f(x) \neq 0 \Leftrightarrow x \in U_f$$

So we are done.

1.1.3 Affine algebraic varieties

1.27. Let k be an algebraically closed field. Let X be an affine algebraic variety on k^n . Let $I(X)$ be the corresponding ideal. The quotient ring

$$P(X) = k[t_1, \dots, t_n] / I(X)$$

is the ring of polynomial functions on X , because 2 polynomials g, h define the same polynomial function on X if and only if $g - h$ vanishes at every point of X , that is, if and only if $g - h \in I(X)$. Let ξ_i be the image of t_i in $P(X)$. The $\{\xi_i\}_{i \in [n]}$ are the coordinate functions on X : if $x \in X$, then $\xi(x)$ is the i -th coordinate on x . $P(X)$ is generated as a k -algebra by the $\{\xi_i\}_{i \in [n]}$, and is called the **coordinate ring** of X . As in 1.26., for each $x \in X$, let $\mathfrak{m}_x$ be the ideal of all $f \in P(X)$ s.t. $f(x) = 0$; it is a maximal ideal of $P(X)$. Hence, let $\tilde{X} = \text{Max}(P(X))$, we have defined a mapping $\mu : X \rightarrow \tilde{X}$, namely $x \mapsto \mathfrak{m}_x$. Show that μ is a bijection.

Proof. Injectivity is easy: Let x, y be distinct points in X . Then $\exists i \in [n]$ s.t. $x_i \neq y_i$ and hence $\xi - x_i \in \mathfrak{m}_x$, but $\notin \mathfrak{m}_y$. To show surjectivity, take any $\mathfrak{m} \in \tilde{X}$, since $P(x)$ is a finitely generated k -algebra, so is $P(X)/\mathfrak{m}$. Then, by Zariski's lemma, $P(X)/\mathfrak{m}$ is a finitely generated k -vector space. Note that $P(X)/\mathfrak{m}$ is a field, so $[P(X)/\mathfrak{m} : k]$ is finite. But since k is algebraically closed, $P(X)/\mathfrak{m}$ has to be isomorphic to k . Consider the ring homomorphism:

$$\begin{aligned} \phi_x : P(X) &\rightarrow k \\ \xi_i &\mapsto \xi_i(x) \end{aligned}$$

where $\mathfrak{m} = \ker(\phi_x) = \langle \xi_1 - x_1, \dots, \xi_n - x_n \rangle \subseteq \mathfrak{m}_x$. But since $\mathfrak{m}$ is maximal, they have to be equal.

1.28. Let $f_1, \dots, f_m$ be elements of $k[t_1, \dots, t_n]$. They determine a polynomial mapping $\phi : k^n \rightarrow k^m$ defined by: if $x \in k^n$, the coordinates of $\phi(x)$ are $f_1(x), \dots, f_m(x)$. Let X, Y be affine algebraic varieties in k^n, k^m respectively. A mapping $\phi : X \rightarrow Y$ is said to be **regular** if ϕ is the restriction to X of a polynomial mapping from k^n to k^m . If η is a polynomial function on Y , then $\eta \circ \phi$ is a polynomial function on X . Hence ϕ induces a k -algebra homomorphism $P(Y) \rightarrow P(X)$, $\eta \mapsto \eta \circ \phi$. Show that in this way we obtain a one-to-one correspondence between

the regular mappings $X \rightarrow Y$ and the k -algebra homomorphisms $P(Y) \rightarrow P(X)$.

Proof. Given a regular map $\phi : X \rightarrow Y$, define the pullback:

$$\begin{aligned}\phi^* : P(Y) &\rightarrow P(X) \\ f &\mapsto f \circ \phi\end{aligned}$$

This is a k -algebra homomorphism. (Concretely, $\phi^*(\bar{y}_i) = \phi_i$, the i -th coordinate function of ϕ). Conversely, let $\theta : P(Y) \rightarrow P(X)$ be a k -algebra homomorphism. Set $f_i := \theta^*(\bar{y}_i) \in P(X)$ and define:

$$\begin{aligned}\phi : X &\rightarrow k^m \\ x &\mapsto (f_1(x), \dots, f_m(x))\end{aligned}$$

We must check $\phi(X) \subseteq Y$. If $g \in I(Y)$, then $\bar{g} = 0 \in P(Y)$, hence

$$0 = \theta(\bar{g}) = \theta(g(\bar{y}_1, \dots, \bar{y}_m)) = g(f_1, \dots, f_m)$$

Thus for every $x \in X$,

$$g(f_1(x), \dots, f_m(x)) = 0$$

Since this holds for all $g \in I(Y)$, we get $\phi(x) \in Y$. Hence $\phi : X \rightarrow Y$ is well-defined.

Since each $f_i \in P(X)$ is represented by a polynomial on k^n , the map ϕ is the restriction of a polynomial map $k^n \rightarrow k^m$. Thus ϕ is regular.

It remains to show that $\phi^* = \theta$. Since $P(Y)$ is generated by $\bar{y}_1, \dots, \bar{y}_m$, every $\eta \in P(Y)$ has the form

$$\eta = h(\bar{y}_1, \dots, \bar{y}_m)$$

Then

$$\phi^*(\eta) = h(f_1, \dots, f_m) = h(\theta(\bar{y}_1), \dots, \theta(\bar{y}_m)) = \theta(\eta)$$

Hence $\phi^* = \theta$.

Finally, if two regular maps induce the same homomorphism, then their pullbacks agree on each coordinate function $\bar{y}_i$. Therefore their coordinate functions are equal, so the maps themselves are equal. Thus $\phi \mapsto \phi^*$ gives a one-to-one correspondence between regular maps $X \rightarrow Y$ and k -algebra homomorphisms $P(Y) \rightarrow P(X)$.

2 Chapter 2 Modules

2.1 Notes and Comments

Remark 11. Comments on [AM69] top of page 19.

Let $f : M \rightarrow N$ be an A -module homomorphism. If M' is a submodule of M s.t. $M' \subseteq \ker(f)$, then f gives rise to a homomorphism $\bar{f} : M/M' \rightarrow N$, defined as: $\bar{f} \circ \pi = f$. And $\ker(\bar{f}) = \ker(f)/M'$. In particular, if $M' = \ker(f)$, then $\bar{f}$ is an isomorphism: $M/\ker(f) \cong \text{Im}(f)$.

$$\begin{array}{ccc} M & \xrightarrow{\pi} & M/M' \\ \downarrow f & \swarrow \bar{f} & \\ N & & \end{array} \qquad \begin{array}{ccc} M & \xrightarrow{\pi} & M/\ker f \\ \downarrow f & \swarrow \cong & \\ \text{Im } f & & \end{array}$$

This fact gives us a way to decompose $f : M \rightarrow N$.

Exercise 2.2.

(i). $\text{Ann}(M + N) = \text{Ann}(M) \cap \text{Ann}(N)$

Note that for any $a \in A$, $a(M + N) = 0 \Leftrightarrow aM = 0$ and $aN = 0$. $\Rightarrow$ is because $M + N \supseteq M, N$; $\Leftarrow$ is because $aM + aN = a(M + N)$. Therefore, $a \in \text{Ann}(M + N)$ if and only if $a \in \text{Ann}(M)$ and $a \in \text{Ann}(N)$.

(ii). $(N : P) = \text{Ann}((N + P)/N)$.

$\forall a \in A, a \in (N : P) \Leftrightarrow aP \subseteq N \Leftrightarrow aP + aN = a(P + N) \subseteq aN + N = N \Leftrightarrow a \in \text{Ann}((P + N)/N)$.

Proposition 2.4. Let M be a finitely generated A -module, let $\mathfrak{a}$ be an ideal of A , let $\phi \in \text{Hom}_A(M, M)$ s.t. $\phi(M) \subseteq \mathfrak{a}M$. Then $\exists a_1, \dots, a_n \in \mathfrak{a}$ s.t.

$$\phi^n + a_1\phi^{n-1} + \dots + a_n = 0$$

Proof. Let $x_1, \dots, x_n$ generate M . Since $\phi(M) \subseteq \mathfrak{a}M$, for each x_i we have: $\phi(x_i) = \sum_{j=1}^n a_{ij}x_j$. i.e., if we let matrix $\mathbf{A} := (a_{ij})_{n \times n}$ and $\mathbf{x} = (x_1, \dots, x_n)^T$, then $\mathbf{A}\mathbf{x} = \phi(\mathbf{x})$. So we have:

$$\begin{bmatrix} a_{11} - \phi & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} - \phi \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} =: \mathbf{A}^\# \mathbf{x} = 0$$

This implies $\det(\mathbf{A}^\#) = 0$ (The zero map) since $\mathbf{x} \neq \mathbf{0}$ (we may assume $M \neq 0$). Expanding out the det yields the conclusion.

Remark 12. This is a generalization of the Cayley-Hamilton theorem. If we let A be a field, and M be a finite-dimensional A -vector space, then we can choose $x_1, \dots, x_n$ to be a basis of M . Given any linear transformation $\phi : M \rightarrow M$, let $\mathbf{A}$ be the corresponding matrix of ϕ w.r.t. the chosen basis. i.e., $\mathbf{A}v = \phi(v), \forall v \in M$. That is to say:

$$\phi^n + a_1\phi^{n-1} + \dots + a_n = 0 \Leftrightarrow \mathbf{A}^n + a_1\mathbf{A}^{n-1} + \dots + a_n = 0$$

where the coefficients $a_1, \dots, a_n$ are exactly those in the characteristic polynomial of $\mathbf{A}$, i.e.

$$\chi_{\mathbf{A}}(\lambda) := \det(\lambda\mathbf{I} - \mathbf{A})$$

Therefore,

$$\chi_{\mathbf{A}}(\lambda) = 0 \Leftrightarrow \chi_{\mathbf{A}}(\mathbf{A}) = 0$$

► But wait, why can't we just say: by definition of $\chi_{\mathbf{A}}$, we have $\chi_{\mathbf{A}}(\mathbf{A}) = \det(\mathbf{A}\mathbf{I} - \mathbf{A}) = 0$? (Think of it!)

Remark 13. Comments on Corollary 2.5. This statement is saying, Suppose M is a finitely generated A -module, and $\mathfrak{a} \triangleleft A$ s.t. $M = \mathfrak{a}M$. Then, by Prop 2.4., we may take $\phi = \text{id}_M$ and have: $\exists a_1, \dots, a_n \in \mathfrak{a}$ s.t.

$$0 = \text{id}^n + a_1\text{id}^{n-1} + \dots + a_n$$

This means, if we let $x = 1 + a_1 + \dots + a_n$, then x acting on M would always get zero. And note that $x - 1 \in \mathfrak{a}$.

Nakayama's lemma (V.1) [AM69] Prop 2.6. *Let M be a finitely generated A -module, $\mathfrak{a} \subseteq \mathfrak{J}$. Then $\mathfrak{a}M = M$ implies $M = 0$.*

Proof 1. Let x be chosen as in the previous remark. Then by Prop 1.9., x is a unit. Thus, $0 = xM = x^{-1}xM = M$.

Proof 2. Let $u_1, \dots, u_n$ be a minimal set that generates M . Since $u_1 \in M = \mathfrak{a}M$, we have $u_1 = a_1u_1 + \dots + a_nu_n$ where $a_i \in \mathfrak{a}$, $i \in [n]$, hence $(1 - a_1)u_1 = a_2u_2 + \dots + a_nu_n$. By Prop 1.9. we know $1 - a_1$ is a unit. So u_1 can be generated by $u_2, \dots, u_n$, contradicting the minimality of the generating set.

Nakayama's lemma (V.2) [Gat13] Corollary 3.27. *Let M be a finitely generated A -module, $\mathfrak{a} \triangleleft A$ s.t. $\mathfrak{a}M = M$. Then there is always an element in $\mathfrak{a}$ that acts on M like the unity. i.e., $\exists a \in \mathfrak{a}$ s.t. $am = m, \forall m \in M$.*

Proof. Again, by the previous remark, $\exists a_1, \dots, a_n \in \mathfrak{a}$ s.t.

$$\text{id} = -(\text{id}^{n-1}a_1 + \dots + a_n)$$

Let $y = -(a_1 + \dots + a_n) \in \mathfrak{a}$, then y acting on M behaves the same as the id_M .

Corollary 2.7. *Let M be a finitely generated A -module, $\mathfrak{a} \subseteq \mathfrak{J}$, N s submodule of M . Then $M = \mathfrak{a}M + N$ implies $M = N$.*

Proof. Follow the book. First, I claim that $\mathfrak{a}(M/N) = (\mathfrak{a}M + N)/N^{\textcircled{\mathfrak{a}}}$. To see this, consider the canonical projection: $\pi : M \rightarrow M/N$, we have:

$$\pi(\mathfrak{a}M) = \mathfrak{a}\pi(M) = \mathfrak{a}(M/N)$$

and the kernel of π restricted on $\mathfrak{a}M$ is $\mathfrak{a}M \cap N$. Hence

$$\mathfrak{a}M/(\mathfrak{a}M \cap N) \cong \mathfrak{a}(M/N)$$

Applying the (second) isomorphism theorem as a bridge, we have:

$$\mathfrak{a}(M/N) \cong \mathfrak{a}M/(\mathfrak{a}M \cap N) \cong (\mathfrak{a}M + N)/N$$

In fact, we have

$$\mathfrak{a}(M/N) = \pi(\mathfrak{a}M) = \{x + N : x \in \mathfrak{a}M\} = (\mathfrak{a}M + N)/N$$

Then, by our assumption, $(\mathfrak{a}M + N)/N = M/N = \mathfrak{a}(M/N)$ and apply Nakayama's lemma to M/N .

Let A be a local ring, $k = A/\mathfrak{m}$ be its residue field. Let M be a finitely generated A -module. $M/\mathfrak{m}M$ is annihilated by $\mathfrak{m}$ (To see this, apply $\textcircled{\mathfrak{a}}$), hence is naturally an $A/\mathfrak{m}$ -module (see page 19 of [AM69]), i.e., a k -vector space, and as such is finite-dimensional (see remark 16).

The next proposition shows that, a basis of $M/\mathfrak{m}M$ (as a k -vector space) can be lifted to a generating set of M . ([CHA23])

Proposition 2.8. *Suppose x_i ($i \in [n]$) be elements of M whose image in $M/\mathfrak{a}M$ form a basis of this vector space. Then $\{x_i\}$ generates M .*

Proof. Let $N = \langle x_1, \dots, x_n \rangle$, it suffices to show $N = M$. Note that the composite map

$$N \xrightarrow{\text{id}} M \longrightarrow M/\mathfrak{m}M$$

where $n \mapsto n + \mathfrak{m}M$ is surjective. Hence $M = \mathfrak{m}M + N$, then apply Corollary 2.7..

Remark 14. AN IMPORTANT COROLLARY OF NAKAYAMA'S LEMMA *Let $(A, \mathfrak{m})$ be a local ring and $k := A/\mathfrak{m}$. Let N be a f.g. A -module. Then:*

$$N \neq 0 \iff N/\mathfrak{m}N \neq 0$$

Proof. $\implies$: Nakayama. $\impliedby$: immediate.

Proposition 2.9.

(i). Let $C_1 :$

$$M' \xrightarrow{u} M \xrightarrow{v} M'' \longrightarrow 0$$

be a sequence of A -modules and homomorphisms. Then C_1 is exact $\Leftrightarrow \forall A$ -modules N , the sequence $C_2 :$

$$0 \longrightarrow \text{Hom}(M'', N) \xrightarrow{v_*} \text{Hom}(M, N) \xrightarrow{u_*} \text{Hom}(M', N)$$

is exact.

Proof. $\Leftarrow$: Clearly v_* is injective $\Rightarrow v$ is surjective, so C_1 is exact at M'' . To show C_1 is exact at M , we need to show $\ker(v) = \text{Im}(u)$. Let $N = M''$, then $u_*(v_*(\text{id}_{M''})) = \text{id}_{M''} \circ v \circ u = 0$. This implies $v \circ u = 0$, hence $\text{Im}(u) \subseteq \ker(v)$. Let $N = \text{coker}(u) = M/\text{Im}(u)$, and let $\phi \in \text{Hom}(M, N)$ be the projection. Then $u_*(\phi) = \phi \circ u = 0$, hence $\phi \in \ker(u_*) = \text{Im}(v_*)$. i.e., $\exists \psi \in \text{Hom}(M'', N)$ s.t. $v_*(\psi) = \psi \circ v = \phi$. This implies $\ker(\phi) \supseteq \ker(v)$, but $\ker(\phi) = \text{Im}(u)$, so we are done.

$\Rightarrow$: v is surjective $\Rightarrow v_*(f) = f \circ v = 0$ if and only if $f = 0$, thus v_* is injective. It remains to show C_2 is exact at $\text{Hom}(M, N)$. For any $\psi \in \text{Hom}(M'', N)$, we have $0 = \psi \circ v \circ u = u_*(v_*(\psi))$ since C_1 is exact, hence $\text{Im}(v_*) \subseteq \ker(u_*)$. To show $\text{Im}(v_*) \supseteq \ker(u_*)$, take any $\phi \in \ker(u_*)$, then $u_*(\phi) = \phi \circ u = 0$, hence $\ker(\phi) \supseteq \text{Im}(u) = \ker(v)$.

$$\begin{array}{ccccccc} M' & \xrightarrow{u} & M & \xrightarrow{v} & M'' & \longrightarrow & 0 \\ & & \downarrow \phi & \nearrow \psi & & & \\ & & N & & & & \end{array}$$

Note that v is surjective, so $M'' = \text{Im}(v) = M/\ker(v)$. Thus, by applying the remark above, we can always decompose ϕ into $\psi \circ v = v_*(\psi)$ for some $\psi \in \text{Hom}(M'', N)$. Consequently, $\phi \in \text{Im}(v_*)$, as desired.

(ii). Let C_3 :

$$0 \longrightarrow N' \xrightarrow{u} N \xrightarrow{v} N''$$

be a sequence of A -modules and homomorphisms. Then C_3 is exact $\Leftrightarrow \forall A$ -modules M , the sequence C_4 :

$$0 \longrightarrow \text{Hom}(M, N') \xrightarrow{u_*} \text{Hom}(M, N) \xrightarrow{v_*} \text{Hom}(M, N'')$$

is exact.

Proof. $\Rightarrow$: u is injective $\Rightarrow \forall f \in \text{Hom}(M, N')$, $u_*(f) = u \circ f = 0$ if and only if $f = 0$, thus u_* is injective. It remains to show $\text{Im}(u_*) = \ker(v_*)$. Take any $\phi \in \text{Hom}(M, N')$, we have $0 = v \circ u \circ \phi = v_*(u_*(\phi))$, hence $\text{Im}(u_*) \subseteq \ker(v_*)$. To show $\text{Im}(u_*) \supseteq \ker(v_*)$, take any $\phi \in \ker(v_*)$, then $v \circ \phi = 0$, hence $\text{Im}(\phi) \subseteq \text{Im}(u)$. It remains to find a $\psi \in \text{Hom}(M, N')$ s.t. $\phi = u \circ \psi$.

$$\begin{array}{ccccccc} 0 & \longrightarrow & N' & \xrightarrow{u} & N & \xrightarrow{v} & N'' \\ & & \nwarrow \psi & & \uparrow \phi & & \\ & & & & M & & \end{array}$$

Note that u is injective, so let $\psi = u^{-1} \circ \phi$, we are done.

$\Leftarrow$: Clearly u_* is injective $\Rightarrow u$ is injective. It remains to show $\ker(v) = \text{Im}(u)$. Take $M = N'$ and $\psi = \text{id}_{N'}$. Then $0 = v_*(u_*(\phi)) = v \circ u$, hence $\text{Im}(u) \subseteq \ker(v)$. Take $M = \ker(v)$ and $\phi = \text{id}_M$. Then $\phi \in \text{Im}(u_*)$, i.e., $\exists \psi \in \text{Hom}(N, N')$ s.t. $u_*(\psi) = u \circ \psi = \phi$. This implies $\text{Im}(\phi) \subseteq \text{Im}(u)$, but $\text{Im}(\phi) = \ker(v)$, so we are done.

$$\begin{array}{ccccccc}
& & 0 & & 0 & & 0 \\
& & \downarrow & & \downarrow & & \downarrow \\
0 & \cdots \cdots \cdots & \ker \alpha & \xrightarrow{\bar{u}} & \ker \beta & \xrightarrow{\bar{v}} & \ker \gamma \\
& & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & M' & \xrightarrow{u} & M & \xrightarrow{v} & M'' \longrightarrow 0 \\
& & \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma \\
0 & \longrightarrow & N' & \xrightarrow{u'} & N & \xrightarrow{v'} & N'' \longrightarrow 0 \\
& & \downarrow & & \downarrow & & \downarrow \\
& & \text{coker } \alpha & \xrightarrow{\bar{u}'} & \text{coker } \beta & \xrightarrow{\bar{v}'} & \text{coker } \gamma \cdots \cdots \cdots 0 \\
& & \downarrow & & \downarrow & & \downarrow \\
& & 0 & & 0 & & 0
\end{array}$$

Proposition 2.10. The Snake Lemma:

Let $\delta : \ker \gamma \rightarrow \text{coker } \alpha$ be defined as in the book. We need to verify that it is well-defined:

Take arbitrary $x \in \ker \gamma$ (embedded in M''). Let $a_1, a_2 \in M$ s.t. $v(a_1) = v(a_2) = x$, and let $y_1, y_2 \in N'$ s.t. $y_i = u'^{-1}(\beta(a_i)), i = 1, 2$. Then $y_1 - y_2 = u'^{-1}(\beta(a_1)) - u'^{-1}(\beta(a_2)) = u'^{-1}(\beta(a_1 - a_2))$. Note that $0 = v(a_1) - v(a_2) = v(a_1 - a_2)$, so $a_1 - a_2 \in \ker(v) = \text{Im}(u)$. Thus $u'^{-1}(\beta(a_1 - a_2)) = \alpha(u^{-1}(a_1 - a_2)) \in \text{Im}(\alpha)$, and hence $\bar{y}_1 = \bar{y}_2$ in $\text{coker } \alpha$, proving that δ is a well-defined map. And it is straightforward to see that δ is a homomorphism, so we are done.

Now we verify that the induced sequence (dashed, gray) is exact:

At $\ker \alpha : \bar{u}$ is injective since u is, done.

At $\ker \beta : \text{Clearly } \text{Im}(\bar{u}) \subseteq \ker(\bar{v})$. To show the opposite, take $\forall x \in \ker(\bar{v})$, we have $x \in \ker(v)$, and hence $x \in \text{Im}(\bar{u})$ thanks to the injectivity of u .

At $\ker \gamma : \text{Take any } x \in \ker \beta \text{ (embedded in } M\text{)}. \text{ Then } \beta(x) = 0, \text{ hence } u'^{-1}(\beta(x)) = 0. \text{ Thus, } \delta(\bar{v}(x)) = 0 \text{ thanks to the commutativity of the diagram. Consequently, } \text{Im}(\bar{v}) \subseteq \ker(\delta). \text{ To show the opposite, take arbitrary } x \in \ker \delta \text{ (embedded in } M''). \text{ Let } a \in M \text{ s.t. } v(a) = x. \text{ Since } \delta(x) = 0, u'^{-1}(\beta(a)) \in \text{Im}(\alpha), \text{ i.e., } \exists b \in M' \text{ s.t. } \alpha(b) = u'^{-1}(\beta(a)). \text{ Thus, by the comm. of the diagram, } 0 = \beta(a) - \beta(u(b)) = \beta(a - u(b)), \text{ implying } a - u(b) \in \ker \beta. \text{ Therefore, } \bar{v}(a - u(b)) = v(a - u(b)) = x - 0 = x \Rightarrow x \in \text{Im}(\bar{v}), \text{ done.}$

At $\text{coker } \alpha : \text{Take } \forall x \in \ker(\gamma). \text{ Let } a \in M \text{ s.t. } v(a) = x. \text{ Since } \beta(a) \in \text{coker } \beta, \text{ we have } \bar{u}'(\delta(x)) = 0. \text{ Thus, } \text{Im}(\delta) \subseteq \ker(\bar{u}'). \text{ To show the opposite, take arbitrary } \bar{x} \in \ker(\bar{u}'), \text{ then } u'(x) \in \text{Im}(\beta). \text{ Let } a \in M \text{ s.t. } \beta(a) = u'(x). \text{ Note that } v'(u'(x)) = 0, \text{ so } v'(\beta(a)) = 0, \text{ which implies } v(a) \in \ker(\gamma) \text{ by the comm. of the diagram. Therefore, } x \in \text{Im}(\delta) \text{ by the definition of } \delta.$

At $\text{coker } \beta : \text{Clearly } \text{Im}(\bar{u}') \subseteq \ker(\bar{v}')$. Conversely, take $\bar{x} \in \ker(\bar{v}')$, then $x \in N$ satisfying $v'(x) \in \text{Im}(\gamma)$. Thus, $\exists b \in M''$ s.t. $\gamma(b) = v'(x)$. We may take $a \in M$ s.t. $v(a) = b$ since v is surjective, then $v'(x - \beta(a)) = v'(x) - v'(b) = 0$, implying $x - \beta(a) \in \ker(v') = \text{Im}(u')$. Let

$\pi : N \rightarrow \text{coker } \beta$ be the projection. So $\exists y \in N'$ s.t. $\pi(u'(y)) = \bar{x}$, hence $\overline{u'}(\bar{y}) = \bar{x}$, done.

At $\text{coker } \gamma : \bar{v}'$ is injective since v' is.

Proposition 2.14.

(a). Let $f : (x, y) \mapsto y \otimes x; g : (y, x) \mapsto x \otimes y$ (check the bilinearity). So f induces a A -linear map $\bar{f} : x \otimes y \mapsto y \otimes x$. Similarly, g induces a A -linear map $\bar{g} : y \otimes x \mapsto x \otimes y$. Therefore, $\bar{f} \circ \bar{g} = \text{id}_{N \otimes M}; \bar{g} \circ \bar{f} = \text{id}_{M \otimes N}$, implying $\bar{f}$ and $\bar{g}$ are isomorphisms.

$$\begin{array}{ccc} M \oplus N & \xrightarrow{f} & N \otimes M \\ \phi \downarrow & \nearrow \bar{f} & \\ M \otimes N & & \end{array} \quad \begin{array}{ccc} N \oplus M & \xrightarrow{g} & M \otimes N \\ \psi \downarrow & \nearrow \bar{g} & \\ N \otimes M & & \end{array}$$

(c). Let $f : ((x, y), z) \mapsto (x \otimes z, y \otimes z)$ (Check the bilinearity). So f induces a homomorphism:

$$\begin{aligned} \bar{f} : (M \oplus N) \otimes P &\rightarrow (M \otimes P) \oplus (N \otimes P) \\ (x, y) \otimes z &\mapsto (x \otimes z, y \otimes z) \end{aligned}$$

Now we need to find the inverse of $\bar{f}$. Consider:

$$\begin{aligned} \delta : (M \otimes P) \oplus (N \otimes P) &\rightarrow (M \oplus N) \otimes P \\ (x \otimes z_1, y \otimes z_2) &\mapsto (x, 0) \otimes z_1 + (0, y) \otimes z_2 \end{aligned}$$

It is easy to check that δ is the inverse (both right and left) of $\bar{f}$, hence $\bar{f}$ is an isomorphism.

$$\begin{array}{ccc} (M \oplus N) \otimes P & \xrightarrow{f} & (M \otimes P) \oplus (N \otimes P) \\ \phi \downarrow & \nearrow \bar{f} & \\ (M \oplus N) \otimes P & & \end{array}$$

Note that at first glance we may think that $\bar{f}$ is not surjective, but it actually is! (Think of it.)

(d). Let $f : (a, x) \mapsto ax$ (Check the bilinearity). So f induces a A -linear map $\bar{f} : a \otimes x \mapsto ax$.

$$\begin{array}{ccc} A \oplus M & \xrightarrow{f} & M \\ \phi \downarrow & \nearrow \bar{f} & \\ A \otimes M & & \end{array}$$

Let $\delta : M \rightarrow A \otimes M$ where $\delta(x) = 1 \otimes x$, we then have:

$$\begin{aligned}\delta \circ \bar{f} : a \otimes x &\mapsto ax \mapsto 1 \otimes ax = a \otimes x \\ \bar{f} \circ \delta : x &\mapsto 1 \otimes x \mapsto x\end{aligned}$$

Therefore, δ is the inverse (both right and left) of $\bar{f}$, hence $\bar{f}$ is an isomorphism.

Exercise 2.15. We mimic the proof of Prop 2.14.(b) given in the book. Fix $z \in P$, let $f_z : M \oplus N \rightarrow M \otimes_A (N \otimes_B P)$ be given by: $(x, y) \mapsto x \otimes_A (y \otimes_B z)$. Check the A -bilinearity:

$$\begin{aligned}f_z(ax_1 + x_2, y) &= (ax_1 + x_2) \otimes_A (y \otimes_B z) \\ &= ax_1 \otimes_A (y \otimes_B z) + x_2 \otimes_A (y \otimes_B z) \\ &= a(x_1 \otimes_A (y \otimes_B z)) + x_2 \otimes_A (y \otimes_B z) \\ &= af_z(x_1, y) + f_z(x_2, y) \\ f_z(x, ay_1 + y_2) &= x \otimes_A ((ay_1 + y_2) \otimes_B z) \\ &= x \otimes_A (ay_1 \otimes_B z + y_2 \otimes_B z) \\ &= x \otimes_A (a(y_1 \otimes_B z) + y_2 \otimes_B z) \\ &= a(x \otimes_A (y_1 \otimes_B z)) + x \otimes_A (y_2 \otimes_B z) \\ &= af_z(x, y_1) + f_z(x, y_2)\end{aligned}$$

Therefore, f_z induces a A -linear map: $\bar{f}_z : M \otimes_A N \rightarrow M \otimes_A (N \otimes_B P)$. Now consider $g : (M \otimes_A N) \oplus P \rightarrow M \otimes_A (N \otimes_B P)$, given by: $(t, z) \mapsto \bar{f}_z(t)$. Check the B -bilinearity:

$$\begin{aligned}g(bt_1 + t_2, z) &= \bar{f}_z(bt_1 + t_2) = b\bar{f}_z(t_1) + \bar{f}_z(t_2) = bg(t_1, z) + g(t_2, z) \\ g(t, bz_1 + z_2) &= \overline{f_{bz_1+z_2}}(t) = \overline{f_{bz_1+z_2}}(x \otimes_A y) \text{ for some } x, y \in M, N, \text{ resp.} \\ &= x \otimes_A (y \otimes_B (bz_1 + z_2)) \\ &= x \otimes_A (b(y \otimes_B z_1) + y \otimes_B z_2) \\ &= b(x \otimes_A (y \otimes_B z_1)) + x \otimes_A y \otimes_B z_2 \\ &= b\bar{f}_{z_1}(t) + \bar{f}_{z_2}(t) = bg(t, z_1) + g(t, z_2)\end{aligned}$$

Therefore, g induces a B -linear map: $\bar{g} : (M \otimes_A N) \otimes_B P \rightarrow M \otimes_A (N \otimes_B P)$.

$$\begin{array}{ccc} M \oplus N & \xrightarrow{f_z} & M \otimes_A (N \otimes_B P) \\ \downarrow \phi & \nearrow \bar{f}_z & \\ M \otimes_A N & & \end{array} \quad \begin{array}{ccc} (M \otimes_A N) \oplus P & \xrightarrow{g} & M \otimes_A (N \otimes_B P) \\ \downarrow \psi & \nearrow \bar{g} & \\ (M \otimes_A N) \otimes_B P & & \end{array}$$

Similarly we can also construct a A -linear map $\bar{h} : M \otimes_A (N \otimes_B P) \rightarrow (M \otimes_A N) \otimes_B P$ and it is easy to see that $\bar{h}$ is the inverse (both right and left) of $\bar{g}$, hence $M \otimes_A (N \otimes_B P) \cong (M \otimes_A N) \otimes_B P$.

Exercise 2.20. For any $f : P_1 \rightarrow P_2$ where P_i $i = 1, 2$ are B -module and f is injective, it suffices to show that $f \otimes \text{id}_{M_B} : P_1 \otimes_B M_B \rightarrow P_2 \otimes_B M_B$ is injective (Prop 2.19.(iii)). Note that we have:

$$P_i \otimes_B M_B = P_i \otimes_B (B \otimes_A M) \cong (P_i \otimes_B B) \otimes_A M \cong P_i \otimes_A M$$

where the $\cong$ s are canonical as in Prop 2.14. and Exercise 2.15. Note that $f \otimes \text{id}_M : P_1 \otimes_A M \rightarrow P_2 \otimes_A M$ is injective since M is flat, hence by composing, we have:

$$x \otimes_B (b \otimes_A m) \mapsto (x \otimes_B b) \otimes_A m \mapsto bx \otimes_A m \xrightarrow{f \otimes \text{id}_M} bf(x) \otimes_A m \mapsto (f(x) \otimes_B b) \otimes_A m \mapsto f(x) \otimes_B (b \otimes_A m)$$

is injective, which is exactly $f \otimes \text{id}_{M_B}$.

Algebras

Let $f : A \rightarrow B$, $g : A \rightarrow C$ be ring homomorphisms. An A -**algebra homomorphism** $h : B \rightarrow C$ is a ring homomorphism which is also an A -module homomorphism. Verify that h is an A -algebra homomorphism if and only if $h \circ f = g$. To avoid any ambiguity in the following, I use $\diamond$ to denote ring action (i.e., A -scalar multiplication) and use $\cdot$ to denote the usual multiplication in a ring.

$\Rightarrow$: Consider $h(a \diamond 1_B)$ for any $a \in A$. We have:

$$h(a \diamond 1_B) = h(f(a) \cdot 1_B) = h(f(a))$$

On the other hand,

$$h(a \diamond 1_B) = a \diamond h(1_B) = g(a) \cdot h(1_B) = g(a) \cdot 1_C = g(a)$$

So we have $g = h \circ f$, as desired.

$\Leftarrow$: To show h is an A -module homomorphism, we need to show $h(a \diamond b_1 + b_2) = a \diamond h(b_1) + h(b_2)$, $\forall a \in A, b_1, b_2 \in B$, but since we assume h is a ring homomorphism, it suffices to show $h(a \diamond b) = a \diamond h(b)$. Indeed, we have

$$h(a \diamond b) = h(f(a) \cdot b) = h(f(a)) \cdot h(b) = g(a) \cdot h(b) = a \diamond h(b)$$

2.2 Exercises

2.2.1 Tensor product and exact sequence

2.1. Show that $(\mathbb{Z}/m\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z}) = 0$ if m, n are co-prime.

Proof. It suffices to show $\forall x \otimes_{\mathbb{Z}} y$ is zero; hence, it suffices to show $1 \otimes_{\mathbb{Z}} 1 = 0$. Note that since m, n are co-prime, $\exists s, t \in \mathbb{Z}$ s.t. $sm + tn = 1$. Thus,

$$1 \otimes_{\mathbb{Z}} 1 = (sm + tn) \otimes_{\mathbb{Z}} 1 = sm \otimes_{\mathbb{Z}} 1 + 1 \otimes_{\mathbb{Z}} tn = 0 + 0 = 0$$

as we desired.

Remark 15. More generally, we have $A/\mathfrak{a} \otimes_A A/\mathfrak{b} = 0$ if $\mathfrak{a}, \mathfrak{b}$ are co-prime. This is because $\exists a \in \mathfrak{a}, b \in \mathfrak{b}$ s.t. $a + b = 1$. So,

$$\bar{x} \otimes_A \bar{y} = (a + b)(\bar{x} \otimes_A \bar{y}) = \bar{a}\bar{x} \otimes_A \bar{y} + \bar{x} \otimes_A \bar{b}\bar{y} = 0 + 0 = 0$$

for any $\bar{x} \in A/\mathfrak{a}, \bar{y} \in A/\mathfrak{b}$.

2.2. Let M be an A -module, $\mathfrak{a} \triangleleft A$ be an arbitrary ideal. Show that $(A/\mathfrak{a}) \otimes_A M$ is isomorphic to $M/\mathfrak{a}M$.

Proof. Follow the hint. We naturally have the exact sequence:

$$0 \longrightarrow \mathfrak{a} \xrightarrow{\text{id}_A} A \xrightarrow{\phi} A/\mathfrak{a} \longrightarrow 0$$

Tensor with M , we get the exact sequence:

$$\mathfrak{a} \otimes_A M \xrightarrow{\text{id}_A \otimes \text{id}_M} A \otimes_A M \xrightarrow{\phi \otimes \text{id}_M} A/\mathfrak{a} \otimes_A M \longrightarrow 0$$

Note that $A \otimes_A M$ is canonically isomorphic to M by $a \otimes x \mapsto ax$. Hence it suffices to show $\text{Im}(\text{id}_A \otimes \text{id}_M) = \mathfrak{a}M$. Note that $\text{Im}(\text{id}_A \otimes \text{id}_M)$ is generated by all $a \otimes x$ where $a \in \mathfrak{a}, x \in M$, and so is $\mathfrak{a}M$, hence we are done.

Remark 16. Now we can deduce the conclusion in the previous remark of 2.1. by this: It suffices to show that $\mathfrak{a}(A/\mathfrak{b}) = A/\mathfrak{b}$, provided that $\mathfrak{a}, \mathfrak{b}$ are co-prime. Suppose $a + b = 1$ where $a \in \mathfrak{a}, b \in \mathfrak{b}$. Then, for any $\bar{y} \in A/\mathfrak{b}$, $\bar{y} = (a + b)\bar{y} = \bar{a}\bar{y} + 0 \in \mathfrak{a}M$. The converse direction is immediate.

Alternatively, we have:

$$A/\mathfrak{a} \otimes_A A/\mathfrak{b} \cong \frac{A/\mathfrak{b}}{\mathfrak{a}(A/\mathfrak{b})} \stackrel{\text{Cor 2.7.}}{=} \frac{A/\mathfrak{b}}{(\mathfrak{a} + \mathfrak{b})/\mathfrak{b}} = \frac{A/\mathfrak{b}}{A/\mathfrak{b}} = 0$$

2.3. Let A be a local ring, M, N finitely generated A -modules. Prove that if $M \otimes N = 0$, then $M = 0$ or $N = 0$.

Proof. Follow the hint. Let $k = A/\mathfrak{m}$ be the residue field where $\mathfrak{m}$ is the maximal ideal.

$$M \otimes_A N = 0 \Rightarrow (k \otimes_A M) \otimes_A (k \otimes_A N) = 0 \Rightarrow M_k \otimes_k N_k = 0 \Rightarrow M_k = 0 \text{ or } N_k = 0$$

Suppose $M_k = 0$. By 2.2. we have $M_k = A/\mathfrak{m} \otimes_A M \cong M/\mathfrak{m}M = 0$, hence $M = 0$ by Nakayama's lemma.

2.4. Let M_i ($i \in I$) be any family of A -modules, and let M be their direct sum. Prove that M is flat if and only if each M_i is flat.

Proof. Straightforward, using Prop 2.19.(ii) and Prop 2.14.(iii).

2.5. Prove that $A[x]$ is a flat A -algebra.

Proof. First, note that A is flat since $A \otimes_A M \cong M$ for any A -module M . Then, note that:

$$A[x] = A \oplus Ax \oplus Ax^2 \oplus \cdots \cong \bigoplus_{i \in \mathbb{N}} A_i \text{ as an } A\text{-module}$$

and apply 2.4.

2.6. For any A -module M , let $M[x]$ denote the set of all polynomials in x with coefficients in M . Show that $M[x] \cong A[x] \otimes_A M$.

Proof. Note that $A[x]/\langle x \rangle \cong A$. So by 2.2. we have $A \otimes_{A[x]} M[x] \cong M[x]/(\langle x \rangle M[x]) \cong M$. Thus,

$$A[x] \otimes_A M \cong A[x] \otimes_A (A \otimes_{A[x]} M[x]) \cong (A[x] \otimes_A A) \otimes_{A[x]} M[x] \cong A[x] \otimes_{A[x]} M[x] \cong M[x]$$

as desired.

2.7. Let $\mathfrak{p} \triangleleft A$ be prime. Show that $\mathfrak{p}[x]$ is prime in $A[x]$. If $\mathfrak{m}$ is maximal in A , is $\mathfrak{m}[x]$ maximal in $A[x]$?

Proof. Note that $A/\mathfrak{p}$ is a A -module, so by 2.6. and 2.2. we have:

$$(A/\mathfrak{p})[x] \cong A[x] \otimes_A A/\mathfrak{p} \cong A[x]/(\mathfrak{p}A[x]) = A[x]/\mathfrak{p}[x]$$

This is a A -module isomorphism. But that does not really help since what we need is a ring isomorphism. Consider instead, the ring homomorphism:

$$A[x] \twoheadrightarrow (A/\mathfrak{p})[x]$$

where the kernel is $\mathfrak{p}[x]$. Then apply 1.2.(b).

► Can we directly show that the above isomorphism is a A -algebra isomorphism? (Think of it).

For the second part. Take $A = \mathbb{Z}$ and $\mathfrak{m} = 2\mathbb{Z}$, but $\mathbb{Z}[x]/2\mathbb{Z}[x] \cong \mathbb{Z}_2[x]$ is clearly not a field by 1.2.(a).

2.8. (a). If M, N are flat A -module, then so is $M \otimes_A N$.

Proof. Apply Prop 2.19.(ii) and Prop 2.14.(ii).

(b). If B is a flat A -algebra, and N is a flat B -module, then N is flat as an A -module.

Proof. Take any (short) exact sequence:

$$0 \longrightarrow P_1 \longrightarrow P_2 \longrightarrow P_3 \longrightarrow 0$$

where each P_i is an A -module. Since B is a flat A -algebra, the sequence

$$0 \longrightarrow P_1 \otimes_A B \longrightarrow P_2 \otimes_A B \longrightarrow P_3 \otimes_A B \longrightarrow 0$$

is exact. Note that the $P_i \otimes_A B$ is naturally a B -module, $i \in [3]$, and since N is a flat B -module, the sequence

$$0 \longrightarrow P_1 \otimes_A B \otimes_B N \longrightarrow P_2 \otimes_A B \otimes_B N \longrightarrow P_3 \otimes_A B \otimes_B N \longrightarrow 0$$

is exact. But this is equivalent of saying:

$$0 \longrightarrow P_1 \otimes_A N \longrightarrow P_2 \otimes_A N \longrightarrow P_3 \otimes_A N \longrightarrow 0$$

is exact, which shows the flatness of N , as an A -module.

Remark 17. To do the next 2 exercises, I first claim and prove a very useful lemma ([[Gat13](#)] Exercise 3.5):

Let N be a submodule of a module M over a ring A . Then

(a). *If N and M/N are finitely generated, then so is M .*

Proof. Let $N = \langle n_1, \dots, n_k \rangle$ and $M/N = \langle \overline{m_1}, \dots, \overline{m_l} \rangle$. Then, $\forall m \in M, \exists r_1, \dots, r_l \in A$ s.t.

$$\overline{m} = r_1 \overline{m_1} + \dots + r_l \overline{m_l}.$$

i.e.,

$$\begin{aligned} m + N &= r_1(m_1 + N) + \dots + r_l(m_l + N) \\ &= r_1 m_1 + \dots + r_l m_l + N \end{aligned}$$

This implies:

$$m - (r_1 m_1 + \dots + r_l m_l) \in N.$$

Since N is generated by $n_1, \dots, n_k$, $\exists s_1, \dots, s_k \in A$ s.t.

$$m - (r_1 m_1 + r_2 m_2 + \dots + r_l m_l) = s_1 n_1 + s_2 n_2 + \dots + s_k n_k.$$

Therefore, $M = \langle m_1, \dots, m_l, n_1, \dots, n_k \rangle$.

(b). *If M is finitely generated, then so is M/N .*

Proof. Let $M = \langle m_1, \dots, m_k \rangle$. Then, $\forall \overline{m} \in M/N, \exists r_1, \dots, r_k \in A$ s.t.

$$\begin{aligned}\overline{m} &= m + N \\ &= r_1 m_1 + \dots + r_k m_k + N \\ &= r_1(m_1 + N) + \dots + r_k(m_k + N) \\ &= r_1 \overline{m_1} + \dots + r_k \overline{m_k}\end{aligned}$$

Therefore, $M/N = \langle \overline{m_1}, \dots, \overline{m_k} \rangle$.

(c). If M is finitely generated, N need not be finitely generated.

Proof. Consider the case where $A = \mathbb{Z}[x_1, x_2, \dots]$ with infinitely many indeterminate and M is A itself as a A -module. Note that M is finitely generated by $\{1_A\}$ (It is a ring!) And let $N = \langle x_1, x_2, \dots \rangle$ as a A -submodule of M . Then N is obviously not finitely generated.

2.9. Let $0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$ be an exact sequence of A -modules. If M' and M'' are finitely generated, so is M .

Proof. Suppose the sequence

$$0 \longrightarrow M' \xrightarrow{f} M \xrightarrow{g} M'' \longrightarrow 0$$

is exact, and $M' = \langle x_1, \dots, x_n \rangle$. This implies $\text{Im } f = \langle f(x_1), \dots, f(x_n) \rangle$ is finitely generated. Note that we can rewrite $\text{Im } g \cong M / \ker g$ as

$$M'' \cong M / \text{Im } f$$

by the exactness at M and M'' . It follows that M is finitely generated since M'' and $\text{Im } f$ are.

2.10. Let M be an A -module, N a finitely generated A -module, and let $u : M \rightarrow N$ be a homomorphism. Let $\mathfrak{a} \subseteq \mathfrak{J}$. If the induced homomorphism $M/\mathfrak{a}M \rightarrow N/\mathfrak{a}N$ is surjective, then u is surjective.

Proof 1. Let $v : N \rightarrow \text{coker } u$ be the projection. Consider the exact sequence:

$$M \xrightarrow{u} N \xrightarrow{v} \text{coker } u \longrightarrow 0$$

Tensor with $A/\mathfrak{a}$ as an A -module, we have

$$A/\mathfrak{a} \otimes_A M \xrightarrow{\text{id}_{A/\mathfrak{a}} \otimes u} A/\mathfrak{a} \otimes_A N \xrightarrow{\text{id}_{A/\mathfrak{a}} \otimes v} A/\mathfrak{a} \otimes_A \text{coker } u \longrightarrow 0$$

is an exact sequence by Prop 2.18.. This implies, by 2.2., the sequence

$$M/\mathfrak{a}M \xrightarrow{\overline{u}} N/\mathfrak{a}N \xrightarrow{\overline{v}} \text{coker } u/\mathfrak{a} \text{coker } u \longrightarrow 0$$

is exact. Since $\overline{u}$ is surjective, $\ker \overline{v} = N/\mathfrak{a}N$. It follows that $\text{coker } u/\mathfrak{a} \text{coker } u = 0$, i.e., $\text{coker } u = \mathfrak{a} \text{coker } u$. Note that $\text{coker } u$ is finitely generated since N is, so $\text{coker } u = 0$ by Nakayama's lemma, i.e., u is surjective.

Proof 2. Consider the composite homomorphism:

$$M \xrightarrow{\pi} M/\mathfrak{a}M \xrightarrow{\bar{u}} N/\mathfrak{a}N$$

where

$$m \mapsto \bar{m} \mapsto u(m) + \mathfrak{a}N$$

Thus, $N = \mathfrak{a}N + \text{Im}(u)$ by surjectivity of $\bar{u} \circ \pi$. Since N is finitely generated, this implies, by Corollary 2.7., $\text{Im}(u) = N$, i.e., u is surjective.

2.11. Let A be a non-zero ring. $\phi : A^m \rightarrow A^n$ (as a homomorphism between A -modules.)

If ϕ is surjective, then $m \geq n$.

Proof. Follow the hint. Let $\mathfrak{m}$ be a maximal ideal of A . Then:

$$f := \text{id} \otimes_A \phi : (A/\mathfrak{m}) \otimes_A A^m \rightarrow (A/\mathfrak{m}) \otimes_A A^n$$

is a linear transformation between vector spaces of dimension m and n resp., over the field $k = A/\mathfrak{m}$. Note that f is surjective since ϕ and id are, hence $m \geq n$.

If ϕ is injective, then $m \leq n$.

Proof. I learned the following from [hs]. Suppose $m > n$, we can extend ϕ to $A^m \rightarrow A^m$ by forcing the last $m - n$ coordinates to always be zero (this does not change the injectivity). Now, since A^m is finitely generated and $\phi(A^m) \subseteq AA^m$ (here we regard A itself as an ideal), by Prop 2.4., $\exists a_1, \dots, a_k \in A$ satisfying

$$p := \phi^k + a_1\phi^{k-1} + \dots + a_k = 0$$

We may set the degree k to be the lowest, and then the constant term a_k must be non-zero. Because if not, we could cancel ϕ of degree one thanks to the injectivity of ϕ (and note that $A \neq 0$ so ϕ cannot be the zero map). Consequently, this contradicts the minimality of the degree. But then, if we let p act on $x := (0, \dots, 0, 1) \in A^m$, we get $(-, \dots, -, a_k) = (0, \dots, 0, 0)$, a contradiction.

2.12. Let M be a finitely generated A -module and $\phi : M \rightarrow A^n$ a surjective homomorphism. Show that $\ker \phi$ is finitely generated.

Proof. Follow the hint. Let $e_1, \dots, e_n$ be a basis of A^n and choose $u_i \in M$ s.t. $\phi(u_i) = e_i$. Let $N = \langle u_1, \dots, u_n \rangle$. Suppose $x \in N \cap \ker \phi$, then $x = a_1u_1 + \dots + a_nu_n$ for some $a_i \in A$ and $\phi(x) = a_1e_1 + \dots + a_ne_n = 0$. This implies all the coefficients a_i are zero, hence $x = 0$. Thanks to the surjectivity of ϕ , it also follows that $N \cong A^n$ by restricting ϕ to N . So we have

$$(N + \ker \phi) / \ker \phi \cong N / (N \cap \ker \phi) \cong N \cong A^n \cong M / \ker \phi$$

Consequently, $N + \ker \phi = M$ and therefore, $M = N \oplus \ker \phi$. And since M and N are finitely generated, so is $\ker \phi$.

2.13. Let $f : A \rightarrow B$ be a ring and let N be a B -module. Regarding N as an A -module

by restriction of scalars, form the B -module $N_B = B \otimes_A N$. Show that the homomorphism $g : N \rightarrow N_B$ which maps y to $1 \otimes y$ is injective and that $g(N)$ is a direct summand of N_B .

Proof. Define p as the hint suggests (verify that it is a homomorphism). The composite homomorphism:

$$N \xrightarrow{g} N_B \xrightarrow{p} N$$

is clearly the id_N . Consequently, g is injective. Suppose $b \otimes y \in \text{Im}(g) \cap \ker p$, then $b \otimes y = 1 \otimes by = 1 \otimes 0 = 0$. And any $b \otimes y \in N_B$ can be written as $b \otimes y = 1 \otimes by + 0 \otimes by \in \text{Im}(g) + \ker p$. Therefore, $N_B = \text{Im}(g) \oplus \ker p$.

Remark 18. Generally, If we have A -module homomorphisms $f : N \rightarrow M; g : M \rightarrow N$ s.t. $g \circ f = \text{id}_N$. Then $M = \text{Im } f \oplus \ker g$. *Proof.* Suppose $f(x) \in \text{Im } f \cap \ker g$, then $x = g(f(x)) = 0$, hence $f(x) = 0$, which means $\text{Im } f \cap \ker g = 0$. $\forall m \in M$, let $x = g(m)$, then $g(m - f(x)) = x - x = 0$, so $m - f(x) \in \ker g$ and hence $m \in \text{Im } f + \ker g$, which means $M = \text{Im } f + \ker g$. Thus, $M = \text{Im } f \oplus \ker g$.

2.2.2 Direct limits

2.14. We shall construct an A -module M called the **direct limit** of the direct system $\mathbf{M} = (M_i, \mu_{ij})$. Let $C = \bigoplus_{i \in I} M_i$, and identify each M_i with its canonical image in C . Let D be the submodule of C generated by all elements of the form $x_i - \mu_{ij}(x_i)$ where $i \leq j$ and $x_i \in M_i$. Let $M = C/D$, let $\mu : C \rightarrow M$ be the projection, and let μ_i be the restriction of μ to M_i .

The module M , or more correctly the pair consisting of M and the family of homomorphisms $\mu_i : M_i \rightarrow M$, is called the **direct limit** of the direct system $\mathbf{M}$, and is written $\varinjlim M_i$. From the construction, it is clear that $\mu_i = \mu_j \circ \mu_{ij}$ whenever $i \leq j$. Because

$$\mu_j(\mu_{ij}(x_i)) = \overline{\mu_{ij}(x_i)} = \overline{x_i} = \mu_i(x_i)$$

2.15. In the situation of 2.14, show that every element in M can be written as $\mu_i(x_i)$ for some $i \in I$ and some $x_i \in M_i$. Show also that if $\mu_i(x_i) = 0$ then $\exists j \geq i$ s.t. $\mu_{ij}(x_i) = 0$ in M_j .

Proof. For the first part, note that every element in M can be written as $\mu(c)$ for some $c \in C$, so pick $\forall m = \mu(c) \in M$ where $c = \sum_{i=1}^n \iota_{r_i}(x_{r_i})$, $r_i \in I$. Pick $j \in I$ that dominates all the $r_i, i \in [n]$, then

$$m = \mu(c) = \mu\left(\sum_{i=1}^n \iota_{r_i}(x_{r_i})\right) = \sum_{i=1}^n \mu_{r_i}(x_{r_i}) = \sum_{i=1}^n \mu_j(\mu_{r_i j}(x_{r_i})) = \mu_j\left(\sum_{i=1}^n \mu_{r_i j}(x_{r_i})\right) = \mu_j(x_j)$$

for some $x_j \in M_j$.

For the second part, since $\mu_i(x_i) = 0 \in M$, we know that $x_i \in C$ can be written as a finite A -linear combination of generators of D , i.e.

$$\iota_i(x_i) = \sum_{k=1}^n r_k (\iota_{\alpha_k}(x_{\alpha_k}) - \mu_{\alpha_k \beta_k}(\iota_{\alpha_k}(x_{\alpha_k})))$$

By the directness of the system, $\exists j \in$ the index set I s.t. j dominates all the indexes appearing above. And from now on we can restrict our index set to $J := \{\alpha_k, \beta_k\}_{k \in [n]} \cup \{i, j\}$ and $C := \bigoplus_{l \in J} M_l$. Define $\theta_j : C \rightarrow M_j$ as:

$$\iota_l(x_l) \mapsto \mu_{lj}(x_l)$$

Note that θ_j can be thought of projecting everything to the j -th coordinate. Consequently, $\theta_j(D) = 0$ since j dominates all indexes in J . Therefore, θ_j can be factored through:

$$C \xrightarrow{\mu} C/D \xrightarrow{\overline{\theta_j}} M_j$$

Also, note that:

$$M_j \xrightarrow{\iota_j} C \xrightarrow{\mu} C/D \xrightarrow{\overline{\theta_j}} M_j \text{ is the identical map}$$

Hence we have

$$\overline{\theta_j} \circ \mu(\iota_i(x_i)) = \overline{\theta_j} \circ \mu\left(\sum_{k=1}^n r_k(\iota_{\alpha_k}(x_{\alpha_k})) - \mu_{\alpha_k \beta_k}(\iota_{\alpha_k}(x_{\alpha_k}))\right)$$

The RHS is evidently zero, whereas for the LHS,

$$\overline{\theta_j} \circ \mu(\iota_i(x_i)) = \overline{\theta_j} \circ \mu_j(\mu_{ij}(x_i)) = \overline{\theta_j} \circ \mu \circ \iota_j(\mu_{ij}(x_i)) = \mu_{ij}(x_i)$$

So we are done.

2.16. Show that the direct limit is characterized (up to isomorphism) by the following property: Let N be an A -module, and for each $i \in I$ let $\alpha_i : M_i \rightarrow N$ be an A -module homomorphism s.t. $\alpha_j \circ \mu_{ij} = \alpha_i$ whenever $i \leq j$. Then $\exists!$ homomorphism $\alpha : M \rightarrow N$ s.t. $\alpha_i = \alpha \circ \mu_i$ for all $i \in I$.

Proof. First, we show the existence of such $\alpha : M \rightarrow N$. The idea is to construct a homomorphism $f : C \rightarrow N$ s.t. $D \subseteq \ker f$, so that f naturally induces a homomorphism $M \rightarrow N$. And then try to show the induced homomorphism satisfies the requirements. Now, define f by:

$$\iota_i(x_i) \mapsto \alpha_i(x_i)$$

Take any generator $\iota_i(x_i) - \iota_j \circ \mu_{ij}(x_i)$, $i \leq j$, we have:

$$f(\iota_i(x_i) - \iota_j \circ \mu_{ij}(x_i)) = \alpha_i(x_i) - \alpha_j \circ \mu_{ij}(x_i) = \alpha_i(x_i) - \alpha_i(x_i) = 0$$

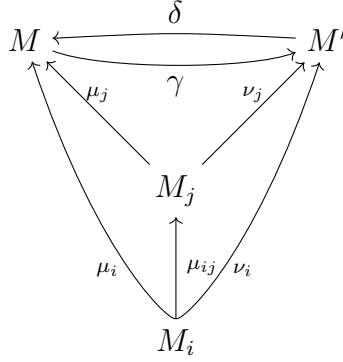
Thus, f induces a homomorphism from M to N , namely α . Take $\forall i \in I$ and $x_i \in M_i$, we have:

$$\alpha \circ \mu_i(x_i) = \alpha \circ \mu(\iota_i(x_i)) = f(\iota_i(x_i)) = \alpha_i(x_i)$$

So indeed, $\alpha_i = \alpha \circ \mu_i$. Next, we show the uniqueness of α . Suppose $\beta : M \rightarrow N$ satisfying $\alpha_i = \beta \circ \mu_i$, then by 2.15. we know every element in M can be written as $\mu_i(x_i)$ for some $i \in I, x_i \in M_i$. Thus, take any $\mu_i(x_i) \in M$, we have

$$\beta(\mu_i(x_i)) = \alpha_i(x_i) = f(\iota_i(x_i)) = \alpha(\mu_i(x_i))$$

Hence $\beta = \alpha$. Finally, we need to show that $\varinjlim M_i$ is unique up to isomorphism. Suppose $\exists$ another A -module M' over the same direct system that also satisfies the above properties, then by the universal property we just proved, $\exists!$ homomorphism, namely $\gamma : M \rightarrow M'$ s.t. $\nu_i = \gamma \circ \mu_i$, $\forall i \in I$. Symmetrically, by applying the universal property on M' we know, $\exists!$ homomorphism, namely $\delta : M' \rightarrow M$ s.t. $\mu_i = \delta \circ \nu_i$ $\forall i \in I$. Consequently, the following diagram commutes:



It remains to show that $\gamma \circ \delta = \text{id}_{M'}$ and $\delta \circ \gamma = \text{id}_M$. Again, take any $\mu_i(x_i) \in M$, we have:

$$\delta \circ \gamma \circ \mu_i(x_i) = \delta \circ \nu_i(x_i) = \mu_i(x_i)$$

Hence $\delta \circ \gamma = \text{id}$, and $\gamma \circ \delta = \text{id}$ follows from symmetry.

Remark 19. The above property for $\varinjlim M_i$ is exactly the universal property of the **colimit** object of the system (M_i, μ_{ij}) in categorical language. And as we saw from the proof, the colimit, if it exists, is unique up to unique isomorphism. See [CHA23] for more discussion.

2.17. Let $\{M_i\}_{i \in I}$ be a family of submodules of an A -module, say P , s.t. for each $i, j \in I, \exists k \in I$ satisfying $M_i + M_j \subseteq M_k$. Define $i \leq j$ to mean $M_i \subseteq M_j$ and let $\mu_{ij} : M_i \subseteq M_j$ be the embedding of M_i in M_j . Show that:

$$\varinjlim M_i = \sum M_i = \bigcup M_i$$

In particular, any A -module is the direct limit of its finitely generated submodules.

Proof. Preconditions: (i). It is easy to see that (M_i, μ_{ij}) forms a direct system by our setting; hence, the direct limit $\varinjlim M_i$ does exist. (ii). The union of submodules are not in general a module, but in here, obviously we don't need to worry about this.

Then, note that for each $i \in I$, we may define $\alpha_i : M_i \rightarrow N \subseteq P$ as the embedding map for both $N = \sum M_i$ or $\bigcup M_i$, so that $\alpha_i = \alpha_j \circ \mu_{ij}$ whenever $i \leq j$.

$$\begin{array}{ccccc}
 & \sum M_k & \xleftarrow{\phi} & \varinjlim M_k & \xrightarrow{\phi} & \bigcup M_k \\
 \alpha_i \nearrow & & & & & \nwarrow \alpha_j \\
 M_i & \xrightarrow{\mu_{ij}} & M_j & & & M_i \xrightarrow{\mu_{ij}} M_j
 \end{array}$$

Therefore, by the universal property of $\varinjlim M_i$, we know that $\exists!$ homomorphism, say $\phi : \varinjlim M_k \rightarrow N$ s.t. $\alpha_i = \phi \circ \mu_i, \forall i \in I$. We want to show that ϕ is an isomorphism. Surjectivity is clear since every element in N is from M_i for some $i \in I$. Injectivity: $\phi(\mu_i(x_i)) = 0 = \alpha_i(x_i)$ implies $x_i = 0$ since α_i s are inclusions. Hence $\mu_i(x_i) = 0$.

Why is $\varinjlim M_i$ a submodule of P ? Informally, we can regard it as sticking all the M_i together and canceling out the distinction between the same element in different coordinates in $\bigoplus M_i$. More precisely, construct a map:

$$\alpha : \varinjlim M_i \rightarrow P \quad \text{where} \quad \mu_i(x_i) \mapsto x_i$$

which is well-defined and injective. So we can regard $\varinjlim M_i$ as a submodule of P via the embedding α and $\text{Im}(\alpha) = \bigcup M_i$.

Finally, P is the direct limit of its finitely generated submodules follows immediately from the fact that P is the union of its finitely generated submodules.

2.18. Let $\mathbf{M} = (M_i, \mu_{ij})$, $\mathbf{N} = (N_i, \nu_{ij})$ be direct systems of A -modules over the same direct set, say I . Let M, N be the direct limits and μ_i, ν_i the associated homomorphisms.

A **homomorphism** $\Phi : \mathbf{M} \rightarrow \mathbf{N}$ is by definition a family of A -module homomorphisms $\phi_i : M_i \rightarrow N_i$ s.t. $\phi_j \circ \mu_{ij} = \nu_{ij} \circ \phi_i$ whenever $i \leq j$. Show that Φ defines a unique homomorphism $\phi = \varinjlim \phi_i : M \rightarrow N$ s.t. $\phi \circ \mu_i = \nu_i \phi_i$, for all $i \in I$.

Proof. There is nothing to prove once we draw the commutative diagram following the description.

2.19. A sequence of direct systems and homomorphisms:

$$\mathbf{M} \longrightarrow \mathbf{N} \longrightarrow \mathbf{P}$$

is **exact** if the corresponding sequence of modules and module homomorphisms is exact for each $i \in I$. Show that the sequence $M \longrightarrow N \longrightarrow P$ of direct limits is exact.

Proof. For any $i, j \in I, i \leq j$, we have the following commutative diagram:

$$\begin{array}{ccccc}
 M & \xrightarrow{\quad \phi \quad} & N & \xrightarrow{\quad \psi \quad} & P \\
 \swarrow \mu_j & & \swarrow \nu_j & & \swarrow \xi_j \\
 & M_j & \xrightarrow{\quad \phi_j \quad} & N_j & \xrightarrow{\quad \psi_j \quad} & P_j \\
 \uparrow \mu_i & \uparrow \mu_{ij} & \searrow \nu_i & \uparrow \nu_{ij} & \searrow \xi_i & \uparrow \xi_{ij} \\
 M_i & \xrightarrow{\quad \phi_i \quad} & N_i & \xrightarrow{\quad \psi_i \quad} & P_i
 \end{array}$$

Take any $\mu_i(x_i) \in M$, we know $\psi_i \circ \phi_i(x_i) = 0$ by the exactness at N_i . It follows that $\xi_i \circ \psi_i \circ \phi_i(x_i) = 0$, hence $\psi \circ \phi(\mu_i(x_i)) = 0$ thanks to the comm. of the diagram, which implies $\text{Im } \phi \subseteq \ker \psi$. To show $\text{Im } \phi \supseteq \ker \psi$, take any $\nu_i(y_i) \in N$ s.t. $\psi(\nu_i(y_i)) = 0 \in P$. Let $z_i = \psi_i(y_i) \in P_i$, then $\xi_i(z_i) = 0 \in P$. So, by 2.15. we may assume $\xi_{ij}(z_i) = 0 \in P_j$. Thus, by the exactness at N_j , $y_j := \nu_{ij}(y_i) \in \text{Im } \phi_j$. So let $x_j \in M_j$ s.t. $\phi_j(x_j) = y_j$, then $\nu_i(y_i) = \nu_j(y_j) = \phi(\mu_j(x_j)) \in \text{Im } \phi$, as desired. Though this can be proved easily by diagram chasing as above, I believe that there exists a deeper understanding behind.

2.20. Let N be any A -module. Show that:

$$\varinjlim (M_i \otimes N) \cong (\varinjlim M_i) \otimes N$$

Proof. Follow the hint. Let $P := \varinjlim (M_i \otimes N)$, $M := (\varinjlim M_i)$, $\psi : P \rightarrow M \otimes N$ be the unique homomorphism by the universal property of P , $\phi : M \otimes N \rightarrow P$ be the unique homomorphism induced by the bilinear map $g : M \oplus N \rightarrow P$ where $(\mu_i(x_i), n) \mapsto \epsilon_i(x_i \otimes n)$ (check the bilinearity).

$$\begin{array}{ccc}
P & \xrightarrow{\psi} & M \otimes N \\
\epsilon_i \swarrow & & \nearrow \mu_i \otimes \text{id} \\
& M_i \otimes N &
\end{array}
\qquad
\begin{array}{ccc}
& M \oplus N & \\
\swarrow & & \searrow g \\
M \otimes N & \xrightarrow{\phi} & P
\end{array}$$

Then, it suffices to show that $\psi \circ \phi = \text{id}$ and $\phi \circ \psi = \text{id}$. Indeed, we have

$$\begin{aligned}
\psi \circ \phi(\mu_i(x_i) \otimes n) &= \psi \circ g(\mu_i(x_i), n) = \psi \circ \epsilon_i(x_i \otimes n) = \mu_i \otimes \text{id}(x_i \otimes n) = \mu_i(x_i) \otimes n \\
\phi \circ \psi(\epsilon_i(x_i \otimes n)) &= \phi \circ \mu_i \otimes \text{id}(x_i \otimes n) = \phi(\mu_i(x_i) \otimes n) = g(\mu_i(x_i), n) = \epsilon_i(x_i \otimes n)
\end{aligned}$$

So we are done.

2.21. Let $\{A_i\}_{i \in I}$ be a family of rings indexed by a directed set I . Regarding A_i as a $\mathbb{Z}$ -module we can form the direct limit $A := \varinjlim A_i$ w.r.t. the valid system (A_i, α_{ij}) . Show that A inherits a ring structure from A_i so that the mappings $A_i \rightarrow A$ are ring homomorphisms. If $A = 0$, show that $A_i = 0$ for some $i \in I$.

Proof. The key point is how to define multiplication in A . The idea is to project the indexes to a higher one, say k , thanks to the directness of the index set, so that we can do arithmetic in A_k and then project to A via α_k . Precisely, consider $(A, +, \cdot)$ where $\cdot$ is defined to be:

$$\alpha_i(x_i) \cdot \alpha_j(x_j) := \alpha_k(\alpha_{ik}(x_i) \alpha_{jk}(x_j)) \text{ for some } k \geq i, j$$

This map $A \times A \rightarrow A$ is well-defined, i.e., independent of the choice of k because we always have

$$\begin{aligned}
\alpha_{k'}(\alpha_{ik'}(x_i) \alpha_{jk'}(x_j)) &= \alpha_{k'}((\alpha_{kk'} \circ \alpha_{ik})(x_i))(\alpha_{kk'} \circ \alpha_{jk})(x_j)) \\
&= \alpha_{k'}(\alpha_{kk'}(\alpha_{ik}(x_i) \alpha_{jk}(x_j))) = \alpha_k(\alpha_{ik}(x_i) \alpha_{jk}(x_j))
\end{aligned}$$

whenever $k' \geq k$. Now, check the arithmetic: associativity of $\cdot$ and distributivity are obviously satisfied by projecting to a common A_k ; $1_A := \alpha_i(1_i)$ is independent of i and acts as the unity of A since

$$\alpha_i(1_i) \cdot \alpha_j(x_j) = \alpha_k(\alpha_{ik}(1_i) \alpha_{jk}(x_j)) = \alpha_k(1_k \alpha_{jk}(x_j)) = \alpha_k(\alpha_{jk}(x_j)) = \alpha_j(x_j)$$

for any $k \geq i, j$. And it is straightforward to check: the $\{\alpha_i\}_{i \in I}$ are ring homomorphisms. If $A = 0$, then $\forall i \in I, \alpha_i(1_i) = 0$. Fix an arbitrary i , consequently, by 2.15., $\exists j \geq i$ s.t. $\alpha_{ij}(1_i) = 1_j = 0 \Rightarrow A_j = 0$.

2.22. Let (A_i, α_{ij}) be a direct system of rings and let $\mathfrak{R}_i$ be the nilradical of A_i . Show that $\varinjlim \mathfrak{R}_i$ is the nilradical of $\varinjlim A_i$. If each A_i is an integral domain, then $\varinjlim A_i$ is an integral domain.

Proof. Take any nilpotent element in $\varinjlim A_i$, say $\alpha_i(x_i)$ for some $i \in I$. Then $\alpha_i(x_i)^n = \alpha_i(x_i^n) = 0$ for some $n \in \mathbb{N}$. By 2.15., this means $\exists j \geq i$ s.t. $0 = \alpha_{ij}(x_i^n) = \alpha_{ij}(x_i)^n$, hence $\alpha_{ij}(x_i) \in \mathfrak{R}_j$ and $\alpha_i(x_i) = \alpha_j(\alpha_{ij}(x_i)) \in \varinjlim \mathfrak{R}_i$. Conversely, take any element in $\varinjlim \mathfrak{R}_i$, say $\alpha_l(x_l)$

for some $l \in I$, then $x_l \in \mathfrak{N}_l \Rightarrow x_l^m = 0$ for some $m \in \mathbb{N} \Rightarrow \alpha_l(x_l^m) = \alpha_l(x_l)^m = 0 \Rightarrow \alpha_l(x_l)$ is nilpotent in $\varinjlim A_i$. Therefore, $\varinjlim \mathfrak{N}_i$ is the nilradical of $\varinjlim A_i$.

To show the second part, suppose $\varinjlim A_i$ is not an integral domain, then $\exists \alpha_i(x_i), \alpha_j(x_j) \in \varinjlim A_i$ s.t. $\alpha_i(x_i)\alpha_j(x_j) = \alpha_k(\alpha_{ik}(x_i)\alpha_{jk}(x_j)) = 0$ for some $k \geq i, j$. This implies, again by 2.15., $\exists l \geq k$ s.t. $0 = \alpha_{kl}(\alpha_{ik}(x_i)\alpha_{jk}(x_j)) = \alpha_{il}(x_i)\alpha_{jl}(x_j) \Rightarrow A_l$ is not an integral domain, as desired.

2.23. Let $\{B_\lambda\}_{\lambda \in \Lambda}$ be a family of A -algebras. For each finite subset $J \subseteq \Lambda$ let B_J denote the tensor product (over A) of the B_λ for $\lambda \in J$. If J' is another finite subset and $J \subseteq J'$, there is a canonical A -algebra homomorphism $B_J \rightarrow B_{J'}$. Let B denote the direct limit of the rings B_J as J runs through all finite subset of Λ . The ring B has a natural A -algebra structure for which the homomorphisms $B_J \rightarrow B$ are A -algebra homomorphisms. The A -algebra B is the **tensor product** of the family of $\{B_\lambda\}_{\lambda \in \Lambda}$.

Proof. This problem aims to set up the definition of the tensor product of an arbitrary family of A -algebras $\{B_\lambda\}_{\lambda \in \Lambda}$. For any pair of finite subsets J, J' of Λ where $J \subseteq J'$, define

$$s_{JJ'} : B_J := \bigotimes_{\lambda \in J} B_\lambda \rightarrow B_{J'} := \bigotimes_{\kappa \in J'} B_\kappa$$

to be the map tensor with the identity maps on the extra factors in $J' - J$. (Check this is an A -algebra homomorphism). Thus, the collection $\{B_J\}_{J \subseteq \Lambda, \text{ finite}}$ forms a direct system of rings, denoted as B . We define

$$\bigotimes_{\lambda \in \Lambda} B_\lambda := \varinjlim_{J \subseteq \Lambda, \text{ finite}} B_J = B$$

We need to show that the ring B has a natural A -algebra structure for which the homomorphisms $\beta_J : B_J \rightarrow B$ are A -algebra homomorphisms. Define the ring action $\diamond : A \times B \rightarrow B$ in B as:

$$a \diamond \beta_J(x_J) = \beta_J(a \cdot x_J)$$

Well-definedness of the action: Suppose $\beta_J(x_J) = \beta_{J'}(y_{J'})$ in B . By the property of the direct system, there exists a finite $K \supseteq J \cup J'$ s.t.

$$s_{JK}(x_J) = s_{J'K}(y_{J'}) \quad \text{in } B_K$$

Then for any $a \in A$ we have

$$s_{JK}(a \cdot x_J) = a \cdot s_{JK}(x_J) = a \cdot s_{J'K}(y_{J'}) = s_{J'K}(a \cdot y_{J'})$$

since s_{JK} and $s_{J'K}$ are A -algebra homomorphisms. Applying β_K and using $\beta_K \circ s_{JK} = \beta_J$ and $\beta_K \circ s_{J'K} = \beta_{J'}$, we obtain

$$\beta_J(a \cdot x_J) = \beta_{J'}(a \cdot y_{J'})$$

Thus the action $\diamond$ is well defined. Now we check A -algebra axioms and A -linearity of β_J . For $a, b \in A$ and $x_J \in B_J$.

$$(a + b) \diamond \beta_J(x_J) = \beta_J((a + b) \cdot x_J) = \beta_J(a \cdot x_J) + \beta_J(b \cdot x_J) = a \diamond \beta_J(x_J) + b \diamond \beta_J(x_J)$$

Similarly,

$$(ab) \diamond \beta_J(x_J) = a \diamond (b \diamond \beta_J(x_J)), \quad 1 \diamond \beta_J(x_J) = \beta_J(x_J)$$

By definition we also have

$$\beta_J(a \cdot x_J) = a \diamond \beta_J(x_J)$$

so each β_J is A -linear. Since β_J is already a ring homomorphism, it follows that each β_J is an A -algebra homomorphism.

2.2.3 Flatness and Tor

2.24. Let M be an A -module, prove that the following are equivalent:

- (a) M is flat;
- (b) $\text{Tor}_n^A(M, N) = 0$ for all $n > 0$ and all A -modules N ;
- (c) $\text{Tor}_1^A(M, N) = 0$ for all A -modules N

Proof. (a) $\implies$ (b): Recall that M is flat $\iff$ the functor $F(-) := M \otimes_A -$ is (both right and left) exact. Fix an arbitrary A -module N ; after applying $F(-)$ to the projective resolution $(C_\bullet, \epsilon)$, the resulting augmented chain complex is exact. Hence for any $n > 0$, $\text{Tor}_n(M, N) = LF_n(C_\bullet) = H_n(F(C_\bullet)) = 0$. (b) $\implies$ (c) is trivial. (c) $\implies$ (a): Follow the hint. Take any short exact sequence:

$$0 \longrightarrow N' \longrightarrow N \longrightarrow N'' \longrightarrow 0$$

Note that $\exists$ a natural morphism $L_1F(N'') \longrightarrow L_0F(N')$ s.t. the following sequence is exact:

$$\text{Tor}_1(M, N'') = LF_1(M, N'') \longrightarrow LF_0(M, N') \longrightarrow LF_0(M, N) \longrightarrow LF_0(M, N'') \longrightarrow 0$$

Since $\text{Tor}_1(M, N'') = 0$, this sequence is just:

$$0 \longrightarrow M \otimes_A N' \longrightarrow M \otimes_A N \longrightarrow M \otimes_A N'' \longrightarrow 0$$

Therefore, M is flat.

2.25. Let $0 \longrightarrow N' \longrightarrow N \longrightarrow N'' \longrightarrow 0$ be a short exact sequence, with N'' flat. Then N is flat $\iff N'$ is flat.

Proof. First, recall that since we are working in a commutative ring, say A , we have $\text{Tor}_n^A(Q, P) \cong \text{Tor}_n^A(P, Q)$. Hence, by 2.24., Q is flat if and only if $\text{Tor}_n(P, Q) = 0$ for all $n > 0$ and all A -module P . Now, fix an arbitrary A -module M . Consider the long exact sequence induced by the right exact functor $M \otimes_A -$ acting on the short exact sequence $0 \longrightarrow N' \longrightarrow N \longrightarrow N'' \longrightarrow 0$:

$$\cdots \rightarrow \text{Tor}_2(M, N'') \rightarrow \text{Tor}_1(M, N') \rightarrow \text{Tor}_1(M, N) \rightarrow \text{Tor}_1(M, N'') \rightarrow M \otimes N' \rightarrow \cdots$$

Note that N'' is assumed to be flat, so $\text{Tor}_1(M, N'') = \text{Tor}_2(M, N'') = 0$. So we have the following exact sequence:

$$\cdots \rightarrow 0 \rightarrow \text{Tor}_1(M, N') \rightarrow \text{Tor}_1(M, N) \rightarrow 0 \cdots$$

If N' is flat, then $\text{Tor}_1(M, N') = 0$. The exactness at $\text{Tor}_1(M, N)$ forces it to be zero, hence N is flat; Similarly, if N is flat, then $\text{Tor}_1(M, N) = 0$. The exactness at $\text{Tor}_1(M, N')$ forces it to be zero, hence N' is flat.

2.26. Let N be an A -module. Then N is flat $\iff \text{Tor}_1^A(A/\mathfrak{a}, N) = 0$, for all finitely generated ideal $\mathfrak{a} \triangleleft A$.

Proof. Step 1: I claim that N is flat if $\text{Tor}_1(M, N) = 0$ for all f.g. A -modules M . Take any f.g. A -modules M, M' with an injective homomorphism $f : M \rightarrow M'$. Then the following sequence is exact:

$$0 \longrightarrow M \xrightarrow{f} M' \twoheadrightarrow \text{coker } f \longrightarrow 0$$

Apply the functor $N \otimes_A -$ on it, we obtain the following exact sequence:

$$\mathrm{Tor}_1(N, M) \longrightarrow N \otimes_A M \xrightarrow{\mathrm{id} \otimes_A f} N \otimes_A M' \longrightarrow N \otimes_A \mathrm{coker} f \longrightarrow 0$$

Since $\mathrm{Tor}_1(N, M) \cong \mathrm{Tor}_1(M, N) = 0$, $\mathrm{id} \otimes_A f$ is injective. Hence, by Prop 2.19., N is flat.

Step 2: I claim that N is flat if $\mathrm{Tor}_1(M, N) = 0$ for all cyclic A -modules M . That is, in order to get $\mathrm{Tor}_1(M, N) = 0$, instead of checking all f.g. A -module M , it suffices to check all cyclic A -modules. Fix an arbitrary f.g. A -module $M = \langle x_1, \dots, x_n \rangle$. If $n = 1$, we are done. Suppose for any submodule $M' \subset M$ with $n - 1$ generators, $\mathrm{Tor}_1(M', N) = 0$. Pick any $x \in \{x_1, \dots, x_n\}$. Consider the exact sequence induced by the right exact functor $N \otimes_A -$ acting on the short exact sequence $0 \longrightarrow \langle x \rangle \longrightarrow M \longrightarrow M/\langle x \rangle \longrightarrow 0$:

$$\dots \longrightarrow \mathrm{Tor}_1(N, \langle x \rangle) \longrightarrow \mathrm{Tor}_1(N, M) \longrightarrow \mathrm{Tor}_1(N, M/\langle x \rangle) \longrightarrow \dots$$

By our assumption, both $\mathrm{Tor}_1(N, \langle x \rangle)$ and $\mathrm{Tor}_1(N, M/\langle x \rangle)$ are 0, hence $\mathrm{Tor}_1(N, M) = 0$. Note that $\langle x \rangle \cong A/\mathrm{Ann}(x)$, hence N is flat $\iff \mathrm{Tor}(A/\mathfrak{a}, N) = 0$ for all ideals in A .

Step 3: Given any ideal $I \triangleleft A$. Let $\mathcal{F}(I) := \{\mathfrak{a} : \mathfrak{a} \subseteq I, \mathfrak{a} \text{ f.g.}\}$, using the fact

$$A/I \cong \varinjlim_{\mathfrak{a} \in \mathcal{F}(I)} A/\mathfrak{a} \quad \text{and} \quad \mathrm{Tor}_1^A(\varinjlim X_i, N) \cong \varinjlim \mathrm{Tor}_1^A(X_i, N)$$

Now Fix $I \triangleleft A$, suppose we have: $\mathrm{Tor}(A/\mathfrak{a}, N) = 0, \forall \mathfrak{a} \in \mathcal{F}(I)$. Then

$$\mathrm{Tor}_1^A(A/I, N) \cong \mathrm{Tor}_1^A\left(\varinjlim_{\mathfrak{a} \in \mathcal{F}(I)} A/\mathfrak{a}, N\right) \cong \varinjlim_{\mathfrak{a} \in \mathcal{F}(I)} \mathrm{Tor}_1^A(A/\mathfrak{a}, N) = 0$$

That is, it suffices to check all f.g. ideals of A , so we are done.

2.27 Show that the following are equivalent.

- (i) A is absolutely flat.
- (ii) Every principal ideal is idempotent.
- (iii) Every f.g. ideal is a direct summand of A .

Proof. (i) $\implies$ (ii): Fix arbitrary $x \in A$. Consider the functor $A/\langle x \rangle \otimes_A -$ acting on the embedding $\langle x \rangle \rightarrow A$. Since A is absolutely flat, $A/\langle x \rangle \otimes_A \langle x \rangle \rightarrow A/\langle x \rangle \otimes_A A \cong A/\langle x \rangle$ is injective. But $\bar{a} \otimes_A x \mapsto \bar{a} \cdot x = \overline{ax} = \bar{0}$, hence $A/\langle x \rangle \otimes_A \langle x \rangle = 0$. Now consider the exact sequence $E_\bullet$:

$$0 \longrightarrow \langle x \rangle \longrightarrow A \longrightarrow A/\langle x \rangle \longrightarrow 0$$

Apply $\langle x \rangle \otimes_A -$ and obtain the following exact sequence:

$$0 \longrightarrow \langle x \rangle \otimes_A \langle x \rangle \longrightarrow \langle x \rangle \longrightarrow 0 \longrightarrow 0$$

Hence $\langle x \rangle \otimes_A \langle x \rangle \cong \langle x \rangle$. Note that $\langle x \rangle \otimes_A \langle x \rangle \cong \langle x^2 \rangle$ by $x \otimes_A x \mapsto x^2$, hence $\langle x^2 \rangle \cong \langle x \rangle$.

Alternatively, directly Apply $A/\langle x \rangle \otimes_A -$ to $E_\bullet$ and obtain the following exact sequence:

$$0 \longrightarrow A/\langle x \rangle \otimes_A \langle x \rangle \longrightarrow A/\langle x \rangle \longrightarrow A/\langle x \rangle \otimes_A A/\langle x \rangle \longrightarrow 0$$

Recall that by 2.2., $M \otimes_A A/\mathfrak{a} \cong M/\mathfrak{a}M$. Hence $A/\langle x \rangle \otimes_A \langle x \rangle \cong \langle x \rangle/\langle x^2 \rangle$; $A/\langle x \rangle \otimes_A A/\langle x \rangle \cong A/\langle x \rangle$. Hence the following sequence

$$0 \longrightarrow \langle x \rangle/\langle x^2 \rangle \longrightarrow A/\langle x \rangle \longrightarrow A/\langle x \rangle \longrightarrow 0$$

is exact, forcing $\langle x \rangle/\langle x^2 \rangle = 0$.

(ii) $\implies$ (iii): The hint gives a complete proof. (iii) $\implies$ (i): let $\mathfrak{a} \triangleleft A$ f.g. Fix an arbitrary A -module N , by 2.26, it suffices to show that $\text{Tor}_1(A/\mathfrak{a}, N) = 0$. Consider the following exact sequence:

$$0 \longrightarrow \mathfrak{a} \xrightarrow{\iota} A \cong \mathfrak{a} \oplus A/\mathfrak{a} \longrightarrow A/\mathfrak{a} \longrightarrow 0$$

It induces an exact sequence by applying the functor $N \otimes_A -$:

$$\cdots \longrightarrow \text{Tor}_1(N, A) \longrightarrow \text{Tor}_1(N, A/\mathfrak{a}) \longrightarrow N \otimes_A \mathfrak{a} \longrightarrow N \otimes_A (\mathfrak{a} \oplus A/\mathfrak{a}) \longrightarrow N \otimes_A A/\mathfrak{a} \longrightarrow 0$$

Note that $\text{Tor}_1(N, A) = 0$ since A as an A -module is free, hence flat. Also note that $N \otimes_A (\mathfrak{a} \oplus A/\mathfrak{a}) \cong (N \otimes_A \mathfrak{a}) \oplus (N \otimes_A A/\mathfrak{a})$, Thus, we obtain the following exact sequence:

$$0 \longrightarrow \text{Tor}_1(N, A/\mathfrak{a}) \longrightarrow N \otimes_A \mathfrak{a} \longrightarrow (N \otimes_A \mathfrak{a}) \oplus (N \otimes_A A/\mathfrak{a}) \longrightarrow \cdots$$

So it suffices to show that $\text{id}_N \otimes_A \iota$ is injective. Note that $\exists r : A \rightarrow \mathfrak{a}$ s.t. $r \circ \iota = \text{id}_{\mathfrak{a}}$. So $(\text{id}_N \otimes_A r) \circ (\text{id}_N \otimes_A \iota) = \text{id}_N \otimes_A (r \circ \iota) = \text{id}_N \otimes_A \text{id}_{\mathfrak{a}} = \text{id}_{N \otimes_A \mathfrak{a}}$, so we are done.

2.28. *Every homomorphic image of an absolutely flat ring is absolutely flat.*

Proof. It suffices to show that $\forall f(x) \in f(A)$, $\exists f(y) \in f(A)$ s.t. $f(x) = f(y)f(x)^2$. This is easy to see by applying $f(-)$ on $x = ax^2$, for some $a \in A$, both sides.

If a local ring is absolutely flat, then it is a field.

Proof. Let $x \in \mathfrak{m}$. Using the notation in the proof of 2.27., $f = 1 - e = 1 - ax$. But by Prop 1.9., this implies f is a unit, hence $A = \langle e \rangle \oplus \langle f \rangle = \langle x \rangle \oplus A$. So $x = 0$, and so is $\mathfrak{m}$.

If A is absolutely flat, every non-unit in A is a zero-divisor.

Proof. Let $x \in A$, non-unit. Then $\exists$ a non-zero submodule $N \subseteq A$ s.t. $A \cong \langle x \rangle \oplus N$. This implies $0 = ax$ for some $a \in A$, as desired.

GO TO Exercise 3.10 - ... to see more on this topic

3 Chapter 3 Rings and Modules of Fractions

3.1 Notes and Comments

Proposition 3.1. *Let $g : A \rightarrow B$ be a ring homomorphism s.t. $g(s)$ is a unit in B , $\forall s \in S$. Then $\exists!$ ring homomorphism $h : S^{-1}A \rightarrow B$ s.t. $g = h \circ f$. i.e., the following diagram commutes.*

$$\begin{array}{ccc} A & \xrightarrow{g} & B \\ f : x \mapsto x/1 \downarrow & \nearrow \exists! h & \\ S^{-1}A & & \end{array}$$

Note that h is uniquely determined by g as: $h(a/s) = g(a) \cdot g(s)^{-1}$.

Moreover, if $\exists$ another ring R with $f' : A \rightarrow R$ that also satisfies the above property, then $S^{-1}A \cong R$. In other words, the above property is a UMP that characterizes $S^{-1}A$.

Remark 20. We may call f here as **canonical localization map**. Note that it is not necessarily injective! (e.g., if some element of S is a zero-divisor/zero).

Localization at a prime ideal $\mathfrak{p}$.

Fix a prime ideal $\mathfrak{p} \triangleleft A$. Let $S = A - \mathfrak{p}$, $f : A \rightarrow S^{-1}A =: A_{\mathfrak{p}}$. Then there is a one-to-one correspondence:

$$\{\text{prime ideals of } A \text{ that are contained in } \mathfrak{p}\} \longleftrightarrow \{\text{prime ideals of } A_{\mathfrak{p}}\}$$

given by $\mathfrak{q} \mapsto \mathfrak{q}^e = S^{-1}\mathfrak{q}$ and $S^{-1}\mathfrak{q} \mapsto (S^{-1}\mathfrak{q})^c = \mathfrak{q}$.

Proof. It's easy to check that $\mathfrak{q}^e = S^{-1}\mathfrak{q}$ and $(S^{-1}\mathfrak{q})^c = \mathfrak{q}$. Then, as the book ([AM69], page 38) has shown, for any ideal $\mathfrak{a} \triangleleft A$, $\mathfrak{a}^e$ would become the whole ring in $A_{\mathfrak{p}}$ if and only if it touches S , hence, since $\mathfrak{q} \subseteq \mathfrak{p}$, $\mathfrak{q}^e$ is a proper ideal, and obviously prime in $A_{\mathfrak{p}}$.

Compare this to the quotient: Let ϕ be any ring homomorphism where the domain is A . There is a one-to-one correspondence between prime ideals in A that contain $\ker \phi$ and prime ideals in $\phi(A)$. In particular, if we take ϕ as the projection $A \rightarrow A/\mathfrak{p}$, then we have the one-to-one correspondence:

$$\{\text{prime ideals of } A \text{ that contain } \mathfrak{p}\} \longleftrightarrow \{\text{prime ideals of } A/\mathfrak{p}\}$$

also given by $\mathfrak{q} \mapsto \mathfrak{q}^e = \bar{\mathfrak{q}}$ and $\bar{\mathfrak{q}} \mapsto \bar{\mathfrak{q}}^c = \mathfrak{q}$.

Therefore, I claim the following slogan:

Localizing at $\mathfrak{p}$ kills all the ideals that are not contained in $\mathfrak{p}$, whereas
quotienting out by $\mathfrak{p}$ kills all the ideals that are contained in $\mathfrak{p}$.

Consequently, the only maximal ideal left in $A_{\mathfrak{p}}$ after localizing at $\mathfrak{p}$ is $\mathfrak{p}^e$, which makes $A_{\mathfrak{p}}$ a local ring.

Proposition 3.7. Note that we have:

$$\begin{aligned}
S^{-1}M \otimes_{S^{-1}A} S^{-1}N &\cong S^{-1}M \otimes_{S^{-1}A} (N \otimes_A S^{-1}A) && \text{(Prop 3.5.)} \\
&\cong (S^{-1}M \otimes_{S^{-1}A} S^{-1}A) \otimes_A N \\
&\cong S^{-1}M \otimes_A N \\
&\cong S^{-1}A \otimes_A M \otimes_A N && \text{(Prop 3.5.)} \\
&\cong S^{-1}(M \otimes_A N) && \text{(Prop 3.5.)}
\end{aligned}$$

In particular,

$$A_{\mathfrak{p}} \otimes_A M \cong M_{\mathfrak{p}} \text{ (Prop 3.5.)} \quad \text{and} \quad M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} N_{\mathfrak{p}} \cong (M \otimes_A N)_{\mathfrak{p}}$$

as $A_{\mathfrak{p}}$ -modules.

Proposition 3.10. (i) $\Rightarrow$ (ii):

$$M \text{ is a flat } A\text{-module} \Rightarrow M_{\mathfrak{p}} \text{ is a flat } A_{\mathfrak{p}}\text{-module} \quad \text{(Prop 3.5. and Ex 2.20.)}$$

(iii) $\Rightarrow$ (i): This proof needs some creativity: Let $N \rightarrow P$ be a A -module homomorphism, $\mathfrak{m}$ be a maximal ideal of A . Then

$$\begin{aligned}
N \rightarrow P \text{ injective} &\Rightarrow N_{\mathfrak{m}} \rightarrow P_{\mathfrak{m}} \text{ injective} && \text{(Prop 3.9.)} \\
&\Rightarrow N_{\mathfrak{m}} \otimes_{A_{\mathfrak{m}}} M_{\mathfrak{m}} \rightarrow P_{\mathfrak{m}} \otimes_{A_{\mathfrak{m}}} M_{\mathfrak{m}} \text{ injective, since } M_{\mathfrak{m}} \text{ flat} && \text{(Prop 2.19.)} \\
&\Rightarrow (N \otimes_A M)_{\mathfrak{m}} \rightarrow (P \otimes_A M)_{\mathfrak{m}} \text{ injective} && \text{(Prop 3.7.)} \\
&\Rightarrow N \otimes_A M \rightarrow P \otimes_A M \text{ injective} && \text{(Prop 3.9.)}
\end{aligned}$$

Hence M is flat by Prop 2.19..

3.2 Exercises

3.2.1 Basics

3.1. Let M be a finitely generated A -module, S a multiplicatively closed set of A . Prove that $S^{-1}M = 0$ if and only if $\exists s \in S$ s.t. $sM = 0$.

Proof. $\Rightarrow$: Let $M = \langle x_1, \dots, x_n \rangle$. Then $x_i/1 = 0 \in S^{-1}M, \forall i \in [n]$. This implies, by definition, $\exists s_i \in S$ s.t. $s_i x_i = 0$. Let $s = \prod_{i=1}^n s_i$, then $sM = 0$. $\Leftarrow$: Take $\forall m/t \in S^{-1}M$, $m/t = (sm)/(st) = 0/st = 0$, hence, $S^{-1}M = 0$.

3.2. Let $\mathfrak{a} \triangleleft A$, $S = 1 + \mathfrak{a}$. Show that $S^{-1}\mathfrak{a}$ is contained in the Jacobson radical of $S^{-1}A$.

Proof. Directly apply Prop 1.9..

Use this result and Nakayama's lemma to give a proof of Corollary 2.5.

Proof. Suppose $M = \mathfrak{a}M$, we need to show that $\exists s \in 1 + \mathfrak{a}$ s.t. $sM = 0$. By 3.1., it suffices to show that $S^{-1}M = 0$. Note that $S^{-1}M$ is finitely generated, $S^{-1}M = (S^{-1}\mathfrak{a})(S^{-1}M)$ (by 3.11.(e)), and $S^{-1}\mathfrak{a}$ is contained in the Jacobson radical of $S^{-1}A$, hence by Nakayama's lemma, $S^{-1}M = 0$.

3.3. Let A be a ring, let S, T be two multiplicatively closed subsets of A , and let U be the image of T in $S^{-1}A$. Show that $(ST)^{-1}A \cong U^{-1}(S^{-1}A)$.

Proof. Let $h : A \rightarrow (ST)^{-1}A$ be the canonical localization map. Consider the composite:

$$A \xrightarrow{f} S^{-1}A \xrightarrow{g} U^{-1}(S^{-1}A)$$

where f, g are also canonical localization maps. Note that $\forall a \in ST$, $g \circ f(a)$ is a unit in $U^{-1}(S^{-1}A)$. Thus, $\exists ! \phi : (ST)^{-1}A \rightarrow U^{-1}(S^{-1}A)$ s.t. $\phi \circ h = g \circ f$. Conversely, since $\forall s \in S$, $h(s)$ is a unit in $(ST)^{-1}A$, we have: $\exists ! \bar{h}$ s.t. h factors through $\bar{h} \circ f$. This implies $\forall f(t) \in U$, $\bar{h}(f(t)) = h(t)$ is a unit in $(ST)^{-1}A$. Thus, $\bar{h}$ further induces a unique homomorphism, say $\psi : U^{-1}(S^{-1}A) \rightarrow (ST)^{-1}A$ s.t. the following diagram (left) commutes:

$$\begin{array}{ccc} A & \xrightarrow{h} & (ST)^{-1}A \\ f \downarrow & \nearrow \bar{h} & \\ S^{-1}A & & \\ g \downarrow & \nearrow \psi & \\ U^{-1}(S^{-1}A) & & \end{array}$$

$$\begin{array}{ccc} A & \xrightarrow{h} & (ST)^{-1}A \\ h \downarrow & \nearrow \psi \circ \phi & \\ (ST)^{-1}A & & \end{array}$$

$$\begin{array}{ccc} S^{-1}A & \xrightarrow{g} & U^{-1}(S^{-1}A) \\ g \downarrow & \nearrow \phi \circ \psi & \\ U^{-1}(S^{-1}A) & & \end{array}$$

Therefore, we have $h = \psi \circ g \circ f = \psi \circ \phi \circ h$. But id also satisfies $h = \text{id} \circ h$. Consequently, by the uniqueness of the map induced by h , $\psi \circ \phi = \text{id}$. Similarly, we have $\phi \circ \psi \circ g = \phi \circ \bar{h} = g$, then the uniqueness of the map induced by g gives $\phi \circ \psi = \text{id}$. Thus, $U^{-1}(S^{-1}A) \cong (ST)^{-1}A$, as desired.

Remark 21. Note that the canonical localization maps are always epimorphisms in the category **ComRing**, even though they are not surjective in general. That is, let $h : A \rightarrow S^{-1}A$ be the

canonical localization map. Then, for any ring R with $\alpha, \beta : S^{-1}A \rightarrow R$, we always have:

$$\alpha \circ h = \beta \circ h \implies \alpha = \beta$$

Proof. Let γ be any ring homomorphism $S^{-1}A \rightarrow R$. Since γ maps units to units, we have $\gamma(h(s))^{-1} = \gamma(h(s)^{-1}) = \gamma(1/s)$, $\forall s \in S$. Consider

$$A \xrightarrow{h} S^{-1}A \xrightarrow{\gamma} R$$

We have:

$$\gamma(a/s) = \gamma(a/1)\gamma(1/s) = \gamma(h(a))\gamma(h(s))^{-1}$$

for any $a/s \in S^{-1}A$. Now, suppose $\alpha \circ h = \beta \circ h$, then

$$\alpha(a/s) = \alpha(h(a))\alpha(h(s))^{-1} = \beta(h(a))\beta(h(s))^{-1} = \beta(a/s)$$

That is, $\alpha = \beta$, as desired.

3.4. Let $f : A \rightarrow B$ be a homomorphism of rings and let S be a multiplicatively closed subset of A . Let $T = f(S)$, show that $S^{-1}B$ and $T^{-1}B$ are isomorphic as $S^{-1}A$ -modules.

Proof. First, we need to clarify how $S^{-1}B$ and $T^{-1}B$ form as $S^{-1}A$ -modules. Pick any $a/x \in S^{-1}A$ and $b/s \in S^{-1}B$, $b'/f(s') \in T^{-1}B$, we may define the ring actions as follows:

$$\frac{a}{x} \diamond \frac{b}{s} := \frac{f(a)b}{xs} \quad \text{and} \quad \frac{a}{x} \diamond \frac{b'}{f(s')} := \frac{f(a)b'}{f(xs')}$$

It's easy to check that the properties for modules hold, resp. Now define

$$\phi : S^{-1}B \longrightarrow T^{-1}B, \quad \frac{b}{s} \mapsto \frac{b}{f(s)}.$$

Well-definedness: If $b/s = b'/s'$ in $S^{-1}B$, then $\exists u \in S$ s.t.

$$f(u)(f(s')b - f(s)b') = 0 \text{ in } B.$$

Since $f(u) \in T$, this is exactly the equality $b/f(s) = b'/f(s')$ in $T^{-1}B$, so ϕ is well-defined. $S^{-1}A$ -linearity: For any $a/x \in S^{-1}A$,

$$\phi\left(\frac{a}{x} \diamond \frac{b}{s}\right) = \phi\left(\frac{f(a)b}{xs}\right) = \frac{f(a)b}{f(xs)} = \frac{a}{x} \diamond \phi\left(\frac{b}{s}\right).$$

Additivity is immediate.

Conversely, define

$$\psi : T^{-1}B \longrightarrow S^{-1}B, \quad \frac{b}{f(s)} \mapsto \frac{b}{s}.$$

Well-definedness, $S^{-1}A$ -linearity, and additivity follow similarly.

Finally,

$$\psi \circ \phi\left(\frac{b}{s}\right) = \psi\left(\frac{b}{f(s)}\right) = \frac{b}{s}, \quad \phi \circ \psi\left(\frac{b}{f(s)}\right) = \phi\left(\frac{b}{s}\right) = \frac{b}{f(s)}.$$

Hence ϕ and ψ are inverse $S^{-1}A$ -module isomorphisms, so $S^{-1}B \cong T^{-1}B$ as $S^{-1}A$ -modules.

3.5. Let A be a ring, suppose that, for each prime ideal $\mathfrak{p}$, $A_{\mathfrak{p}}$ has no nilpotent element $\neq 0$. Show that A has no nilpotent element $\neq 0$.

Proof. Note that A has no nilpotent element $\neq 0 \Leftrightarrow \mathfrak{N}(A) =: \mathfrak{N} = 0$. Similarly, for a fixed $\mathfrak{p} \triangleleft A$, $A_{\mathfrak{p}}$ has no nilpotent element $\neq 0 \Leftrightarrow \mathfrak{N}(A_{\mathfrak{p}}) = 0$. So it suffices to show $\mathfrak{N}(A_{\mathfrak{p}}) = 0, \forall \mathfrak{p} \triangleleft A \implies \mathfrak{N} = 0$. Note that $\mathfrak{N}(A_{\mathfrak{p}}) = \mathfrak{N}_{\mathfrak{p}}$ by Corollary 3.12. Then, by regarding $\mathfrak{N}$ as an A -module and applying Prop 3.8., we are done. And note that the converse is also true.

If each $A_{\mathfrak{p}}$ is an integral domain, is A necessarily an integral domain?

This is not true. Consider $A = k_1 \oplus k_2$ where k_1, k_2 are fields, then A is not an integral domain and A has exact 2 prime ideals, namely $\mathfrak{p} := 0 \times k_2$ and $\mathfrak{q} := k_1 \times 0$. It suffices to show, when localizing, say, at $\mathfrak{p}$, the local ring $A_{\mathfrak{p}}$ is an integral domain. Let $S := A - \mathfrak{p}$, take any $\alpha_1 := (x_1, y_1)/s_1, \alpha_2 := (x_2, y_2)/s_2 \in A_{\mathfrak{p}}$ where $s_1, s_2 \in S$. Suppose that $\alpha_1 \alpha_2 = 0$ in $A_{\mathfrak{p}}$, then $\exists t = (u, y) \in S$ where u is non-zero in k_1 and y^* is arbitrary in k_2 s.t.

$$(ux_1x_2, y^*y_1y_2) = 0 \in A$$

Thus, at least one of x_1, x_2 is zero, w.l.o.g., set $x_1 = 0$. Then if $y_1 = 0$, we are done; if not, then $y^*y_2 = 0$, so we can find an element $t := (v, y^*y_2) \in S$ s.t. $t \cdot (x_1, y_1) = 0 \in A$, which makes $\alpha_1 = 0$ in $A_{\mathfrak{p}}$. Hence we are done.

Remark 22. The following observation is hinted by [ECOH]. Notice that

$$\theta_{\mathfrak{p}} : A_{\mathfrak{p}} \longrightarrow k_1 \quad \frac{(x, y)}{(u, y')} \mapsto \frac{x}{u}$$

is a well-defined ring isomorphism. Therefore, $A_{\mathfrak{p}} \cong k_1$, which is an integral domain.

3.6. Let $A \neq 0$ be a ring and let Σ be the set of all multiplicatively closed subsets $S \subseteq A$ s.t. $0 \notin S$. Show that Σ has maximal elements, and that $S \in \Sigma$ is maximal if and only if $A - S$ is a minimal prime ideal in A .

Proof. The first part can be shown by standard Zorn's lemma arguments. For the second part, $\implies$: Clearly $a, b \in A - S$ implies $ab \in A - S$. The tricky part is to show $A - S$ forms an abelian group under addition, i.e., $a, b \in A - S$ implies $a + b \in A - S$. Here I claim that:

$$\text{If } S \in \Sigma \text{ is maximal, then } a \notin S \Leftrightarrow \exists s \in S, n \in \mathbb{N} \text{ s.t. } sa^n = 0.$$

Proof of the claim: $\Leftarrow$ is trivial. For $\Rightarrow$, suppose $\forall s \in S, n \in \mathbb{N}, sa^n \neq 0$, then $S \cup \{a\} \in \Sigma$, contradicts the maximality of S . Now, suppose $a, b \notin S$, then $\exists s, t \in S; m, n \in \mathbb{N}$ s.t. $sa^m = 0$ and $tb^n = 0$. Thus, $st(a+b)^{m+n} = 0$, and consequently, $a+b \in A - S$. Finally, it remains to show that for any $a \in A, r \in A - S \implies ar \in A - S$. Note that $r \in A - S \implies \exists u \in S, p \in \mathbb{N}$ s.t. $ur^p = 0$, hence $u(ar)^p = 0 \implies ar \in A - S$, as desired.

$\Leftarrow$: This direction is straightforward.

3.7. A multiplicatively closed subset of S of A is said to be **saturated** if

$$xy \in S \Leftrightarrow x \in S \text{ and } y \in S$$

Prove that: (a). S is saturated $\Leftrightarrow A - S$ is a union of prime ideals.

Proof. $\Rightarrow$: To show $A - S$ is a union of prime ideals, it suffices to show that:

$$\forall a \in A - S, \exists \mathfrak{p} \triangleleft A \text{ s.t. } \mathfrak{p} \ni a \text{ and } \mathfrak{p} \cap S = \emptyset$$

We play a similar game here as in the proof of Prop 1.8.: Take any $a \in A - S$, let Γ_a be the collection of all ideals that contain a but do not touch S . Since S is saturated, $ab \notin S$ for all $b \in A$. Consequently, $\langle a \rangle \in \Gamma_a$ and hence Γ_a is not empty. Then, for any ascending chain $\{\mathfrak{a}_i\}_{i \in I}$ in Γ_a ordered by inclusion, $\bigcup_{i \in I} \mathfrak{a}_i$ is clearly an upper-bound in Γ_a . Thus, by Zorn's lemma, Γ_a has maximal elements. Let $\mathfrak{p}_a$ be a maximal element of Γ_a , I claim that $\mathfrak{p}_a$ is prime by the following arguments: Let $x, y \notin \mathfrak{p}_a$, then $\mathfrak{p}_a + \langle x \rangle, \mathfrak{p}_a + \langle y \rangle$ are ideals strictly containing $\mathfrak{p}_a$, so they must touch S . Let $s, t \in (\mathfrak{p}_a + \langle x \rangle) \cap S, (\mathfrak{p}_a + \langle y \rangle) \cap S$, resp., then we have:

$$st \in (\mathfrak{p}_a + \langle x \rangle)(\mathfrak{p}_a + \langle y \rangle) = \mathfrak{p}_a^2 + (\langle x \rangle + \langle y \rangle)\mathfrak{p}_a + \langle xy \rangle \subseteq \mathfrak{p}_a + \langle xy \rangle$$

This implies that $\mathfrak{p}_a + \langle xy \rangle$ touches S since $st \in S$. It follows that $xy \notin \mathfrak{p}_a$, hence $\mathfrak{p}_a$ is prime, as we desire.

$\Leftarrow$: Suppose the contrary that S is not saturated, then $\exists a, b \in A$ s.t.

$$ab \in S \text{ but } a \notin S$$

Consequently, since $a \in A - S$, we know that $\exists$ prime ideal $\mathfrak{q}_a \ni a$ s.t. $\mathfrak{q}_a \cap S = \emptyset$. But this implies $ab \notin S$, a contradiction.

(b). If S is any m.c.s. of A , $\exists!$ smallest saturated m.c.s. $\overline{S} \supseteq S$ and actually,

$$\overline{S} = A - \bigcup_{i \in I} \mathfrak{p}_i$$

where $\{\mathfrak{p}_i\}_{i \in I}$ is the collection of all prime ideals that do not touch S . $\overline{S}$ is called the **saturation** of S . If $S = 1 + \mathfrak{a}$ where $\mathfrak{a}$ is an ideal of A , find $\overline{S}$.

Proof. Let $S' = A - \bigcup_{i \in I} \mathfrak{p}_i$ as described above, we want to show that S' is the smallest saturated m.c.s. containing S . Clearly $S' \supseteq S$, so by part (a), it suffices to show that S' is an m.c.s.. Take any $a, b \in S'$, then $ab \notin A - S'$ because otherwise, it would violate the primeness of some $\mathfrak{p} \subseteq A - S$. Now, if $S = 1 + \mathfrak{a}$, then

$$S \cap \mathfrak{p} \neq \emptyset \Leftrightarrow \exists a \in \mathfrak{a}, p \in \mathfrak{p} \text{ s.t. } 1 + a = p \Leftrightarrow \mathfrak{a} + \mathfrak{p} = \langle 1 \rangle, \text{ i.e. } \mathfrak{a} \text{ and } \mathfrak{p} \text{ are co-prime.}$$

This means $S \cap \mathfrak{p} = \emptyset \Leftrightarrow \mathfrak{a} + \mathfrak{p}$ is contained in a maximal ideal, which is also a prime ideal that does not touch S . Therefore, $A - \overline{S}$ is the union of all maximal ideals that contain $\mathfrak{a}$.

3.8. Let S, T be m.c.s. of A s.t. $S \subseteq T$. Let $\phi : S^{-1}A \rightarrow T^{-1}A$ be the ring homomorphism where $a/s \mapsto a/s$. Show that the following statements are equivalent:

- (a). ϕ is bijective.
- (b). For each $t \in T$, $t/1$ is a unit in $S^{-1}A$.
- (c). For each $t \in T$, $\exists x \in A$ s.t. $xt \in S$.
- (d). T is contained in the saturation of S .

(e). Every prime ideal that touches T also touches S .

Proof. (a) $\implies$ (b): ϕ is bijective $\implies$ the inverse map ϕ^{-1} exists $\implies \forall t \in T, \phi((t/1) \cdot \phi^{-1}(1/t)) = \phi(t/1) \cdot (1/t) = (t/1) \cdot (1/t) = 1 \implies (t/1) \cdot \phi^{-1}(1/t) = 1 \in S^{-1}A \implies t/1$ is a unit in $S^{-1}A$.

(b) $\implies$ (c): $\forall t \in T, \exists a/s \in S^{-1}A$ s.t. $(t/1) \cdot (a/s) = ta/s = 1 \in S^{-1}A$ since $t/1$ is a unit in $S^{-1}A$. This implies $\exists s' \in S$ s.t. $s'ta = s's \in S$, that is, $\exists x \in A$ s.t. $xt \in S$.

(c) $\implies$ (d): Let $\bar{S}$ be the saturation of S . Suppose $T \setminus \bar{S} \neq \emptyset$, then $\exists t \in T$ but $t \notin \bar{S}$. We know $\exists a \in A$ s.t. $at \in S \subseteq \bar{S}$, but this implies $t \in \bar{S}$, contradiction.

(d) $\implies$ (e): By our characterization of $\bar{S}$, every prime ideal that does not touch S must not touch $\bar{S}$, hence not touch T .

(e) $\implies$ (b): Suppose $\exists t \in T$ s.t. $t/1$ is a non-unit in $S^{-1}A$, then by Corollary 1.5., $t/1$ is contained in a maximal ideal of $S^{-1}A$, which is a prime ideal of the form $S^{-1}\mathfrak{p}$ where $\mathfrak{p}$ is a prime ideal in A that does not touch S (by Prop 3.11.), but $t \in \mathfrak{p}$. That is, $\mathfrak{p}$ is an ideal that touches T but not S .

(b) $\implies$ (a): Let σ, τ be the canonical localization map from A to $S^{-1}A, T^{-1}A$, respectively. Note that for any $s \in S$, $\sigma(s) = s/1$ is a unit in $T^{-1}A$ since $S \subseteq T$, and for any $t \in T$, $\tau(t) = t/1$ is a unit in $S^{-1}A$ since the condition of (b). Hence, the following diagrams commute: (Note that ϕ is exactly the homomorphism induced by τ by uniqueness.)

$$\begin{array}{ccc} A & \xrightarrow{\tau} & T^{-1}A \\ \sigma \downarrow & \nearrow \phi & \\ S^{-1}A & & \end{array} \qquad \begin{array}{ccc} A & \xrightarrow{\sigma} & S^{-1}A \\ \tau \downarrow & \nearrow \psi & \\ T^{-1}A & & \end{array}$$

Therefore, we have: $\sigma = \psi \circ \tau = \psi \circ (\phi \circ \sigma) = (\psi \circ \phi) \circ \sigma$. Since σ is an epi (see the remark after 3.3.), we have $\psi \circ \phi = \text{id}$. By symmetry, $\phi \circ \psi = \text{id}$, hence ϕ is an isomorphism.

3.9. The set S_0 of all non-zero-divisors in A is a saturated m.c.s. Hence the set $D = A - S_0$ is a union of prime ideals (see 1.14). Show that every minimal prime ideal of A is contained in D .

Proof. By 3.6. (where we inherit the notations from), it suffices to show that for any maximal element $S \in \Sigma$, $S_0 \subseteq S$. Suppose not, then SS_0 is a m.c.s. strictly containing S , hence not in Σ . Then $\exists ss_0 = 0 \in SS_0$, contradiction.

The ring $S_0^{-1}A$ is called the **total ring of fractions** of A . Show that

(a). S_0 is the largest m.c.s. of A for which the homomorphism $A \rightarrow S_0^{-1}A$ is injective.

Proof. Straightforward checking

(b). Every element in $S_0^{-1}A$ is either a zero-divisor or a unit.

Proof. Note that a/s_0 is a unit $\Leftrightarrow a \in S_0$; a/s_0 is a zero-divisor $\Leftrightarrow a \notin S_0$.

(c). Let A be a ring in which every non-unit is a zero-divisor. Then $f : A \rightarrow S_0^{-1}A$ is a bijection.

Proof. It suffices to show surjectivity. Note that $S_0 \subseteq U$ where U is the collection of units of A . So for each a/s_0 , we have $f(as_0^{-1}) = a/1 \cdot 1/s_0 = a/s_0$.

3.2.2 Absolute flatness, faithful flatness, supports, fibers, etc.

3.10. Let A be a ring.

(i) If A is absolutely flat, then $S^{-1}A$ is absolutely flat.

Proof. Use 2.27. (ii).

(ii) A is absolutely flat $\iff A_{\mathfrak{m}}$ is a field for all maximal ideal $\mathfrak{m}$.

Proof. $\implies$: See 2.28. $\impliedby$: See Prop 3.10.

3.11. Let A be a ring. Show the following are equivalent:

(i) $A/\mathfrak{R}$ is absolutely flat.

(ii) Every prime ideal of A is maximal.

(iii) $\text{Spec}(A)$ is T_1 .

(iv) $\text{Spec}(A)$ is Hausdorff.

Proof.

(i) $\iff$ (ii): Every prime ideal of A is maximal $\iff \forall \mathfrak{m} \triangleleft A$, there is no ideal strictly contained in $\mathfrak{m}^e$, except zero, in $A/\mathfrak{R} \iff \forall \mathfrak{m} \triangleleft A$, $(A/\mathfrak{R})_{\mathfrak{m}^e}$ has no proper ideal $\iff \forall \mathfrak{m} \triangleleft A$, $(A/\mathfrak{R})_{\mathfrak{m}^e}$ is a field $\stackrel{3.10}{\iff} A/\mathfrak{R}$ is absolutely flat.

(ii) $\iff$ (iii): See 1.18.

(iii) $\impliedby$ (iv): Automatically guaranteed.

(iv) $\impliedby$ (i) and (ii): Take any two distinct points $x, y \in \text{Spec}(A/\mathfrak{R})$, i.e., two distinct prime ideals $\mathfrak{p}_x, \mathfrak{p}_y \triangleleft A/\mathfrak{R}$. Let $R = A/\mathfrak{R}$. (ii) implies $\mathfrak{p}_x, \mathfrak{p}_y$ are maximal in R , hence $\mathfrak{p}_x + \mathfrak{p}_y = \langle 1_R \rangle$. i.e., there exists $a \in \mathfrak{p}_x, b \in \mathfrak{p}_y$ s.t. $\langle a \rangle + \langle b \rangle = \langle 1_R \rangle$. Note that by 2.27.(ii), $\langle a \rangle$ is idempotent, hence $\exists e \in \langle a \rangle$ s.t. $e^2 = e$ and $\langle a \rangle = \langle e \rangle$. Let $f := 1_R - e$, then $\langle f \rangle \subseteq \langle b \rangle$. Therefore, we have: $\langle e \rangle \subseteq \mathfrak{p}_x, \langle f \rangle \subseteq \mathfrak{p}_y$, hence $X_e \ni x, X_f \ni y$. Note that $X_e \cap X_f = X_{ef} = X_0 = \emptyset$, so $\text{Spec}(A/\mathfrak{R})$ is Hausdorff. Since $\text{Spec}(A)$ is homeomorphic to $\text{Spec}(A/\mathfrak{R})$, we are done.

If these conditions are satisfied, show that $\text{Spec}(A)$ is compact and totally disconnected (i.e., the only connected subset of $\text{Spec}(A)$ are those consisting of a single point).

Proof. For compactness, see 1.17. Totally disconnectedness follows from $X_e \sqcup X_f = \text{Spec}(R)$.

3.12. Let A be an integral domain. Let $T(-)$ be the torsion submodule of a A -module.

(i) $M/T(M)$ is torsion-free.

Proof. Fix $m \notin T(M)$. We have: $a \cdot (m + T(M)) = 0 \iff a \cdot m \in T(M) \iff \exists a' \neq 0_A$ s.t. $a'a \cdot m = 0$. Since $m \notin T(M)$, this implies $a'a = 0_A$, but A is an integral domain, hence $a = 0$.

(ii) If $f : M \rightarrow N$ is an A -module homomorphism, then $f(T(M)) \subseteq T(N)$.

Proof. Let $m \in T(M)$. Then $\exists a \neq 0_A$ s.t. $am = 0_M$. So $f(am) = af(m) = 0_N$, i.e. $f(m) \in T(N)$.

(iii) If $0 \rightarrow M' \rightarrow M \rightarrow M''$ is exact, then $0 \rightarrow T(M') \rightarrow T(M) \rightarrow T(M'')$ is exact.

Proof. Straightforward check.

(iv) If M is any A -module, then $T(M)$ is the kernel of the mapping $x \mapsto 1 \otimes x$ of M into $K \otimes_A M$, where K is the field of fraction of A .

Proof. Use Prop 3.5.

3.13. Let A be an integral domain, let S be a m.c.s of A . Show that $T(S^{-1}M) = S^{-1}T(M)$. Deduce that the following are equivalent.

(i) M is torsion free.

(ii) $M_{\mathfrak{p}}$ is torsion free for all prime ideal $\mathfrak{p}$.

(iii) $M_{\mathfrak{m}}$ is torsion free for all maximal ideal $\mathfrak{m}$.

Proof. Take any non-zero element $m/s \in T(S^{-1}M)$. Then $\exists$ non-zero $a \in A$ s.t. $a \cdot m/s = (am)/s = 0$. This means $\exists t \in S$ s.t. $tam = 0$. Since $t, a \neq 0$ and A is an integral domain, $ta \neq 0$, hence $m \in T(M)$ and $m/s \in S^{-1}T(M)$. The converse direction is immediate.

(i) $\implies$ (ii) $\implies$ (iii) is trivial. Note that $T(M_{\mathfrak{m}}) = T(M)_{\mathfrak{m}}$, and $T(M)_{\mathfrak{m}} = 0$ for all maximal ideals implies $T(M) = 0$ by Prop 3.8. Hence (iii) $\implies$ (i) follows.

3.14. Let M be a A -module and $\mathfrak{a} \triangleleft A$. Suppose $M_{\mathfrak{m}} = 0$ for all maximal ideal $\mathfrak{m} \supseteq \mathfrak{a}$. Show that $M = \mathfrak{a}M$.

Proof. Note that $M/\mathfrak{a}M$ can be viewed as a $A/\mathfrak{a}$ -module. Then, by Corollary 3.4.(iii), $0 = M_{\mathfrak{m}}/\mathfrak{a}M_{\mathfrak{m}} \cong (M/\mathfrak{a}M)_{\overline{\mathfrak{m}}}$, for all maximal ideal $\mathfrak{m} \supseteq \mathfrak{a}$. This implies $(M/\mathfrak{a}M)_{\overline{\mathfrak{m}}} = 0$ for all maximal ideal $\overline{\mathfrak{m}} \triangleleft A/\mathfrak{a}$. Thus, $M/\mathfrak{a}M = 0$ by Prop 3.8.

3.15. Let F be the A -module A^n . Show that every set of n generators of F is a basis of F .

Proof. Let $F = \langle x_1, \dots, x_n \rangle$. Let $\{e_1, \dots, e_n\}$ be the standard basis. Define $\phi : F \rightarrow F$ by $\phi(e_i) = x_i$. Then ϕ is surjective. Hence, it suffices to show that ϕ is injective. Note that by Prop 3.9. we may assume that A is a local ring. Let $N := \ker \phi$. The following sequence

$$0 \longrightarrow N \longrightarrow F \xrightarrow{\phi} F \longrightarrow 0$$

is exact. Let $\mathfrak{m}$ be the maximal ideal of A , $k := A/\mathfrak{m}$ be the residue field. Apply $A/\mathfrak{m} \otimes_A -$ and the following sequence

$$\cdots \longrightarrow \mathrm{Tor}_1(A/\mathfrak{m}, F) \longrightarrow A/\mathfrak{m} \otimes_A N \longrightarrow A/\mathfrak{m} \otimes_A F \xrightarrow{\mathrm{id} \otimes_A \phi} A/\mathfrak{m} \otimes_A F \longrightarrow 0$$

is exact. Note that $A/\mathfrak{m} \otimes_A F \cong k^n$, and $\mathrm{Tor}_1(A/\mathfrak{m}, F) = 0$ since F is flat. Hence we have the following exact sequence:

$$0 \longrightarrow A/\mathfrak{m} \otimes_A N \longrightarrow k^n \xrightarrow{\mathrm{id} \otimes_A \phi} k^n \longrightarrow 0$$

Note that $\mathrm{id} \otimes_A \phi$ is a surjection between vector spaces; hence, it is injective, forcing $A/\mathfrak{m} \otimes_A N = 0$. Now, note that

- (i). N is f.g. as a A -module by 2.12.
- (ii). $0 = A/\mathfrak{m} \otimes_A N \cong N/\mathfrak{m}N$ by 2.2. Hence $N = \mathfrak{m}N$.
- (iii). $\mathfrak{m} = \mathfrak{J}(A)$.

Therefore, by Nakayama's lemma, $N = 0$, as desired.

Deduce that every set of generators of F has at least n elements.

Proof. Assume for the contradiction that $F := \langle y_1, \dots, y_{n-1} \rangle$. Then we can take any $z \in F$ not in this generating set and have $F := \langle y_1, \dots, y_{n-1}, z \rangle$, hence $\{y_1, \dots, y_{n-1}, z\}$ is a basis of F . But z can be written as a A -linear combination of $\{y_1, \dots, y_{n-1}\}$, which is a contradiction.

3.16. *Let B be a flat A -algebra. Then following are equivalent.*

- (i) $\mathfrak{a}^{ec} = \mathfrak{a}$ for all ideals in A .
- (ii) $\mathrm{Spec}(B) \longrightarrow \mathrm{Spec}(A)$ is surjective.
- (iii) For every maximal ideal $\mathfrak{m} \triangleleft A$ we have $\mathfrak{m}^e \neq \langle 1 \rangle$.
- (iv) If M is any non-zero module, then $M_B := B \otimes_A M \neq 0$.
- (v) For every A -module M , the mapping $x \mapsto 1 \otimes_A x$ of M into M_B is injective.

Proof. (i) $\implies$ (ii) $\implies$ (iii): Trivial. (iii) $\implies$ (iv): Take any non-zero element $x \in M$. Consider the embedding: $0 \longrightarrow \langle x \rangle \longrightarrow M$. Since B is flat, $B \otimes_A \langle x \rangle \longrightarrow M_B$ is injective. Note that $\langle x \rangle \cong A/\mathrm{Ann}(x)$, hence by 2.2., $B \otimes_A \langle x \rangle \cong B/\mathrm{Ann}(x)B$. Since $\mathrm{Ann}(x) \subset \mathfrak{m}$ for some maximal ideal $\mathfrak{m} \triangleleft A$, $\mathrm{Ann}(x)B = (\mathrm{Ann}(x))^e \subset \mathfrak{m}^e \subsetneq B$. Thus, $M_B \cong B/\mathrm{Ann}(x)B \neq 0$. (iv) $\implies$ (v): Let ϕ be the mapping. Then we have the following exact sequence: $0 \longrightarrow \ker \phi \longrightarrow M \longrightarrow M_B$. Since B is flat, tensor with B : $0 \longrightarrow (\ker \phi)_B \longrightarrow M_B \longrightarrow (M_B)_B$ is exact. Now by 2.13., $(\ker \phi)_B = 0$, forcing $\ker \phi = 0$, as we desired. (v) $\implies$ (i): Let $M = A/\mathfrak{a}$. Then $M_B \cong B/\mathfrak{a}B = B/\mathfrak{a}^e$, and the map $M \longrightarrow M_B$ is just induced by $f : A \longrightarrow B$, with kernel $= \mathfrak{a}^{ec}/\mathfrak{a}$. Hence $\mathfrak{a} = \mathfrak{a}^{ec}$ for all ideal $\mathfrak{a} \triangleleft A$. B is said to be **faithfully flat** over A .

3.17. *Let $A \xrightarrow{f} B \xrightarrow{g} C$ be ring homomorphism. IF $g \circ f$ is flat and g is faithfully flat,*

then f is flat.

Proof. Given any injective A -module homomorphism $\phi : M \longrightarrow N$, we need to show that the induced homomorphism

$$\psi := \text{id}_B \otimes \phi : M_B := B \otimes_A M \longrightarrow N_B := B \otimes_A N$$

is also injective. Consider the following commutative diagram:

$$\begin{array}{ccc} M_B & \xrightarrow{x \mapsto 1_C \otimes x} & C \otimes_B M_B \\ \psi \downarrow & & \downarrow \text{id}_C \otimes_B \psi \\ N_B & \xrightarrow{y \mapsto 1_C \otimes y} & C \otimes_B N_B \end{array}$$

Since C is faithfully flat over B via g , by 3.16.(v), the two horizontal morphisms are injective. Moreover, using the canonical isomorphism

$$C \otimes_B (B \otimes_A -) \cong C \otimes_A -,$$

the morphism $\text{id}_C \otimes_B \psi$ identifies with $\text{id}_C \otimes_A \phi$, which is injective since C is flat over A via $g \circ f$. Therefore, ψ is injective, and hence f is flat.

3.18. Let $f : A \longrightarrow B$ be a flat ring homomorphism. Let $\mathfrak{q} \triangleleft B$ prime and $\mathfrak{p} := \mathfrak{q}^c$. Then $f^* : \text{Spec}(B_{\mathfrak{q}}) \longrightarrow \text{Spec}(A_{\mathfrak{p}})$ is surjective.

Proof. Let $S := A \setminus \mathfrak{p}$ and $T := B \setminus \mathfrak{q}$. Since $s \in S$ implies $f(s) \notin \mathfrak{q}$, f induces a map

$$\tilde{f} : A_{\mathfrak{p}} \rightarrow B_{\mathfrak{q}}, \quad \frac{a}{s} \mapsto \frac{f(a)}{f(s)}.$$

Note that $B_{\mathfrak{p}} = f(S)^{-1}B$ and $f(S) \subseteq T$. $T, f(S)$ are two m.c.s. in B with $Tf(S) = T$. By the two-step localization isomorphism (3.3),

$$B_{\mathfrak{q}} = T^{-1}B = (Tf(S))^{-1}B \cong U^{-1}(f(S)^{-1}B) = U^{-1}B_{\mathfrak{p}},$$

where $U = \{t/1 \in B_{\mathfrak{p}} \mid t \in T\}$. Hence $B_{\mathfrak{q}}$ is a localization of $B_{\mathfrak{p}}$. Since f is flat, (3.10) implies that $A_{\mathfrak{p}} \rightarrow B_{\mathfrak{p}}$ is flat, and since localization preserves flatness, $B_{\mathfrak{p}} \rightarrow B_{\mathfrak{q}}$ is flat. By (2.8.ii), the composite $A_{\mathfrak{p}} \rightarrow B_{\mathfrak{q}}$ is flat. Now $A_{\mathfrak{p}}$ is a local ring with maximal ideal $\mathfrak{p}_{\mathfrak{p}}$. For any element $a/s \in \mathfrak{p}_{\mathfrak{p}}$, where $a \in \mathfrak{p}$ and $s \in S$, we have

$$\tilde{f}(a/s) = \frac{f(a)}{f(s)} \in f(\mathfrak{p})B_{\mathfrak{q}} \subseteq \mathfrak{q}_{\mathfrak{q}}.$$

Thus $(\mathfrak{p}_{\mathfrak{p}})^e \subseteq \mathfrak{q}_{\mathfrak{q}} \neq \langle 1 \rangle$. By 3.16(iii), $B_{\mathfrak{q}}$ is faithfully flat over $A_{\mathfrak{p}}$. Therefore, by 3.16(ii), the induced map $f^* : \text{Spec}(B_{\mathfrak{q}}) \rightarrow \text{Spec}(A_{\mathfrak{p}})$ is surjective.

3.19. Let A be a ring, M an A -module. The **support** of M is defined to be the set $\text{Supp}(M)$ of prime ideals $\mathfrak{p}$ of A s.t. $M_{\mathfrak{p}} \neq 0$. Prove the following:

(i) $M \neq 0 \iff \text{Supp}(M) \neq \emptyset$.

Proof. Prop 3.8.

(ii) $V(\mathfrak{a}) = \text{Supp}(A/\mathfrak{a})$.

Proof. Let $\mathfrak{p} \in V(\mathfrak{a})$, i.e., $\mathfrak{p} \supseteq \mathfrak{a}$. We need to show that $(A/\mathfrak{a})_{\mathfrak{p}} \neq 0$. Note that $(A/\mathfrak{a})_{\mathfrak{p}} \cong A_{\mathfrak{p}}/\mathfrak{a}_{\mathfrak{p}}$, and since $\mathfrak{a} \subseteq \mathfrak{p}$, $\mathfrak{a}_{\mathfrak{p}}$ does not collapse to the entire ring when zooming in $A_{\mathfrak{p}}$, hence $\mathfrak{a}_{\mathfrak{p}} \subsetneq A_{\mathfrak{p}}$, as we desired. However, if $\mathfrak{q} \notin V(\mathfrak{a})$, i.e., $\mathfrak{q} \not\supseteq \mathfrak{a}$, $\mathfrak{a}_{\mathfrak{q}}$ would collapse to the entire local ring $A_{\mathfrak{q}}$ through the localization at $\mathfrak{q}$, making $(A/\mathfrak{a})_{\mathfrak{q}} \cong A_{\mathfrak{q}}/\mathfrak{a}_{\mathfrak{q}} = 0$.

(iii) If $0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$ is an exact sequence, then $\text{Supp}(M) = \text{Supp}(M') \cup \text{Supp}(M'')$.

Proof. Take $\mathfrak{p} \notin \text{Supp}(M)$ (so $M_{\mathfrak{p}} = 0$), we need to show that $\mathfrak{p} \notin \text{Supp}(M') \cap \text{Supp}(M'')$. Note that the following sequence is exact:

$$0 \longrightarrow M'_{\mathfrak{p}} \longrightarrow 0 \longrightarrow M''_{\mathfrak{p}} \longrightarrow 0$$

Forcing $M'_{\mathfrak{p}} = M''_{\mathfrak{p}} = 0$, hence $\mathfrak{p} \notin \text{Supp}(M')$ and $\mathfrak{p} \notin \text{Supp}(M'')$, as desired. Conversely, if $\mathfrak{q} \notin \text{Supp}(M') \cap \text{Supp}(M'')$, then the exactness of the following sequence:

$$0 \longrightarrow 0 \longrightarrow M_{\mathfrak{q}} \longrightarrow 0 \longrightarrow 0$$

forces $M_{\mathfrak{q}} = 0$, hence $\mathfrak{q} \notin \text{Supp}(M)$.

(iv) If $M = \sum M_i$, then $\text{Supp}(M) = \cup \text{Supp}(M_i)$.

Proof. We may assume that each M_i is a submodule of M and I is a direct system of inclusion. Then

$$M_{\mathfrak{p}} \cong A_{\mathfrak{p}} \otimes_A M = A_{\mathfrak{p}} \otimes_A \left(\varinjlim M_i \right) \cong \varinjlim \left(A_{\mathfrak{p}} \otimes_A M_i \right) \cong \varinjlim M_{i\mathfrak{p}} = \sum M_{i\mathfrak{p}}$$

Therefore,

$$\mathfrak{p} \notin \text{Supp}(M) \iff M_{\mathfrak{p}} = 0 \iff \sum M_{i\mathfrak{p}} = 0 \iff M_{i\mathfrak{p}} = 0, \forall i \iff \mathfrak{p} \notin \text{Supp}(M_i), \forall i$$

(v) If M is f.g., then $\text{Supp}(M) = V(\text{Ann}(M))$.

Proof. Let $M = \langle x_1, \dots, x_n \rangle$. Then

$$\begin{aligned} V(\text{Ann}(M)) &= V\left(\bigcap_{i=1}^n \text{Ann}(x_i)\right) = \bigcup_{i=1}^n V(\text{Ann}(x_i)) \stackrel{(ii)}{=} \bigcup_{i=1}^n \text{Supp}(A/\text{Ann}(x_i)) = \bigcup_{i=1}^n \text{Supp}(\langle x_i \rangle) \\ &\stackrel{(iv)}{=} \text{Supp}(M) \end{aligned}$$

(vi) If M, N are f.g., then $\text{Supp}(M \otimes_A N) = \text{Supp}(M) \cap \text{Supp}(N)$.

Proof. Use 2.3.

(vii) If M is f.g. and $\mathfrak{a} \triangleleft A$, then $\text{Supp}(M/\mathfrak{a}M) = V(\mathfrak{a} + \text{Ann}(M))$.

Proof. We have

$$\begin{aligned} \text{Supp}(M/\mathfrak{a}M) &= \text{Supp}(A/\mathfrak{a} \otimes_A M) \stackrel{(vi)}{=} \text{Supp}(A/\mathfrak{a}) \cap \text{Supp}(M) \\ &\stackrel{(ii) \text{ and } (v)}{=} V(\mathfrak{a}) \cap V(\text{Ann}(M)) = V(\mathfrak{a} + \text{Ann}(M)) \end{aligned}$$

- (viii) If $f : A \longrightarrow B$ is a ring homomorphism and M is f.g. A -module, then $\text{Supp}(B \otimes_A M) = f^{*-1}(\text{Supp}(M))$.

Proof. (???) Let $\mathfrak{q} \triangleleft B$, $\mathfrak{p} := \mathfrak{q}^c$. Note that

$$(B \otimes_A M)_{\mathfrak{q}} \cong B_{\mathfrak{q}} \otimes_B (B \otimes_A M) \cong B_{\mathfrak{q}} \otimes_A M \cong B_{\mathfrak{q}} \otimes_{A_{\mathfrak{p}}} A_{\mathfrak{p}} \otimes_A M \cong B_{\mathfrak{q}} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}}$$

Thus, $M_{\mathfrak{p}} = 0 \implies (B \otimes_A M)_{\mathfrak{q}} = 0$, i.e., $\text{Supp}(B \otimes_A M) \subseteq f^{*-1}(\text{Supp}(M))$. Conversely, let $\mathbf{M} := (B \otimes_A M)_{\mathfrak{q}} \cong B_{\mathfrak{q}} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}}$, we need to show $M_{\mathfrak{p}} = 0$. We cannot directly use 2.3. since $B_{\mathfrak{q}}$ is not necessarily null, but we can work in $\mathbf{M}/\mathfrak{m}_S \mathbf{M}$ where $\mathfrak{m}_S$ is the maximal ideal in the local ring $S := B_{\mathfrak{q}}$:

$$\mathbf{M}/\mathfrak{m}_S \mathbf{M} \cong S/\mathfrak{m}_S \otimes_S \mathbf{M} \cong S/\mathfrak{m}_S \otimes_S (S \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}}) \cong S/\mathfrak{m}_S \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}}$$

Hence, $0 = \mathbf{M} \implies 0 = \mathbf{M}/\mathfrak{m}_S \mathbf{M}$. Now, $M_{\mathfrak{p}}$ is f.g. since M is; $S/\mathfrak{m}_S$ is a non-zero field. Hence, by 2.3., $M_{\mathfrak{p}} = 0$.

3.20. Let $f : A \longrightarrow B$ be a ring homomorphism. $f^* : \text{Spec}(B) \longrightarrow \text{Spec}(A)$. Show that:

- (i) Every prime ideal of A is a contracted ideal $\iff f^*$ is surjective.

Proof. Trivial.

- (ii) Every prime ideal of B is an extended ideal $\implies f^*$ is injective.

Proof. Take any $\mathfrak{q} \triangleleft B$ prime, then $\mathfrak{q} = \mathfrak{a}^e$ for some $\mathfrak{a} \triangleleft A$. Now, $\mathfrak{q}^c = \langle 0 \rangle$ implies $\mathfrak{a}^{ec} = \langle 0 \rangle$, but $\mathfrak{a}^{ec} \supseteq \mathfrak{a}$, forcing $\mathfrak{a} = \langle 0 \rangle$. Hence, $\mathfrak{q} = \langle 0 \rangle$.

3.21.

- (i) Let $\phi : A \longrightarrow S^{-1}A$ be the canonical localization map. Show that $\phi^* : \text{Spec}(S^{-1}A) \longrightarrow \text{Spec}(A)$ is a homeomorphism of $\text{Spec}(S^{-1}A)$ onto its image in $X = \text{Spec}(A)$. Let this image be denoted by $S^{-1}X$. In particular, if $f \in A$, the image of $\text{Spec}(A_f)$ in X is the basic open set X_f .

Proof. By Prop 3.11(i), every ideal in $S^{-1}A$ is an extended ideal, of the form $S^{-1}\mathfrak{a}$ for some $\mathfrak{a} \triangleleft A$ not touching S . Hence, by 3.20.(ii), ϕ^* is injective and hence $\phi^* : \text{Spec}(S^{-1}A) \longrightarrow S^{-1}X$ is a bijection. We know ϕ^* is continuous, so it suffices to show its inverse is also continuous. Note that $\phi^*(V(\mathfrak{a}^e)) = V(\mathfrak{a}) \cap S^{-1}X$, so we are done. In particular, let $\phi : A \longrightarrow A_f$. $S = \{1, f, f^2, \dots\}$. So $\text{Im } \phi^* = \{\mathfrak{p} \in \text{Spec}(A) \mid f \notin \mathfrak{p}\} = X_f$.

- (ii) Let $f : A \longrightarrow B$ ring homomorphism. $f^* : Y := \text{Spec}(B) \longrightarrow X := \text{Spec}(A)$. Identifying $\text{Spec}(S^{-1}A)$ with its canonical image $S^{-1}X$ in X and $\text{Spec}(S^{-1}B) := \text{Spec}(f(S)^{-1}B)$ with its canonical image $S^{-1}Y$ in Y . Show that $S^{-1}f^* : \text{Spec}(S^{-1}B) \longrightarrow \text{Spec}(S^{-1}A)$ is the restriction of f^* to $S^{-1}Y$, and that $S^{-1}Y = f^{*-1}(S^{-1}X)$.

Proof. We first show that $S^{-1}Y = f^{*-1}(S^{-1}X)$. Indeed, for $\mathfrak{q} \in \text{Spec}(B)$, $\mathfrak{q} \in S^{-1}Y \iff \mathfrak{q} \cap f(S) = \emptyset \iff f^{-1}(\mathfrak{q}) \cap S = \mathfrak{q}^c \cap S = \emptyset \iff f^*(\mathfrak{q}) \in S^{-1}X$. And the following diagram commutes and tells everything:

$$\begin{array}{ccc} \text{Spec}(A) & \xleftarrow{f^*} & \text{Spec}(B) \\ \uparrow \phi_A^* & & \uparrow \phi_B^* \\ \text{Spec}(S^{-1}A) & \xleftarrow{S^{-1}f^*} & \text{Spec}(S^{-1}B) \end{array} \quad \begin{array}{ccc} \mathfrak{q}^c & \xleftarrow{f^*} & \mathfrak{q} \\ \uparrow \phi_A^* & & \uparrow \phi_B^* \\ S^{-1}\mathfrak{q}^c & \xleftarrow{S^{-1}f^*} & f(S)^{-1}\mathfrak{q} \end{array}$$

- (iii) Let $\mathfrak{a} \triangleleft A$, $\mathfrak{b} := \mathfrak{a}^e \triangleleft B$. Let $\bar{f} : A/\mathfrak{a} \rightarrow B/\mathfrak{b}$ be the homomorphism induced by f . If $\text{Spec}(A/\mathfrak{a})$ is identified with its canonical image $V(\mathfrak{a})$ in X , and $\text{Spec}(B/\mathfrak{b})$ with its image $V(\mathfrak{b})$ in Y . Show that $\bar{f}^*$ is the restriction of f^* to $V(\mathfrak{b})$.

Proof. First, we need to check $f^*(V(\mathfrak{b})) \subseteq V(\mathfrak{a})$. Indeed, for any $\mathfrak{q} \supseteq \mathfrak{b}$, $f^*(\mathfrak{q}) = \mathfrak{q}^c \supseteq \mathfrak{b}^c = \mathfrak{a}^{ec} \supseteq \mathfrak{a}$, hence $f^*(\mathfrak{q}) \in V(\mathfrak{a})$. Let $\pi_A : A \twoheadrightarrow A/\mathfrak{a}$; $\pi_B : B \twoheadrightarrow B/\mathfrak{b}$. Then it is easy to check $\pi_B \circ f = \bar{f} \circ \pi_A$, i.e. the following diagram (left) commutes. Hence, by the functorial property, $f^* \circ \pi_B^* = \pi_A^* \circ \bar{f}^*$, i.e., the following diagram (right) commutes. Moreover, by the correspondence theorem, π_A^*, π_B^* are injective.

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \pi_A \downarrow & & \downarrow \pi_B \\ A/\mathfrak{a} & \xrightarrow{\bar{f}} & B/\mathfrak{b} \end{array} \qquad \begin{array}{ccc} \text{Spec}(A) & \xleftarrow{f^*} & \text{Spec}(B) \\ \pi_A^* \uparrow & & \uparrow \pi_B^* \\ \text{Spec}(A/\mathfrak{a}) & \xleftarrow{\bar{f}^*} & \text{Spec}(B/\mathfrak{b}) \end{array}$$

Now, let $\mathfrak{q} \in \text{Spec}(B/\mathfrak{b})$, and regard it as a prime ideal of B containing $\mathfrak{b}$. Under the identification $\text{Spec}(B/\mathfrak{b}) = V(\mathfrak{b})$ and $\text{Spec}(A/\mathfrak{a}) = V(\mathfrak{a})$, we have $\pi_B^*(\mathfrak{q}) = \mathfrak{q}$ and $\pi_A^*(\Delta) = \Delta$. Then, since the diagram commutes

$$\pi_A^*(\bar{f}^*(\mathfrak{q})) = \bar{f}^*(\mathfrak{q}) = f^*(\pi_B^*(\mathfrak{q})) = f^*(\mathfrak{q})$$

Hence $\bar{f}^*$ coincides with the restriction of f^* to $V(\mathfrak{b})$.

- (iv) Let $\mathfrak{p}$ be a prime ideal of A . Take $S = A - \mathfrak{p}$ in (ii) and then reduce mod $S^{-1}\mathfrak{p}$ as in (iii). Deduce that the subspace $f^{*-1}(\mathfrak{p})$ of Y is naturally homeomorphic to $\text{Spec}(B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}) = \text{Spec}(k(\mathfrak{p}) \otimes_A B)$, where $k(\mathfrak{p})$ is the residue field of the local ring $A_{\mathfrak{p}}$. $\text{Spec}(k(\mathfrak{p}) \otimes_A B)$ is called the **fiber** of f^* over $\mathfrak{p}$.

Proof. Denote $\mathfrak{p}A_{\mathfrak{p}}$ as $\mathfrak{p}_{\mathfrak{p}}$. By (ii) and (iii) we have the following commutative diagram:

$$\begin{array}{ccc} \text{Spec}(A) & \xleftarrow{f^*} & \text{Spec}(B) \\ \uparrow & & \uparrow \\ \text{Spec}(A_{\mathfrak{p}}) & \xleftarrow{f_{\mathfrak{p}}^*} & \text{Spec}(B_{\mathfrak{p}}) \\ \uparrow & & \uparrow \\ \text{Spec}(A_{\mathfrak{p}}/\mathfrak{p}_{\mathfrak{p}}) & \xleftarrow{\bar{f}_{\mathfrak{p}}^*} & \text{Spec}(B_{\mathfrak{p}}/\mathfrak{p}_{\mathfrak{p}}) \end{array}$$

Now, by the identifications, elements in $\text{Spec}(A_{\mathfrak{p}}/\mathfrak{p}_{\mathfrak{p}})$ correspond to the elements in $\text{Spec}(A_{\mathfrak{p}})$ that are contained in $\mathfrak{p}$, and elements in $\text{Spec}(A_{\mathfrak{p}})$ correspond to the elements in $\text{Spec}(A)$ that contain $\mathfrak{p}$. Note that the only element in $\text{Spec}(A_{\mathfrak{p}}/\mathfrak{p}_{\mathfrak{p}})$ is 0 since $\text{Spec}(A_{\mathfrak{p}}/\mathfrak{p}_{\mathfrak{p}}) =: k(\mathfrak{p})$ is

a field. Therefore, 0 corresponds to the singleton $\mathfrak{p} \in \text{Spec}(A)$ by the identifications. So we have the following:

$$\begin{array}{ccc}
 \mathfrak{p} & \xleftarrow{f^*} & f^{*-1}(\mathfrak{p}) \\
 \uparrow \text{identification} & & \uparrow \text{identification} \\
 \langle 0 \rangle & \xleftarrow{\overline{f}_{\mathfrak{p}}^*} & \text{Spec}(B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}})
 \end{array}$$

And this identification is a homeomorphism. To show $\text{Spec}(B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}) = \text{Spec}(k(\mathfrak{p}) \otimes_A B)$, We need to first clarify something:

- (a) Just like $A/\mathfrak{a}$ can be viewed as an A -module, $S^{-1}A$ can also be viewed as an A -module by $x \diamond (a/s) := xa/s$. In particular, $A_{\mathfrak{p}}$, therefore, can be viewed as an A -module.
- (b) What is $B_{\mathfrak{p}}$? By definition, $B_{\mathfrak{p}} = f(S)^{-1}B$. Since we know that B can be viewed as an A -module (actually, an A -algebra, with $f : A \rightarrow B$), $B_{\mathfrak{p}}$ is a "fractional" A -module. Hence $\mathfrak{p}B_{\mathfrak{p}}$ is just a submodule of $B_{\mathfrak{p}}$.

Therefore

$$A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \otimes_A B \stackrel{2.2}{\cong} A/\mathfrak{p} \otimes_A A_{\mathfrak{p}} \otimes_A B \cong A/\mathfrak{p} \otimes_A B_{\mathfrak{p}} \stackrel{2.2}{\cong} B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$$

3.22. Let A be a ring and $\mathfrak{p}$ a prime ideal of A . Then the canonical image of $\text{Spec}(A_{\mathfrak{p}})$ in $\text{Spec}(A)$ is equal to the intersection of all the open neighborhoods of $\mathfrak{p}$ in $\text{Spec}(A)$.

Proof. Let $h : A \rightarrow A_{\mathfrak{p}}$ be the canonical localization map. Notice that

1. $\text{Im } h^* = \{\mathfrak{q} \in \text{Spec}(A) \mid \mathfrak{q} \subseteq \mathfrak{p}\}$.
2. We know $\{X_f\}_{f \in A}$ forms an open basis of $\text{Spec}(A)$, so every open neighborhood of $\mathfrak{p}$ contains a X_f where $f \notin \mathfrak{p}$. Therefore, the intersection of all the open neighborhoods of $\mathfrak{p}$ equals $\bigcap_{f \notin \mathfrak{p}} X_f$.

The remaining is just tedious verification.

3.2.3 Presheaves, sheaves, and the constructible topology

3.23. Let $X := \text{Spec}(A)$ and let U be a basic open set in X .

- (i) If $U = X_f$, show that the ring $A(U) := A_f$ depends only on U and not on f .

Proof. We need to show that $X_f = X_g \implies A_f \cong A_g$. Indeed,

$$X_f = X_g \stackrel{1.17}{\iff} \text{rad}(\langle f \rangle) = \text{rad}(\langle g \rangle) \implies f \in \text{rad}(\langle g \rangle) \text{ and } g \in \text{rad}(\langle f \rangle)$$

This implies $f^m = ug$ and $g^n = vf$ for some positive integer m, n and some $u, v \in A$. Now define

$$\phi_{fg} : A_f \rightarrow A_g \quad \frac{a}{f} \mapsto \frac{av}{g^n} \quad \text{and} \quad \phi_{gf} : A_g \rightarrow A_f \quad \frac{a}{g} \mapsto \frac{au}{f^m}$$

Verify that they are well defined and $\phi_{fg} \circ \phi_{gf} = \text{id}$, $\phi_{gf} \circ \phi_{fg} = \text{id}$.

- (ii) Let $U' = X_g$ be another basic open set s.t. $U' \subseteq U$. Show that there is an equation of the form $g^n = uf$ for some positive integer n and some $u \in A$. Use this to define a homomorphism $\rho : A(U) \longrightarrow A(U')$ by $a/f^m \mapsto au^m/g^{mn}$. Show that ρ depends only on U and U' . This homomorphism is called the **restriction** homomorphism.

Proof. Similar to (i), we have:

$$X_g \subseteq X_f \implies V(\langle g \rangle) \supseteq V(\langle f \rangle) \implies V(\text{rad}(\langle g \rangle)) \supseteq V(\text{rad}(\langle f \rangle)) \implies g \in \text{rad}(\langle f \rangle)$$

This implies that $g^n = uf$ for some positive integer n and some $u \in A$. We define the same ρ as in (i).

- (iii) If $U = U'$, then ρ is the identity map.

Proof. See (i).

- (iv) If $U \supseteq U' \supseteq U''$ are basic open sets in X , show that the diagram commutes:

$$\begin{array}{ccc} A(U) & \xrightarrow{\quad} & A(U'') \\ & \searrow & \nearrow \\ & A(U') & \end{array}$$

Proof. Let $U := X_f, U' := X_g$, and $U'' := X_h$. Then it is straightforward to check the commutativity:

$$\begin{array}{ccc} A_f & \xrightarrow{\rho_{fh}} & A_h \\ & \searrow \rho_{fg} & \nearrow \rho_{gh} \\ & A_g & \end{array}$$

- (v) Let $x = \mathfrak{p}$ be a point in X . Show that

$$\varinjlim_{x \in U} A(U) \cong A_{\mathfrak{p}}$$

Proof. Note that $\mathfrak{p} \in X_f \iff f \in A \setminus \mathfrak{p} =: S$. Working in the category of A -algebra, we have

$$\varinjlim A[1/s] \cong S^{-1}A$$

And the desired isomorphism follows.

Remark 23. The assignment of the ring $A(U)$ to each basic open set U of X , and the restriction homomorphisms ρ , satisfying conditions (iii) and (iv) above, constitutes a **presheaf of rings** on the basis of open sets $\{X_f\}_{f \in A}$. (v) says that the **stalk** of this presheaf at $x \in X$ is the corresponding local ring $A_{\mathfrak{p}}$.

3.24. Show that the presheaf of Exercise 23 has the following property: Let $\{U_i\}_{i \in I}$ be a covering of X by basic open sets. For each $i \in I$, let $s_i \in A(U_i)$ be such that, for each pair of indices i, j , the images of s_i and s_j in $A(U_i \cap U_j)$ are equal. Then $\exists! s \in A = A(X)$ whose image in $A(U_i)$ is $s_i, \forall i \in I$.

Proof. Note that X is quasi-compact, so we may assume that the index set I is finite, say $[n]$. So we write

$$X = X_{f_1} \cup \cdots \cup X_{f_n} = X_{f_1^N} \cup \cdots \cup X_{f_n^N}$$

for any positive integer N . This implies $\exists d_1, \dots, d_n \in A$ s.t. $1_A = \sum_{i=1}^n d_i f_i^N$. Note that each $s_i \in A$ can be written as $s_i = b_i / f_i^r$ for some common positive integer r . Now, $s_i = s_j$ in $A_{f_i f_j}$, so

$$\frac{b_i}{f_i^r} = \frac{b_j}{f_j^r}$$

in $A_{f_i f_j}$. That is, there exists a (large enough) integer k s.t. for any $i, j \in [n]$,

$$(f_i f_j)^k (b_i f_j^r - b_j f_i^r) = 0$$

Now, let $N := r + k$ and let $c_i := b_i f_i^k$ for all $i \in [n]$. So

$$\begin{aligned} b_i f_i^k f_j^N &= b_j f_j^k f_i^N \\ c_i f_j^N &= c_j f_i^N \end{aligned}$$

Now, define:

$$s := \sum_{i=1}^n d_i c_i \in A$$

We claim that s is the unique element in A whose image in A_{f_i} is s_i for every $i \in [n]$. Indeed, in A we have:

$$f_j^N s = f_j^N \sum_{i=1}^n d_i c_i = \sum_{i=1}^n d_i f_j^N c_i = \sum_{i=1}^n d_i f_i^N c_j = c_j \sum_{i=1}^n d_i f_i^N = c_j$$

Hence, in A_{f_j} we have:

$$s = \frac{c_j}{f_j^N} = s_j$$

as desired. To see the uniqueness, suppose that both $s, t \in A$ satisfy the requirement. Then $a := s - t$ is zero in each A_{f_i} . That is, for each $i \in [n], \exists m_i$ s.t. $f_i^{m_i} a = 0$. Let m be a common one. Then

$$f_i^m a = 0$$

for all $i \in [n]$. We know $\langle f_1^m, \dots, f_n^m \rangle = A$. That is, $\exists e_1, \dots, e_n \in A$ s.t.

$$1_A = \sum_{i=1}^n e_i f_i^m$$

Thus

$$a = \sum_{i=1}^n e_i f_i^m a = 0$$

So we are done.

Remark 24. This essentially implies that the presheaf is a **sheaf**.

3.25. Let $f : A \longrightarrow B, g : A \longrightarrow C$ be ring homomorphisms and let $h : A \longrightarrow B \otimes_A C$ be defined as $x \mapsto f(x) \otimes g(x)$. Let X, Y, Z, T be the prime spectra of $A, B, C, B \otimes_A C$, resp. Then $h^*(T) = f^*(Y) \cap g^*(Z)$.

Proof. Follow the hint. Let $\mathfrak{p} \in X$, and let $k := A_{\mathfrak{p}}$ be the residue field at $\mathfrak{p}$. By 3.21, the preimage $h^{*-1}(\mathfrak{p})$ is homeomorphic to the spectrum of

$$(B \otimes_A C) \otimes_A k \cong (B \otimes_A k) \otimes_k (C \otimes_A k)$$

Hence,

$$\mathfrak{p} \in h^*(T) \iff (B \otimes_A k) \otimes_k (C \otimes_A k) \neq 0 \iff (B \otimes_A k) \neq 0 \text{ and } (C \otimes_A k) \neq 0$$

But this is equivalent of saying,

$$\mathfrak{p} \in f^*(Y) \cap g^*(Z)$$

for the same reason, as desired.

3.26. Let $(B_{\alpha}, g_{\alpha\beta})$ be a direct system of rings and B the direct limit. For each α , let $f_{\alpha} : A \longrightarrow B_{\alpha}$ be a ring homomorphism s.t. $g_{\alpha\beta} \circ f_{\alpha} = f_{\beta}$ whenever $\alpha \preceq \beta$. The f_{α} induce $f : A \longrightarrow B$. Show that

$$f^*(\text{Spec}(B)) = \bigcap_{\alpha} f_{\alpha}^*(\text{Spec}(B_{\alpha}))$$

Proof. The strategy follows similarly from the previous exercise. Let $\mathfrak{p} \in \text{Spec}(A)$ and let $k := A_{\mathfrak{p}}$ be the residue field at $\mathfrak{p}$. Then $f^{*-1}(\mathfrak{p})$ is homeomorphic to the spectrum of

$$B \otimes_A k = (\varinjlim B_{\alpha}) \otimes_A k \cong \varinjlim (B_{\alpha} \otimes_A k)$$

Hence,

$$\mathfrak{p} \in f^*(\text{Spec}(B)) \iff \varinjlim (B_{\alpha} \otimes_A k) \neq 0 \iff B_{\alpha} \otimes_A k \neq 0$$

for each α by 3.21. But this is equivalent of saying

$$\mathfrak{p} \in f_{\alpha}^*(\text{Spec}(B_{\alpha}))$$

for every α , as desired.

3.27.

(i) Let $f_{\alpha} : A \longrightarrow B_{\alpha}$ be any family of A -algebras and let $f : A \longrightarrow B$ be their tensor product over A (See 2.21). Then

$$f^*(\text{Spec}(B)) = \bigcap_{\alpha} f_{\alpha}^*(\text{Spec}(B_{\alpha}))$$

Proof. Recall that in 2.23 we have

$$B := \bigotimes_{\lambda \in \Lambda} B_{\lambda} = \varinjlim_{J \subseteq \Lambda, J \text{ finite}} B_J$$

Now, for each $B_J = \bigotimes_{\alpha \in J} B_\alpha$, by 3.25 we have

$$f_J^*(\text{Spec}(B_J)) = \bigcap_{\alpha \in J} f_\alpha^*(\text{Spec}(B_\alpha))$$

By 3.26 we have

$$f^*(\text{Spec}(B)) = \bigcap_J f_J^*(\text{Spec}(B_J)) = \bigcap_J \bigcap_{\alpha \in J} f_\alpha^*(\text{Spec}(B_\alpha)) = \bigcap_\alpha f_\alpha^*(\text{Spec}(B_\alpha))$$

as desired.

- (ii) Let $f_\alpha : A \longrightarrow B_\alpha$ be any finite family of A -algebras and let $B := \prod_\alpha B_\alpha$. Define $A \longrightarrow B$ by $f(x) = (f_\alpha(x))$. Then

$$f^*(\text{Spec}(B)) = \bigcup_\alpha f_\alpha^*(\text{Spec}(B_\alpha))$$

Proof. By 1.22., $\text{Spec}(B) = \bigsqcup_\alpha X_\alpha$ where each X_α is canonically isomorphic to $\text{Spec}(B_\alpha)$. Let

$$\pi_\alpha : B \longrightarrow B_\alpha$$

Then we have

$$\begin{aligned} \pi_\alpha^* : \text{Spec}(B_\alpha) &\rightarrow \text{Spec}(B) \\ \mathfrak{q} &\mapsto \pi_\alpha^{-1}(\mathfrak{q}) = \prod_{\beta \neq \alpha} B_\beta \times \mathfrak{q} \end{aligned}$$

By 1.22., for each $\mathfrak{p} \in \text{Spec}(B)$, $\exists! \alpha$ and some $\mathfrak{q} \in \text{Spec}(B_\alpha)$ s.t. $\mathfrak{p} = \pi_\alpha^{-1}(\mathfrak{q})$. Now

$$f^*(\mathfrak{p}) = f^{-1}(\mathfrak{p}) = f^{-1}(\pi_\alpha^{-1}(\mathfrak{q})) = f_\alpha^{-1}(\mathfrak{q}) = f_\alpha^*(\mathfrak{q})$$

Hence $f^*(\text{Spec}(B)) \subseteq \bigcup_\alpha f_\alpha^*(\text{Spec}(B_\alpha))$. The other direction follows immediately.

- (iii) Hence the subsets of $X = \text{Spec}(A)$ of the form $f^*(\text{Spec}(B))$, where $f : A \longrightarrow B$ is a ring homomorphism, satisfy the axioms for closed sets in a topological space. The associated topology is the **constructible topology** on X . It is finer than the Zariski topology.

Proof. To see this, it suffices to show that every closed set in Zariski topology is also closed in the constructible topology. Let $\pi : A \longrightarrow A/\mathfrak{a}$. Note that

$$V(\mathfrak{a}) \cong \pi^*(\text{Spec}(A/\mathfrak{a}))$$

so we are done.

- (iv) Let X_c denote the set X endowed with the constructible topology. Show that X_c is quasi-compact.

Proof. It suffices to show that any collection of closed sets with empty intersection has a finite sub-collection with empty intersection. This can be seen immediately from the proof of 3.27.(i).

3.28. *Continuation of 3.27.* From now on, let ζ denote the Zariski topology and let τ denote the constructible topology, whenever the ground set X is clear.

- (i) *For each $g \in A$, the set X_g is both open and closed in the constructible topology.*

Proof. Since the τ is finer than the ζ , X_g is open in τ ; To see that X_g is closed in τ , let

$$f : A \longrightarrow A_g = S^{-1}A$$

be the canonical localization map, where $S = \{1_A, g, g^2, \dots\}$. Note that X_g consists of all prime ideals of A that do not touch S . Hence, by Proposition 3.11.(iv),

$$X_g = f^*(A_g)$$

Hence, X_g is closed in τ .

- (ii) *Let C' denote the smallest topology on X for which the sets X_g are both open and closed. Let $X_{C'}$ denote the set X endowed with this topology. Show that $X_{C'}$ is Hausdorff.*

Proof. Let $\mathfrak{p}, \mathfrak{q} \in X$ where $\mathfrak{p} \neq \mathfrak{q}$. Take $g \in \mathfrak{q} \setminus \mathfrak{p}$. Then $\mathfrak{q} \in V(\langle g \rangle)$ and $\mathfrak{q} \in X_g$. But since X_g is both open and closed, we are done.

- (iii) *Deduce that the identity map $\text{id} : X_C \longrightarrow X_{C'}$ is a homeomorphism. Hence, a subset $E \subseteq X$ is of the form $f^*(\text{Spec}(B))$ for some $f : A \longrightarrow B$ if and only if it is closed in C' .*

Proof. Note that C is finer than C' . Hence, id is continuous (and bijective, of course). 3.27.(iv) says that X_C is quasi-compact; 3.28.(ii) says that $X_{C'}$ is Hausdorff. Hence, by [Mun00] Theorem 26.6., id is a homeomorphism.

- (iv) *X_C is compact, Hausdorff, and totally disconnected.*

Proof. It remains to show that X_C is totally disconnected. Since $X_C \cong X_{C'}$, it is equivalent to show that $X_{C'}$ is totally disconnected. Let $\mathfrak{p}, \mathfrak{q} \in X$ where $\mathfrak{p} \neq \mathfrak{q}$. By (ii), any subset $E \subseteq X$ containing $\mathfrak{p}$ and $\mathfrak{q}$ can be separated by $X_g \cap E$ and $V(\langle g \rangle) \cap E$ for some $g \in A$. Hence, the only connected subsets are singletons, as desired.

3.29. *Let $f : A \longrightarrow B$ be a ring homomorphism. Show that $f^* : \text{Spec}(B) \longrightarrow \text{Spec}(A)$ is a continuous closed map for the constructible topology.*

Proof. Let $E = g^*(\text{Spec}(C))$ be a closed subset in $\text{Spec}(B)$ where $g : B \longrightarrow C$ is a ring homomorphism. Note that $(g \circ f)^* = f^* \circ g^*$. Hence, $f^*(E)$ is closed in $\text{Spec}(A)$ and f^* is therefore a closed map. The continuity of f^* has been shown in 1.21.(i).

3.30. *Show that the Zariski topology and the constructible topology are the same on $\text{Spec}(A)$ if and only if $A/\mathfrak{A}$ is absolutely flat.*

Proof. Denote $X := \text{Spec}(A)$. Recall that by 3.11., $A/\mathfrak{A}$ is absolutely flat $\iff X_\zeta$ is Hausdorff. Let $\text{id} : X_\tau \longrightarrow X_\zeta$ be the identical map. Since τ is finer than ζ , id is continuous; By 3.27.(iv), X_τ is quasi-compact. Hence, if X_ζ is Hausdorff, then by [Mun00] Theorem 26.6., id is a homeomorphism. The converse direction is immediate.

References

- [AM69] M.F. Atiyah and I.G. Macdonald. *Introduction to Commutative Algebra*. Addison-Wesley series in mathematics. Addison-Wesley Publishing Company, 1969.
- [CHA23] MELODY CHAN. Course notes for math 2520 graduate algebra. 2023.
- [ECOH] LINK EXERCISES, DO CHAPTER, BODY OMISSION, and COMPLETION FIX TENSOR HEIGHT. Fix references.
- [Gat13] Andreas Gathmann. Commutative algebra. *Class Notes TU Kaiserslautern*, 2014:2013, 2013.
- [hs] Balazs Strenner (https://mathoverflow.net/users/11214/balazs_strenner). Atiyah-macdonald, exercise 2.11. MathOverflow. URL:<https://mathoverflow.net/q/47846> (version: 2010-12-01).
- [Mun00] J.R. Munkres. *Topology*. Prentice Hall, 2000.